POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Mabini Campus, Manila, Philippines

INSTRUCTIONAL MATERIAL FOR PATHFIT 1







PHYSICAL ACTIVITY TOWARDS HEALTH AND FITNESS 1

MOVEMENT COMPETENCY TRAINING

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Overview

This Instructional Materials for Physical Activity Towards Health and Fitness (PATHFit) 1: Movement Competency Training is a resource material designed for the students Polytechnic University of the Philippines.

This course reintroduces the fundamental movement patterns that consist of non-locomotor and locomotor skills, which are integrated with core training to meet the demands of functional fitness and physical activity performance. Emphasis will be on exercise regression and progression for the enhancement of fitness and the adaptation of movement competencies to independent physical activity pursuits. In conjunction with fitness and wellness concepts, exercise, and healthy eating principles, the periodic evaluation will be conducted of one's level of fitness and physical activity, as well as eating patterns to monitor one's progress and achievement of personal fitness and dietary goals. Adapted PE is included in this course to ensure the physical fitness of students with additional needs (SWAN).

As the final requirement of the course, the students are expected to develop, implement, and evaluate their own home-based fitness program that is crafted based on their baseline fitness level, SMART goals, and their available resources.

Program Outcomes

Active and Health Living

1. Participate in moderate to vigorous physical activities (MVPAs) in accordance with national and global recommendations on physical activities for health.
2. Adapt movement competencies to independent physical activity pursuits that are health- enhancing and personally rewarding.
3. Monitor progress and evaluate achievement of personal fitness and dietary goals.
4. Devise, apply and assess a range of strategies (i.e., exercise program design; goal-setting; periodic evaluation) to improve one's physical activity performances and those of others.

Advocacy and Action

1. Enhance and advocate for one's personal and others' fitness, safety, and wellness through physical activity participation and/or leadership.
2. Promote practical and creative interventions that will create community connection and contribute to the health and wellness of the school community and its periphery.

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COURSE TITLE	**Physical Activity Towards Health and Fitness 1: Movement Competency Training**
COURSE CODE	PATHFIT 1
CREDIT UNITS	2 Units
COURSE PREREQUISITE	None
COURSE DESCRIPTION	This course reintroduces the fundamental movement patterns that consist of non-locomotor and locomotor skills, which are integrated with core training to meet the demands of functional fitness and physical activity performance. Emphasis will be on exercise regression and progression for the enhancement of fitness and the adaptation of movement competencies to independent physical activity pursuits. In conjunction with fitness and wellness concepts, exercise, and healthy eating principles, the periodic evaluation will be conducted of one's level of fitness and physical activity, as well as eating patterns to monitor one's progress and achievement of personal fitness and dietary goals. Adapted PE is included in this course to ensure the physical fitness of students with additional needs (SWAN).

Institutional Learning Outcomes	Program Outcomes	Course Outcomes
1. Critical and Creative Thinking. Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world. 2. Effective Communication. Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances. 3. Strong Service Orientation. Graduates exemplify strong commitment to service excellence for the people, the clientele, industry and other sectors. 4. Adept and Responsible Use or Development of Technology. Graduates demonstrate optimized and responsible use of state of the art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills. 5. Passion for Lifelong Learning. Graduates perform and function in the society by taking responsibility in their quest for further improvement through lifelong learning. 6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility – Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.	**Active and Health Living** 1. Participate in moderate to vigorous physical activities (MVPAs) in accordance with national and global recommendations on physical activities for health. 2. Adapt movement competencies to independent physical activity pursuits that are health- enhancing and personally rewarding. 3. Monitor progress and evaluate achievement of personal fitness and dietary goals. 4. Devise, apply and assess a range of strategies (i.e., exercise program design; goal-setting; periodic evaluation) to improve one's physical activity performances and those of others. **Advocacy and Action** 1. Enhance and advocate for one's personal and others' fitness, safety, and wellness through physical activity participation and/or leadership. 2. Promote practical and creative interventions that will create community connection and contribute to the health and wellness of the school community and its periphery.	1. Demonstrate appreciation on the importance of regular engagement in moderate to vigorous physical activities and sports 2. Improve fitness from baseline levels through active participation in different moderate to vigorous physical activities 3. Demonstrate movement competencies in performing the different exercise movements 4. Execute proper techniques in performing resistance training and other home-based exercises 5. Evaluate a personal food log based on dietary recommendations 6. Plan, implement, and evaluate personal exercise program

Course Plan					
Week	***Topic***	***Learning Outcomes***	***Methodology***	***Resources***	***Assessment***
Week 1	**Orientation** on: • Course Content • School and Classroom Policies • Subject Requirements and Grading System • Fitness Journal	• Understand the school and classroom policies as well as the grading system • Know the different subject requirements • Recognize one's fitness status through an honest declaration of their fitness condition in their "Fitness Journal"	Lecture Discussion Printed Lectures Video andVisual Presentation	• PUP Student Hand Book 2019 Revised Edition • Revised OBE Syllabus • Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. • Fitness Journal. Carlos, A.G., Dela Cruz L., Lubis F., Enriquez A., & Delfinado J.	Oral Recitation Self-Assessment of the use of PARQ in the Fitness Journal
Week 2	**Chapter 1** **PHYSICAL ACTIVITY AND EXERCISE** • 8 Dimensions of Wellness • Physical Activity and Exercise • Ways in strengthening immune system • Staying Active in the New Normal	• Display understanding on the importance of active lifestyle • Explain the 8 dimension of wellness and its importance in achieving better quality of life. • Recognize the non-communicable diseases (NCDs) as global health problems related to sedentary lifestyle • the different ways to stay active in the New Normal	Lecture Discussion Focus Group Discussion Role Playing	• Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness, Fourteenth Edition. Cengage Learning, USA • Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York • Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. • Puen, D. A. Y., Cobar, A. G. C., Dimarucot, H.C., (2022) Perceived Barriers to Physical Activity of College Students in Manila, Philippines during the COVID-19 Community Quarantine: An Online Survey. Sport Mont, 19(2), 101-106. doi: 10.26773/smj.210617 • Camarador, R.A., Dela Cruz, L. A., Serrano, S.S., Banadera, S.P., (2022) Me and My Fitness During COVID 19 Pandemic: Anthropometric, Physical Activity, and Wellness Lifestyle Habits of Collegiate Students. Polytechnic University of the Philippines, College of Human Kinetics	Written Test (Activity Sheet) Reflective Essay Performance Task (Role Playing)
Week 3	**Chapter 2** **INTRODUCTION TO ANATOMY** • Organ Systems • Anatomical Landmarks, Directions, Body Regions • Bones & Muscles • Joints & Types of Anatomical Movements	• Discuss the importance of the different organ systems • Identify the different anatomical landmarks, directions and body regions that are essentials to physical activities and exercise engagement • Understand the major	Lecture Demonstration Video and Visual Presentation	• Floyd, R.T. (2015). Manual of structural kinesiology. New York, NY : McGraw-Hill Education. • Patton, Kevin T. and Thibodeau, Anthony, Gary A., (2010) Textbook of Anatomy and Physiology, 19th Edition, Mosby Elsevier • Mitchell, Tommy, (2015) Introduction to Anatomy & Physiology, 1st Edition, New Leaf Publishing Group Inc.	Written Test Reflective Essay Performance Test Quiz Bee

		divisions and functions of skeletal system • Know the different major group of muscles and its functions • Apply different anatomical movements		• Martini, Frederic H., Nath, Judi I., Bartholomew, Edwin F., (2015) Fundamentals of Anatomy & Physiology, 10th Edition, Pearson Education Inc.	
Week 4-5	**Chapter 3** **EXERCISE PRESCRIPTION** • Physical Activity Pyramid • Three Phases of Exercise Program • Principles of Training Exercise • The FITT Formula • Exercise Prescription Guidelines • How to Determine Cardio Respiratory Training Zone	• Demonstrate understanding of basic principles on exercise prescription. • Apply the FITT formula to the creation of an exercise program. • Determine their Cardio Respiratory Training Zone • Create and perform exercise workout that are relatively safe to perform	Printed Lectures Video and Visual Presentation Lecture Demonstration	• Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness,Fourteenth Edition. Cengage Learning, USA • Hoeger, W.W.K. (2015). Lifetime physical fitness & wellness : a personalized program (13th ed.). Stamford, Connecticut : Cengage Learning. • Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York • Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. Polytechnic University of the Philippines, College of Human Kinetics	Written Test (Activity Sheet) Reflective Essay Performance Task 1 (Simple Exercise Workout) Performance Task 2 (Cardiorespiratory Exercise Prescription)
Week 6	**Chapter 4** **PHYSICAL FITNESS AND SELF-TESTING ACTIVITIES** • Physical Fitness • Importance of Physical Fitness Assessment • Things to Consider in Self-Testing Activity • Measuring Health-Related Physical Fitness o Body Composition o Cardio Respiratory Fitness o Flexibility o Muscular Strength and Endurance	• Identify the different health-related fitness tests • Explain the importance of physical fitness assessment • Demonstrate proper skills and confidence on administering physical fitness test • Establish baseline data (pre-test score) • Display acceptance and understanding on their strength and weaknesses	Printed Lectures Video and Visual Presentation	• Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness,Fourteenth Edition. Cengage Learning, USA • Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York • Hoeger, W.W.K. (2015). Lifetime physical fitness & wellness : a personalized program (13th ed.). Stamford, Connecticut : Cengage Learning. • Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. Polytechnic University of the Philippines, College of Human Kinetics	Written Test (Activity Sheet) Performance Task (Fitness Tests) Reflective Essay
Week	**Chapter 5** **Non-Locomotor**	• Identify the different types of non-	Data Bases and Search Engines	• Camarador, R., Camarador L., Serrano, J.A., & Mantala,	Written Test

7-8	**Movements** Bracing the Core, Bird Dog Series, Dead Bug Series, Rolling, Plank series, Cobra Stretch, Squat series, scapular protraction & retraction and other movements, Lunges	locomotor movements • Execute the non-locomotor movements effectively • Design a specific activity workout using the given movement • Apply basic exercise principles in creating sample exercise workout	Printed Lectures Video and Visual Demonstration	J. (2020) Instructional Material on Movement Competency Training. Polytechnic University of the Philippines, College of Human Kinetics • Mangubat, A.S., Tolitol, M.B., Urbiztondo, S.M.M. & Vergara, L.A. (2016). Health-optimizing physical education (HOPE) 1 : fitness. Quezon City : Vibal Group, Inc.	(Activity Sheet) Reflective Essay Return Demonstration
Week 9	**MIDTERM** Skill Demonstration	• Demonstrate the different fundamental movement skills proficiently	Video and Visual Presentation		Performance Task (Skill Demonstration)
Week 10-12	**FUNDAMENTAL MOVEMENT PATTERNS** **Locomotor Movements** Walk, Run, Hop, Leap, Skip, Jump, Slide, Baby Crawl, Inch Worm, Gallop, Grapevine, Side Step, Crab Crawl	• Identify the different types of locomotors movements • Execute the locomotor movements effectively • Design a specific activity workout using the given movement • Apply basic exercise principles in creating sample exercise workout	Data Bases and Search Engines Printed Lectures Video and Visual Demonstration		Written Test (Activity Sheet) Reflective Essay Return Demonstration Performance Task (Skill Demonstration)
Week 13-14	**Chapter 6** **BASIC STRENGTH TRAINING AND OTHER HOME EXERCISES** • Health Benefits of Muscle Fitness Exercise • Basic Safety Guidelines in Engaging Muscular Strength Exercises • Sample Strength Training Exercises ○ Upper and Lower extremities, and Core Exercises • Other Training Methods and Types of Equipment • Physical Activities at home and Other Suggested Exercises	• Identify the different physical activities at home and their benefits • Discuss the importance of strength training exercises for the core, upper extremities, and lower extremities. • Perform properly the different exercises at home to enhance core muscles, upper body muscles, and lower body muscles.	Data Bases and Search Engines Printed Lectures Video and Visual Demonstration	• Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. Polytechnic University of the Philippines, College of Human Kinetics • Cissik, John (2019) Strength and Conditioning A Concise Introduction, 2nd Edition. Routledge, ,USA.	Written Test Reflective Essay Performance Task (Skill Demonstration)
Week 15-17	**The FITNESS CHALLENGE** • General Guidelines on Fitness Plan Creation and Implementation • Setting SMART Goals • Suggested Fitness Workouts: Walking Program: 10,000 Steps Challenge, Aerobic Dance Fitness, Body Weight Exercises, Yoga Exercises, Jump Rope Exercise, and others	• Come up with a SMART exercise program based on their goals and objectives supported by healthy eating habits • Improve fitness level based on the baseline (pre-test) score • Demonstrate physical literacy in performing the different activities following the safety guidelines	Focus Group Discussion Collaborative Work	• Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness,Fourteenth Edition. Cengage Learning, USA • Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York • Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement	Journal Writing Formative Peer/Group Assessment Performance Task (Implementation and monitoring of fitness challenge)

	• Comprehensive Personal Fitness Plan Worksheet			Competency Training. Polytechnic University of the Philippines, College of Human Kinetics	
Week 18	**FINALS** Submission of video and/or written Documentary Report	• Demonstrate appreciation of the benefits of engaging in physical activities and exercise programs.	Video and/or Written Documentary Report	• Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. Polytechnic University of the Philippines, College of Human Kinetics	Documentary Report

References

Printed Books/Module

- *Camarador, R., Camarador L., Serrano, J.A., & Mantala, J. (2020) Instructional Material on Movement Competency Training. Polytechnic University of the Philippines, College of Human Kinetics*
- *Cissik, John (2019) Strength and Conditioning A Concise Introduction, 2nd Edition. Routledge, ,USA.*
- *Floyd, R.T. (2015). Manual of structural kinesiology. New York, NY : McGraw-Hill Education.*
- *Hoeger, W.W.K. (2015). Lifetime physical fitness & wellness : a personalized program (13th ed.). Stamford, Connecticut : Cengage Learning.*
- *McKinley, M.P., O'Loughlin, V.D. & O'Brien, E.P. (2017). Human anatomy (15th ed.). New York, New York : McGraw Hill Education.*
- *PUP Student Hand Book 2019 Revised Edition*

e-books/module

- *Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness, Fourteenth Edition. Cengage Learning, USA*
- *Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York*
- *Patton, Kevin T. and Thibodeau, Gary A. Anthony's (2010) Textbook of Anatomy and Physiology, 19th Edition, Mosby Elsevier*
- *Mangubat, A.S., Tolitol, M.B., Urbiztondo, S.M.M. & Vergara, L.A. (2016). Health-optimizing physical education (HOPE) 1 : fitness. Quezon City : Vibal Group, Inc. Mitchell, Tommy, (2015) Introduction to Anatomy & Physiology, 1st Edition, New Leaf Publishing Group Inc.*
- *Martini, Frederic H., Nath, Judi I., Bartholomew, Edwin F., (2015) Fundamentals of Anatomy & Physiology, 10th Edition, Pearson Education Inc.*

Research Studies

- *Bellen, J. J. B., & Camarador, R. A. Health status and lifestyle habits of college freshmen students: Basis for a proposed intervention program. International Journal of Health, Physical Education & Computer Science in Sports, 37(1), 57.*
- *Camarador, R.A., Dela Cruz, L. A., Serrano, S.S., Banadera, S.P., (2022) Me and My Fitness During COVID 19 Pandemic: Anthropometric, Physical Activity, and Wellness Lifestyle Habits of Collegiate Students. Polytechnic University of the Philippines, College of Human Kinetics*
- *Puen, D. A. Y., Cobar, A. G. C., Dimarucot, H. C., (2021) Perceived Barriers to Physical Activity of College Students in Manila, Philippines during the COVID-19 Community Quarantine: An Online Survey. Sport Mont, 19(2), 101-106. doi: 10.26773/smj.210617*
- *Santos, M. E. (2015). Alternative Home Upper Body Workout Using Water Bottles For Selected College Freshmen Students of Angeles University Foundation. International Journal of Physical Education, Fitness and Sports, 4(2), 15-21. doi:10.26524/1523*

Course Grading System

Class Standing
70%

- *Quizzes*
- *Recitation*
- *Projects/Assignments/Seatwork/Special Report*

Midterm / Final Examinations
30%

100%

Midterm Grade + Final Term Grade = *FINAL GRADE*
2

Section 8. *Grading System*

8.1 Students shall be graded or marked in accordance with the following system.

Grade/Mark	Percentage/ Equivalence	Description
1.0	97-100	Excellent
1.25	94-96	Excellent
1.5	91-93	Very Good
1.75	88-90	Very Good
2.0	85-87	Good
2.25	82-84	Good
2.5	79-81	Satisfactory
2.75	76-78	Satisfactory
3.0	75	Passing
5.0	65-74	Failure
Inc.		Incomplete
W		Withdrawn

Top 10 Guidelines for Digital Citizenship

1. *Protect your online privacy*
2. *Respect the online privacy of others.*
3. *Protect your property.*
4. *Respect the property of others.*
5. *Respect the rules, values, and policies of your family, religion, community, and school.*
6. *Understand the values of other cultures, religions, and communities.*
7. *Build a positive online reputation and portfolio of work.*
8. *Use online communications in constructive ways.*
9. *Evaluate the accuracy of any information you find or receive online - or share online.*
10. *Maintain a healthy balance between your online activities and relationships with your physical world activities and relationships.*

Consultation Time

Chapter 1

PHYSICAL ACTIVITY AND EXERCISE

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. demontrate undesrtanding on the impact of phsycal inactivity in different countries;
2. recognize the importance of phsyical activity and exercise to have a strong and healthy body and mind, most especially in time of crisis; and
3. discuss ways on how to strengthen immune system to fight corona virus.

Course Outline:

- Overview
- Course Materials: Discussion
  - 8 Dimensions of Wellness
  - Physical Activity and Exercise
  - Ways in strengthening immune system
  - Staying Active in the New Normal
- Summary
- Suggested Media at Home
- Activities and Assessments
  - Reflective Questions
  - Knowledge Check
  - Performance Task
- References

II. OVERVIEW

With our modern world today that highly depends in technology, having different modes of transportation and an environment that does not support and promote active lifestyle, it became more challenging for the people to engage in daily physical activity. According to Lancet Global Health, 1.4 billion people are physically inactive or more than one in four adults globally are not physically active. In the number, women (23%) are less active compared to men (32%), while countries with high economy (37%) are more inactive compared to middle income (26%) and low income countries (16%). It can be implied that the higher economy the higher cases of physical inactivity.

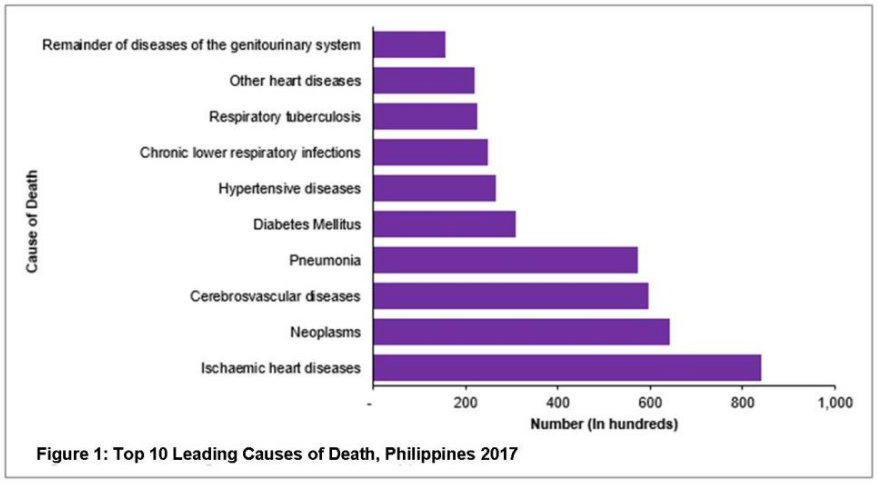


Figure 1: Top 10 Leading Causes of Death, Philippines 2017

Physical inactivity is increasing in many countries today. It can partly blame due to lack of physical activity participation during leisure time and an increase in sedentary behavior in the workplace. Physical inactivity increases people's risk of having non-communicable diseases such as, cardiovascular disease,

stroke, diabetes, cancer, and mental health conditions. Global health is being influenced by population-ageing, rapid unplanned urbanization, and globalization, all of which result in unhealthy environments and behaviors (WHO, 2018).

World Health Organization (2016) estimates that around 15.2 million people die worldwide due to ischaemic heart disease and stroke, which remains the top 2 leading causes of death globally for the past 15 years. Likewise, as shown in figure 1, ischaemic heart diseases (84,120 or 14.5 percent) was also top cause of death in the Philippines. In addition, cerebrovascular diseases, diabetes, hypertensive, and other heart diseases which are all non-communicable diseases were also part of the top 10 causes of death according to Philippine Statistics Authority, 2017. Thus, the need for all countries to increase the priority to provide programs for all individuals of all ages to have an active life style and establish environment that will support physical activity engagement since having regular physical activity is main key to prevent and treat non-communicable diseases (NCDs).

It is common to all that for us to improve our quality of life, participation in fitness programs or other physical activities is a must. However, we need to realize that with these alone are not always sufficient to lessen the possible risk of having diseases and won't eve guarantee a healthier body and mind. For example, a student who always participate in Zumba dance, jog 30-45 minutes three times a week, and engage in strength training regularly might be considered with an excellent fitness level. However, without a family and friend support, that might cause emotional instability, and excessive stress. Also, bad eating habits, and drinking too much alcohol will still put an individual at risk for cardiovascular disease and other chronic disease of which they may not be aware of. Keeping an optimal level of wellness is very important to have a higher quality of life. It is a significant matter because everything we do and every emotion we feel relates to our well-being.

Though we are aware of unhealthy habits that we have, most especially Filipinos, we are still satisfied with our life as long as we are free from symptoms of possible diseases. We just came to realize that we need a positive change until we had serious health problems. Nevertheless, healthy lifestyle habits of today will dictate the health and well-being of tomorrow.

Wellness is no longer looked as simply the absence of disease but the presence of purpose in life, joyful relationships, active involvement in fulfilling work and play, a healthy body and mind, living in a sustainable environment, and happiness. It is an active process of becoming conscious of and creating choices moving toward a healthy and higher quality of life. Moreover, it also defined as the integration of many different components that expand one's potential to live ("quality of life") and work effectively to make a significant contribution to society.

III. COURSE MATERIALS: DISCUSSION

A. The Eight Dimensions of Wellness

There are eight dimensions of wellness: physical, emotional, social, mental, environmental, financial, occupational, and spiritual. Each dimension is interrelated to each other in which one often affects the others. Thus, attention must be given to all these dimensions. A good example is when your parents do not have a regular decent work; definitely you are not financially stable and may cause problems within the family since it is a basic necessity.



FIGURE 2

Having all the dimensions, it is not necessarily to be equally balanced. We should set our goals to achieve personal harmony and satisfaction that is true to us. We may have our own priorities, aspirations, styles, and personal outlooks to live life to its full potential.

1. Physical wellness is associated with being healthy. It is the ability of oneself to promote proper care of our physical health for optimum development and proper management of health problems. To be considered physically well individual, you must be physically active, has proper nutrition, engage in regular exercise program, maintain ideal body weight, practice safe sex, get enough sleep, nonuse of illegal drugs, avoid alcohol drinking, lessen exposure to environmental pollutants, and seeks medical care if necessary. Moreover, physically well people also display good health-related fitness level.

2. Emotional wellness contains awareness and acceptance of your own feelings. It is the ability to accept limitations, attain emotional stability, express emotions appropriately, cope up with stress, accept and adjust to changes, and appreciate life even with failures and regrets. An emotionally well individual has the ability to response positively by making best decisions and moving forward during failures or crisis. Emotional wellness empowers you to move forward with confidence and motivation instead of spending most of the time and effort worrying about your problems. It also entails happiness, which provides meaning and joy to your life. Though there is no single formula for achieving happiness, according to studies, happiness can be achieve mainly through family love and support.

3. Intellectual wellness, or mental wellness, entails that; you can apply and share the knowledge and skills you learned with others engage in innovative and mentally-stimulating works and create more opportunities to grow and learn. Individuals who are mentally fit are open-minded for new ideas and accept mistakes committed by others. Because of curiosity, they are motivated to learn more and explore new concepts and understandings. To have optimal level of intellectual wellness, it stimulates exploration. In this way, it provides you to know and discover some issues that constitute critical thinking, problem solving, creativity and innovation. Through academics, personal hobbies, and cultural and community involvement, you can develop intellectual wellness.

4. Social wellness refers to our relationships with others and how we interact with them.
One of the trademarks of social wellness is the ability to interact to others with harmony, both within and outside our family. Moreover, with positive self-esteem it gives you the comfort and confidence to be open, friendly, and warm to others. Social wellness entails building healthy relationships and genuine connection with people around you. It also comprises on how to balance the unique needs of romantic relationships with other parts of your life as well as your social life with your professional or academic lives.

5. Environmental wellness refers to the impact of our environment on our well-being. It indicates a lifestyle that promotes and protects healthy environment. The main principle of environmental wellness is respect for our environment including wildlife and its ecosystem. It does not require you to join environmental activities or join in any environmental organization but to practice good habits to preserve and promote a healthy environment like Leave No Trace (LNT) principle while travelling and four R's (Refuse, Reduce, Reuse, Recycle) principle. These are more realistic and good enough. Always remember that our health is negatively affected if we are living in a polluted and unsafe environment. A good example is when you are living in an environment like a squatter or slam areas and which your health is at stake due to unhealthy environment brought by different kinds of pollution and over population.

6. Occupational wellness motivates us to go for work in which we will gain personal development and satisfaction. It gives us career options and encouragement to pursue opportunities to grow more and enjoy what we love most. This dimension of wellness

acknowledges the importance of fulfillment, personal development and sense of meaning through work. It is not always about high salary, top position or remarkable working conditions. Any work can bring occupational wellness if it provides any kind of supports that are very much important to the individual. Salary might be the top priority or most important factor to consider for others, but others might give greater value on opportunity for development, recognition for achievement, and/or good working relationship with boss and co-workers. Individuals who are occupationally well have their own "ideal" job, which permits them to strive harder to succeed.

7. Spiritual wellness provides a unifying power that integrates all dimensions of wellness. Basic characteristics of spiritual people include a sense of meaning and direction in life and a relationship to a higher being. Pursuing these avenues may lead to personal freedom, including prayer, faith, love, closeness to others, peace, joy, fulfillment, and unselfishness.

There are quite number of studies that claim positive relationship among spiritual wellness, emotional wellness, and self-fulfillment. Prayer is an indication of our spirituality and communication with a higher power. Several studies conclude the positive effects of prayer on health. Most of these have associated prayer with humility, love, sincerity, and compassion to positive health results. Other studies have shown faster healing time and less difficulties for those who are praying compared with patients who were not.

Spiritual wellness not only allows you to seek meaning and purpose in human existence, but it also allows you to appreciate your life experiences for what they are. When you find meaning in your life experiences, you will be able to develop a harmony with your inner self and the outside world. In simpler terms, spiritual wellness grants you balance. Nevertheless, spiritual wellness does not require you to have a religious practice, though it may be only way you may take to lift and develop your spiritual wellness.

8. Financial Wellness. Financial Wellness contains the process of learning how to successfully manage financial expenditures. It entails the ability to have financial resources to provide essential needs and financial literacy on how to manage personal and family expenses. Money is very significant in everybody's lives, for without it, it will largely affects your way of living mainly your health and academic endeavor. More often, financial stress usually leads as one of the causes of stress in college students. Budgeting financial resources for projects, on the job training, thesis, transportation and food allowance, and other academic works are some of the important things to deal with in college life. Thus, learning how to maximize your financial resources and developing smart habits will help you to lessen possible financial stress in the future and achieve your goals.

B. Physical Activity and Exercise

Promoting physical activity and having regular exercise means improving one's daily lives. According to research, people who were living in a sedentary life can improve their health by regularly including moderate levels of activity throughout the day. Moreover, additional evidences suggest that the more you increase the intensity level, with variety of physical activities, the more health benefits you may achieve. Physical activity (PA) is bodily movement produced by skeletal muscles. It utilizes energy and provides positive health benefits. It can be light or moderate to vigorous intensity. Examples of daily light physical activity are walking, taking the stairs, grocery, gardening, and other household chores.

Exercise is a physical activity that is planned, structured, and repetitive that aims to maintain or improve components of physical fitness. It is usually assumed as an activity that requires a moderate to vigorous intensity effort. Some examples are jogging, running, swimming, biking, aerobic activities, and strength training.

In contrary, physical inactivity indicates a level of activity that is lower than the recommended physical activity to improve health. It means there is an extreme light consumption of energy throughout the day such as walking casually, perform self-care, or do other light work that are far to contribute in our overall health. On the other hand, scientific evidence explains that strenuous physical activity is not needed to achieve health benefits. Having regular moderate activities provide significant benefits in health and well-being for the people who are not physically active. While for individuals who are already moderately active, having moderate to vigorous level of physical activity will even provide greater health benefits. Brisk walking or jogging, playing recreational sports, dancing, and gardening are some moderate activities that you may engage in.

C. Ways in Strengthening Immune System

Healthy Eating Habits

Proper nutrition is essential to overall health and wellness. Good nutrition means that a person's diet supplies all the essential nutrients for healthy body functioning, including normal tissue growth, repair, and maintenance. The diet should also provide enough substrates to produce the energy necessary for work, physical activity, and relaxation. Nutrients should be obtained from a wide variety of sources.

Diet and nutrition often play a crucial role in the development and progression of chronic diseases. A diet high in saturated fat and trans fat increases the risk for diseases of the cardiovascular system, including atherosclerosis, coronary heart disease (CHD), and strokes. In sodium-sensitive individuals, high salt intake has been linked to high blood pressure. Up to 50 percent of all cancers may be diet related. Obesity, diabetes, and osteoporosis also have been associated with faulty nutrition. A healthy diet encompasses a wide range of benefits, with positive impacts on nutrition, overall health, economy as well as the environment. According to the World Health Organization (WHO), a healthy diet emphasizes regular intake of vegetables, fruits, whole grains, root crops, fat-free or low-fat milk, lean meats, poultry, fish, egg, beans and nuts. It is also low in saturated fats, trans fats, cholesterol, sodium and added sugars.

Coronavirus-related concerns may affect dietary habits, leading to higher calorie consumption that could promote weight gain. Limiting trips to the grocery store, skipping fresh fruits and vegetables in favor of stocking up on calorie-dense, non-perishable foods, and financial difficulties may cause people to opt for less expensive, ultra-processed food options. And all the extra hours you're spending at home may make you more likely to snack, "stress-bake," or prepare high-calorie, comfort-food meals. Additionally, many people are coping with fear, anxiety, stress, financial concerns, sadness, boredom, and isolation — all of which can have a negative effect on diet, physical health, and mental well-being. The great importance of a healthy diet during this self-isolation period also cannot be underestimated.

A high percentage of COVID-19 deaths worldwide are associated with one or more chronic conditions. It is also evident that older people are at a higher risk for severe illness with this pandemic. Nutrition is not a cure for COVID-19, but it is a modifiable contributor to the development of chronic disease, which is highly associated with COVID-19 severe illness and deaths. A well-balanced diet strengthens the immune system and reduces the risk of chronic disease and infectious diseases. It is clear in these challenging times that optimizing nutrition is also important, not only for ourselves but also for every patient/client that we treat. Every physiotherapist should be aware of the benefits of healthy eating and be able to provide sound nutritional guidance to their patients, especially those with chronic

disease. Having knowledge about nutritional interventions that may help prevent chronic conditions and their associated risks is now more important than ever.

Eating a healthy diet is very important during the COVID-19 pandemic. What we eat and drink can affect our body's ability to prevent, fight and recover from infections. While no foods or dietary supplements can prevent or cure COVID-19 infection, healthy diets are important for supporting immune systems. Good nutrition can also reduce the likelihood of developing other health problems, including obesity, heart disease, diabetes and some types of cancer.

For babies, a healthy diet means exclusive breastfeeding in the first six months, with the introduction of nutritious and safe foods to complement breastmilk from age 6 months to 2 years and beyond. For young children, a healthy and balanced diet is essential for growth and development. For older people, it can help to ensure healthier and more active lives.

Tips for maintaining a healthy diet:

1. Eat a variety of food, including fruits and vegetables

- Every day, eat a mix of wholegrains like wheat, maize and rice, legumes like lentils and beans, plenty of fresh fruit and vegetables , with some foods from animal sources (e.g. meat, fish, eggs and milk).
- Choose wholegrain foods like unprocessed maize, millet, oats, wheat and brown rice when you can; they are rich in valuable fibre and can help you feel full for longer.
- For snacks, choose raw vegetables, fresh fruit, and unsalted nuts.

2. Cut back on salt

- Limit salt intake to 5 grams (equivalent to a teaspoon) a day.
- When cooking and preparing foods, use salt sparingly and reduce use of salty sauces and condiments (like soy sauce, stock or fish sauce).
- If using canned or dried food, choose varieties of vegetables, nuts and fruit, without added salt and sugars.
- Remove the salt shaker from the table, and experiment with fresh or dried herbs and spices for added flavor instead.
- Check the labels on food and choose products with lower sodium content.

3. Eat moderate amounts of fats and oils

- Replace butter, ghee and lard with healthier fats like olive, soy, sunflower or corn oil when cooking.
- Choose white meats like poultry and fish which are generally lower in fats than red meat; trim meat of visible fat and limit the consumption of processed meats.
- Select low-fat or reduced-fat versions of milk and dairy products.
- Avoid processed, baked and fried foods that contain industrially produced trans-fat.
- Try steaming or boiling instead of frying food when cooking.

4. Limit sugar intake

- Limit intake of sweets and sugary drinks such as fizzy drinks, fruit juices and juice drinks, liquid and powder concentrates, flavoured water, energy and sports drinks, ready-to-drink tea and coffee and flavoured milk drinks.
- Choose fresh fruits instead of sweet snacks such as cookies, cakes and chocolate. When other dessert options are chosen, ensure that they are low in sugar and consume small portions.
- Avoid giving sugary foods to children. Salt and sugars should not be added to complementary foods given to children under 2 years of age, and should be limited beyond that age.

5. Stay hydrated: Drink enough water

Good hydration is crucial for optimal health. Whenever available and safe for consumption, tap water is the healthiest and cheapest drink. Drinking water instead of sugar-sweetened beverages is a simple way to limit your intake of sugar and excess calories.

6. Avoid hazardous and harmful alcohol use

Alcohol is not a part of a healthy diet. Drinking alcohol does not protect against COVID-19 and can be dangerous. Frequent or excessive alcohol consumption increases your immediate risk of injury, as well as causing longer-term effects like liver damage, cancer, heart disease and mental illness. There is no safe level of alcohol consumption.

7. Breastfeed babies and young children

Breastmilk is the ideal food for infants. It is safe, clean and contains antibodies which help protect against many common childhood illnesses. Babies should be breastfed exclusively during the first 6 months of life, as breast milk provides all the nutrients and fluids they need.

- From 6 months of age, breast milk should be complemented with a variety of adequate, safe and nutrient-dense foods. Breastfeeding should continue under babies at 2 years of age or beyond.

Women with COVID-19 can breastfeed if they wish to do so and should take infection prevention and control measures.

Having Enough Sleep

When it comes to your health, sleep plays an important role. While more sleep won't necessarily prevent you from getting sick, skimping on it could adversely affect your immune system, leaving you susceptible to a bad cold or case of the flu. To keep yourself sniffle-free this season, here's what you need to know.

Sleep and Cytokines

Without sufficient sleep, your body makes fewer cytokines, a type of protein that targets infection and inflammation, effectively creating an immune response. Cytokines are both produced and released during sleep, causing a double whammy if you skimp on shut-eye. Chronic sleep loss even makes the flu vaccine less effective by reducing your body's ability to respond.

Stock Up on Naps

To stay healthy, especially during the influenza season, get the recommended seven to eight hours of sleep a night. This will help keep your immune system in fighting shape, and also protect you from other health issues including heart disease, diabetes, and obesity. If your sleep schedule is interrupted by a busy workweek or other factors, try to make up for the lost rest with naps. Taking two naps that are no longer than 30 minutes each —one in the morning and one in the afternoon—has been shown to help decrease stress and offset the negative effects that sleep deprivation has on the immune system. If you can't swing a half-hour nap during the workday, try grabbing a 20-minute siesta on your lunch hour, and another right before dinner.

Other Healthy Tactics

Of course, there's more to boosting your immunity and guarding against illness than getting ample sleep. It's also important to practice smart stay-healthy strategies such as washing your hands with soap regularly, avoiding close contact with people who are obviously under the weather, and talking with your doctor about getting an annual flu shot. And remember: Even if you do come down with a case of seasonal sniffles, you'll be able to bounce back faster if your body is well rested. There is also an evidence that suggests regular exercise

helps you fall asleep faster and improves sleep quality — and getting a good night's sleep has also been found to boost your immune system.

Fatigue, sleep deprivation and stress are common in healthcare professionals especially during the COVID-19 pandemic. Sleep and the body's immune system are inextricably linked. Sleep deprivation potentially impairs the body's immune response, negatively affects inflammatory homeostasis and may also lead to an increased risk of infection and worsen the clinical conditions caused by a disease such as COVID-19. It is imperative to address sleep health for the sake of healthcare professionals and their patients particularly the stresses imposed during a pandemic.

D. Stay Active in This New Normal

Source: https://www.benendenhospital.org.uk/health-news/helpful-tips/stay-active-in-this-new-normal/

Regular exercise and physical activity promote strong muscles and bones. It improves respiratory, cardiovascular health, and overall health. Staying active can also help you maintain a healthy weight, reduce your risk for type 2 diabetes, heart disease, and reduce your risk for some cancers.

According to the NHS website, exercise is the miracle cure we've always had, but for too long we've neglected to take our recommended dose.

To stay healthy, adults should try to be active every day and aim to achieve at least 150 minutes of physical activity over a week through a variety of activities.

For most people, the easiest way to get moving is to make activity part of everyday life, like walking or cycling instead of using the car to get around. However, the more you do, the better, and taking part in activities such as sports and exercise will make you even healthier. For any type of activity to benefit your health, you need to be moving quick enough to raise your heart rate, breathe faster and feel warmer. This level of effort is called moderate intensity activity. If you're working at a moderate intensity you should still be able to talk but you won't be able to sing the words to a song.

An activity where you have to work even harder is called vigorous intensity activity. There is substantial evidence that vigorous activity can bring health benefits over and above that of moderate activity. You can tell when it's vigorous activity because you're breathing hard and fast, and your heart rate has gone up quite a bit. If you're working at this level, you won't be able to say more than a few words without pausing for a breath.

Focus on what you want to do

Being active is also about finding out what activity works for you. It doesn't have to be a fully-fledged personal training program, but just focusing on what you want to do a bit more or less of each day can make a difference.

By being active, you will burn calories that you store from eating throughout the day and—it can be as easy as walking the dog or as rigorous as running a marathon. Providing opportunities for children to be active early on puts them on a path to better physical and mental health. It's never too late to jumpstart a healthy lifestyle.

Many people find that being active fits naturally into spending time with those they live with such as family, housemates, and pets and this helps with keeping motivated.

Being active together

Being active with others has social and mental health benefits, and some people simply find it more fun. When exercising outdoors, remember to follow all current government COVID-

19 health protection advice, including practicing social distancing at all times. Exercising over a video call is another great way to stay active with others, but remotely – so you get all the benefits of moving and being connected to other people, while still practicing social distancing and protecting your health. And there's no restriction on who you can get active with online – you can get moving with friends, family that don't live with you, and even your work colleagues.

Digital tools to help you move

Some tools have been designed to track the activities you already do. Public Health England is encouraging adults to download the free 'Active 10' app to help complete your 10 minutes of brisk walking daily and to gradually build this up to multiple 'Active 10s'. The app is the first of its kind to show how much brisk walking you're doing and when you are doing this at a high enough intensity and long enough to get health benefits.

You may have grown up loving sport, or you may prefer to get moving by doing a fun activity instead. Happily, there are options out there that may work for you depending on how you are able to move. If you have a health condition, the symptoms of your condition may mean you have to undertake activities you used to do in a more modified way, or else try something new.

So, whether is walking for a set amount of time at lunch time, or 10-minute seated yoga workout or gardening - set yourself a new routine and build in new habits. Repeat the bits you enjoy daily and keep trying new activities that inspire and motivate you.

Summary

- Nowadays, people have become more sedentary due to various ways that lighten up their everyday life and workload. This physical inactivity increases people's risk of having non-communicable diseases such as, cardiovascular disease, stroke, diabetes, cancer, and mental health conditions. Thus, promoting different physical activities and having regular exercise can truly help improve one's daily lives and maintain good health and fitness.
- Unfortunately, these mandated restrictions on travel and directives against participating in outdoor activities affect peoples' mental and physical health, especially those who are used to regular outdoor physical activities. But currently, the greatest risk of COVID-19 infection is exposure. That's why it is imperative that we find creative ways to be active and exercise while staying at home, maintaining social distancing and doing proper hygienic countermeasures.
- Proper nutrition, healthy eating habits, correct food handling, having enough sleep, and staying active are the different ways to strengthen our immune system amid this pandemic.

Suggested Media at Home

- **Pinggang Pinoy** https://www.fnri.dost.gov.ph/index.php/116-pinggang-pinoy

- **COVID-19: Importance of home exercise while being isolated** https://www.youtube.com/watch?v=p9Dw-4ycMQQ&fbclid=IwAR3EkZAI9cIxoeH5swmaya7Q97_mkJfr7of0FAnt85vG9DntMQUoA5rXegI&app=desktop

- **Exercise, Immune Function, and COVID 19** https://www.youtube.com/watch?v=COyEDxl5N2A&fbclid=IwAR1iNeGJFjIBJQWexOvLhUHh6Espl5OnMzVp9uB-GxOavsWyE0l_wWfXsqs&app=desktop

- **COVID-19: Tips for Getting Good Sleep** https://www.youtube.com/watch?v=JRGJ5m3VHXo&fbclid=IwAR2tEOh8JTILrPsoHI6f_6enkYWKcTOteVyHP6XYlCirGiOEaQUC9KIulEc&app=desktop

IV. ACTIVITIES/ASSESSMENTS

In this section, a variety of suggested activities and assessments were provided. It provides activities to assess the appreciation (reflective questions) and understanding (knowledge check) of the students on the topics provided. It also suggests options for practical application (a performance task) by providing a performance activity that includes the rubrics or criteria for rating. Remember that as a teacher, you have the academic freedom to choose what activities you would like to assign to your students, modify some of the tasks, or provide additional or other activities that you think will be more suitable for our dear students, recognizing the different capabilities and situations they have right now.

Activity Worksheet 1. *Reflective Questions*

Name:_____ Section:_____ Date:_____ Group no._____

During the new or next normal, what do you think is the importance of physical activities in your own personal lifeand and also with your family? (10points)

What are the ways that you are doing in order to cope effectively in this pandemic situation? Explain your answer. (10points)

Activity Worksheet 2. Knowledge Check

Name:_____ Section:_____ Date:_____ Group no._____

I. Encircle the letter of the best answer that corresponds to the following questions or statements.

1. It is the ability to accept limitations, cope up with stress, accept and adjust to changes, and appreciate life even with failures and regrets.
   a. Spiritual Wellness b. Emotional Wellness c. Physical Wellness d. Health

2. _____refers to our relationships with others and how we interact with them**.**
   a. Spiritual Wellness b. Emotional Wellness c. Social Wellness d. Fitness

3. Below are all top leading causes of deaths in the world except_____.
   a. Heart Diseases b. Stroke c. Tuberculosis d. Diabetes

4. This is due to lack of physical activity participation during leisure time and an increase in sedentary behavior.
   a. COVID19 b. Hyperventilation c. Lockdowns d. Physical inactivity

5. The following are the diseases one can acquire during physical inactivity except _____.
   a. cancer b. dengue c. hypertension d. stroke

6. It is a physical activity that is planned, structured, and repetitive that aims to maintain or improve components of physical fitness.
   a. dance b. exercise c. sports d. yoga

7. This is the recommended physical activity session by the WHO for adults and elderly.
   a. 30 min/day of moderate-to-vigorous PAs c. 60 min/day of moderate-to-vigorous PAs
   b. 75 min/wk of vigorous PAs d. 150 min/wk of vigorous PAs

8. These are the activities that can help us maintain our health and fitness while being mandated to stay home during the COVID19 pandemic.
   a. aerobics b. home-based exercises c. running events d. sports participation

9. It is the recommended intake of salt per day.
   a. 5 grams b. 10 grams c. 15 grams d. 1 tablespoon

10. It is a type of protein that targets infection and inflammation that is produced and released during sleep.
    a. 5 grams b. 10 grams c. 15 grams d. 1 tablespoon

Performance Task 1

My Daily Food Intake

Name:_____ Section:_____ Date:_____ Group no._____

Instruction: Record your daily food intake for 7 days and its quantity. *(Example: Breakfast: 2 cups of white rice and 2 pcs of hotdog with 1glass of milk/1glass of water).* Also provide the time of intake (6:30am) and any form of hyration such as water, sports drink or energy drink, etc.

Date Started:_____

DAY 1	FOOD/SNACKS/QUANTITY	TIME
Breakfast		
AM Snack		
Lunch		
PM Snack		
Dinner		
Mid-night Snack		
DAY 2	**FOOD/SNACKS/QUANTITY**	**TIME**
Breakfast		
AM Snack		
Lunch		
PM Snack		
Dinner		
Mid-night Snack		
DAY 3	**FOOD/SNACKS/QUANTITY**	**TIME**
Breakfast		
AM Snack		
Lunch		
PM Snack		
Dinner		
Mid-night Snack		
DAY 4	**FOOD/SNACKS/QUANTITY**	**TIME**
Breakfast		
AM Snack		

Lunch		
PM Snack		
Dinner		
Mid-night Snack		
DAY 5	**FOOD/SNACKS/QUANTITY**	**TIME**
Breakfast		
AM Snack		
Lunch		
PM Snack		
Dinner		
Mid-night Snack		
DAY 6	**FOOD/SNACKS/QUANTITY**	**TIME**
Breakfast		
AM Snack		
Lunch		
PM Snack		
Dinner		
Mid-night Snack		
DAY 7	**FOOD/SNACKS/QUANTITY**	**TIME**
Breakfast		
AM Snack		
Lunch		
PM Snack		
Dinner		
Mid-night Snack		

Date Ended :_____

Recorded by: _____

Signature over printed name

Instruction

After the activity kindly answer the following questions:

1. Does your daily food intake follows the ideal food guide for Filipinos/Pinggang Pinoy? If yes, how do you manage? If no, Why?

   Do you think being healthy or physically fit nowadays is very hard to achieve? What do you think are the factors that affect in achieving a healthy lifestyle?

2. With our current situation now, being lockedown becasue of COVID2019, what is your plan to maintain a healthy lifestyle?

3. Based on this activity, what did you realize?

Performance Task 2

Dimensions of Wellness

Group Leader :_____ Section: _____
Members: _____ Date: _____
_____ Group no. _____
_____

Activity: Role Playing/Phantomime (By Group)

Instruction: With a proportional distribution, divide the entire class into 4-8 groups. Prepare a lottery where the 8 dimensions of wellness (can be a combination of 2 dimensions) will be picked and the genre of the play (e.g., drama, suspense, horror, action, etc.)

Suggested Lottery for Combined Dimensions

(1) Physical & Spiritual **(2)** Environmental & Financial **(3)**Social & Spiritual **(4)** Occupational & Mental

Suggested Genre of Play

Action, Horror, Drama, Comedy, Thriller, & Love Story

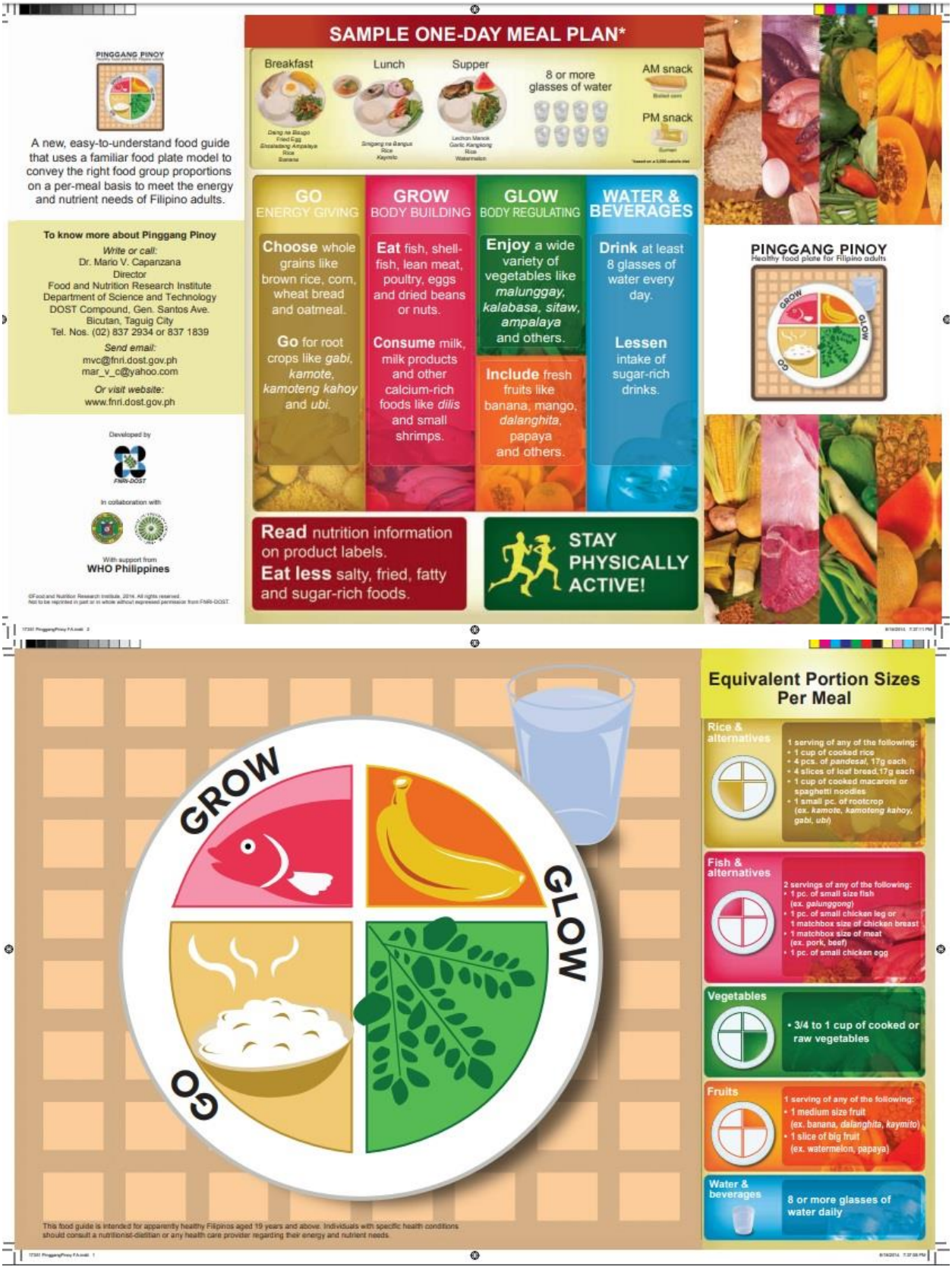
The leader of each group will pick from the lottery for the dimension and genre of play. They are not allowed to announce or inform anyone of the result of the lottery. They will be tasked with developing and presenting a role play, or phantomime (if using a mask), demonstrating how the dimensions of wellness may affect our daily lives. The challenge here is to deliver the play well, and after the presentation, the audience will be asked to guess the dimensions tackled and the genre. A merit will be awarded to a student who can answer the questions. After the guessing period, a representative from the class will be given 2-3 minutes to explain their play. The class will be graded by the teacher (50%) and students (50%) using the following criteria/rubrics:

50% Delivery of the Play
Clarity of the Message, Delivery of the actors and actresses,
30% Cooperation and Teamwork
10% Story Play
10% Explanation after the Play

To make the activity more fun and challenging. During the tasking, announce that there will be an award at the end of the presentation. Additional merit can be given with the following awards: best actor and actress; best supporting actor and actress; best director; best play or picture; and best story play.

Comments/Suggestions:_____
_____
_____
_____
_____
_____
_____

Pinggang Pinoy



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Chapter 2

INTRODUCTION TO ANATOMY

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. demontrate explain the importance of studying anatomy;
2. define anatomy, describe the origins of anatomical terms, and explain the significance of Terminologia Anatomica (International Anatomical Terminology);
3. use anatomical terms to describe body regions, and relative positions;
4. know the different major group of bones and muscles and its functions; and
5. apply the different anatomical movements.

Course Outline:

- Overview
- Course Materials: Discussion
    - Organ Systems
    - Anatomical Landmarks, Directions, Body Regions
    - Bones & Muscles
    - Joints & Types of Anatomical Movements
- Suggested Media at Home
- Activities and Assessments
    - Reflective Questions
    - Knowledge Check
    - Performance Task
- References

II. OVERVIEW

Anatomy is the study of body structures. You will discover how your body works under normal and abnormal conditions. The knowledge on this chapter will serve as the foundation for understanding all other basic life sciences.

Anatomy is the oldest medical science. Egyptian drawings from 1600 BCE illustrated basic knowledge of blood vessels. These drawings show that people have always been fascinated with the human body. Since that time, imaging techniques for studying the human body have evolved, enabling us to describe the locations and functions of body parts.

People have always been interested in the inner workings of the human body. Many medical terms have Greek roots, as do many other anatomical terms and phrases that originated more than 1500 years ago. For example, the term anatomy is derived from Greek roots that mean “a cutting open.” Anatomy is the study of internal and external body structures and their physical relationships among other body parts.

III. COURSE MATERIALS: DISCUSSION

A. The Organ Systems

The human body consists of 11 organ systems. The organs of each system contribute to a particular function. However, some organs belong to more than one system.

Specifically, the pharynx is part of both the respiratory and the digestive systems, and the male urethra belongs to both the reproductive and urinary systems.

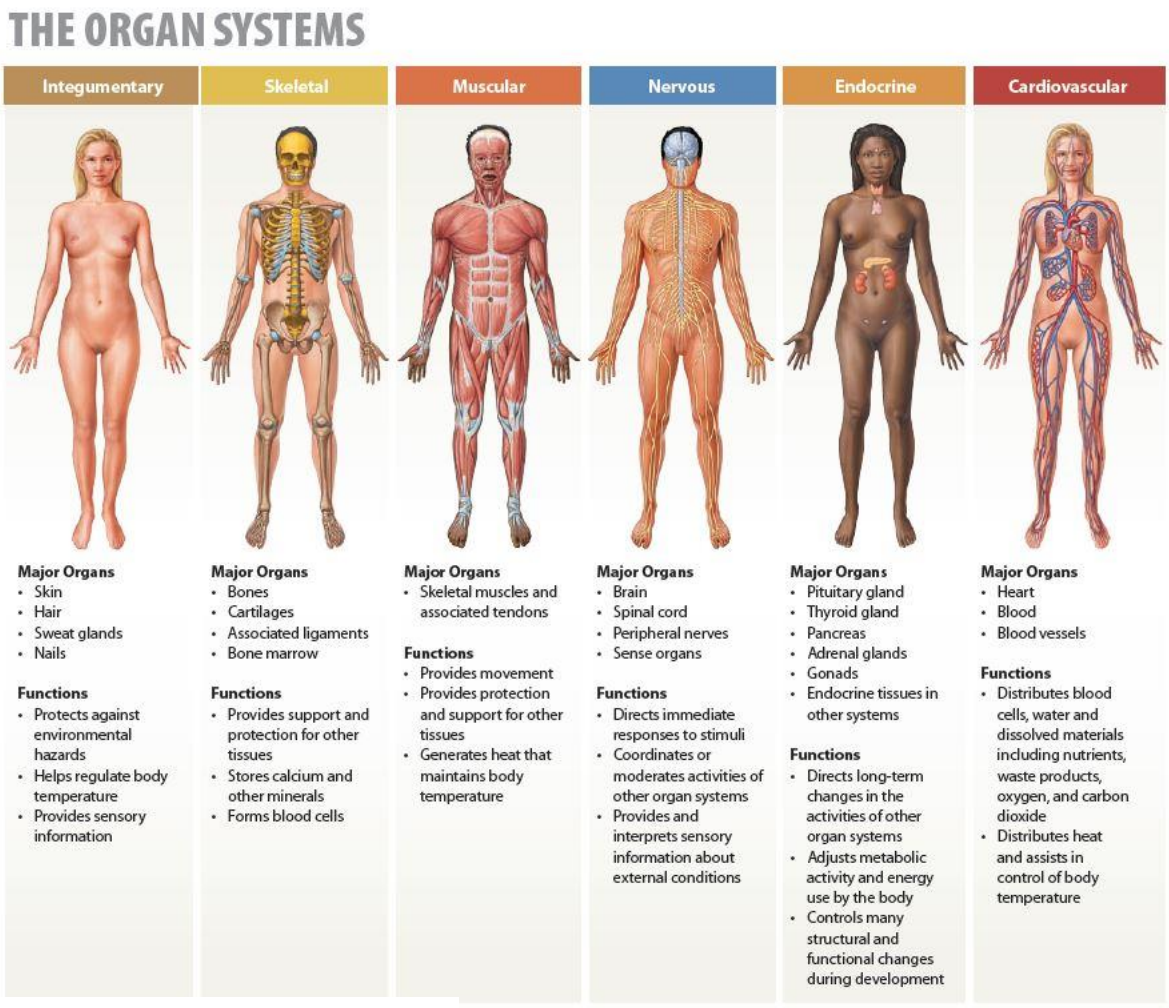


Figure 1: The Organ Systems of the Human Body (Part 1)

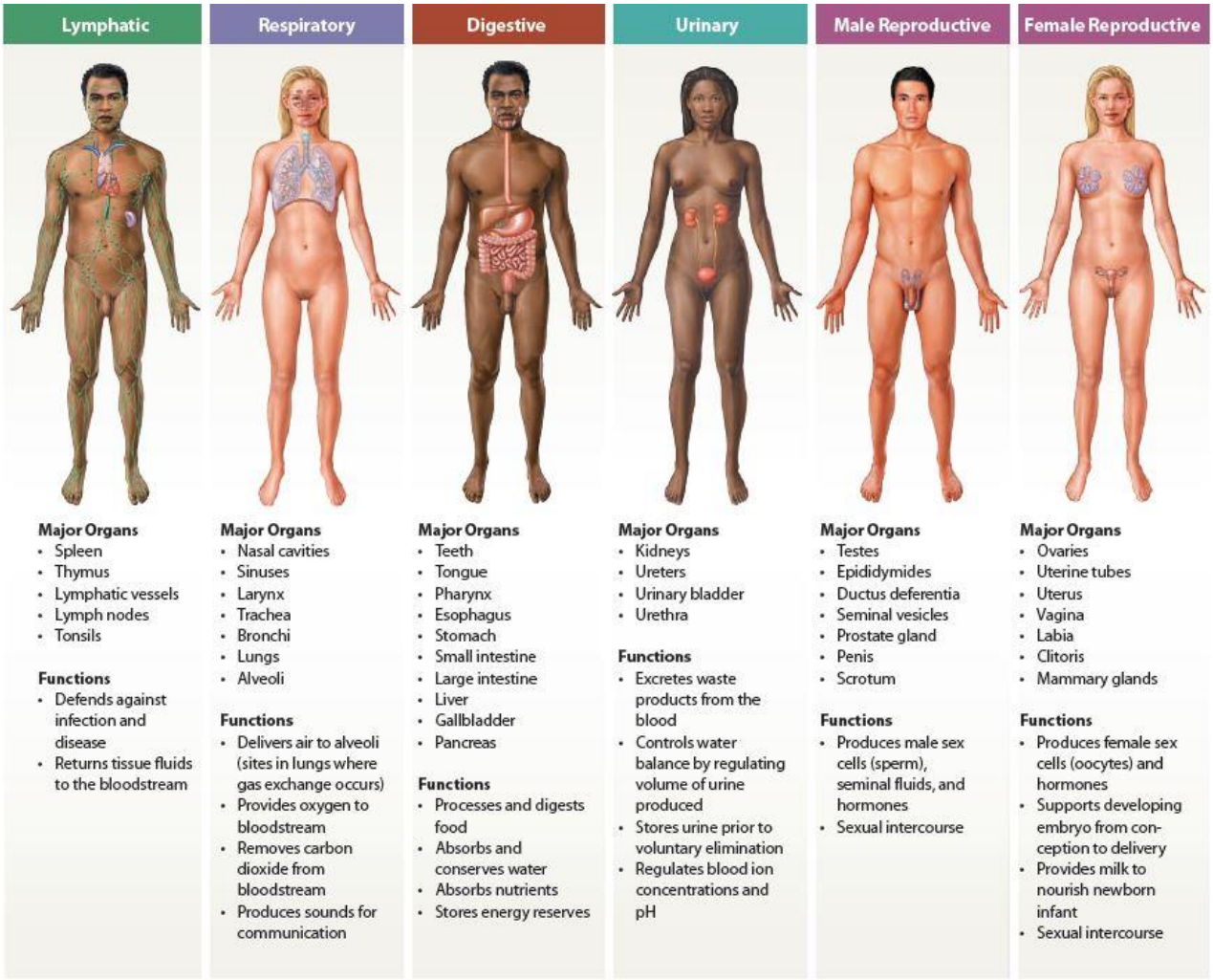


Figure 2: The Organ Systems of the Human Body (Part 2)

B. Anatomical Terms, Body Regions, Anatomical Positions and Directions

1. Anatomical Landmarks

The standard anatomical reference for the human form is the anatomical position. When the body is in this position, the hands are at the sides with the palms facing forward, and the feet are together. A person lying down in the anatomical position is said to be supine when face up, and prone when face down.

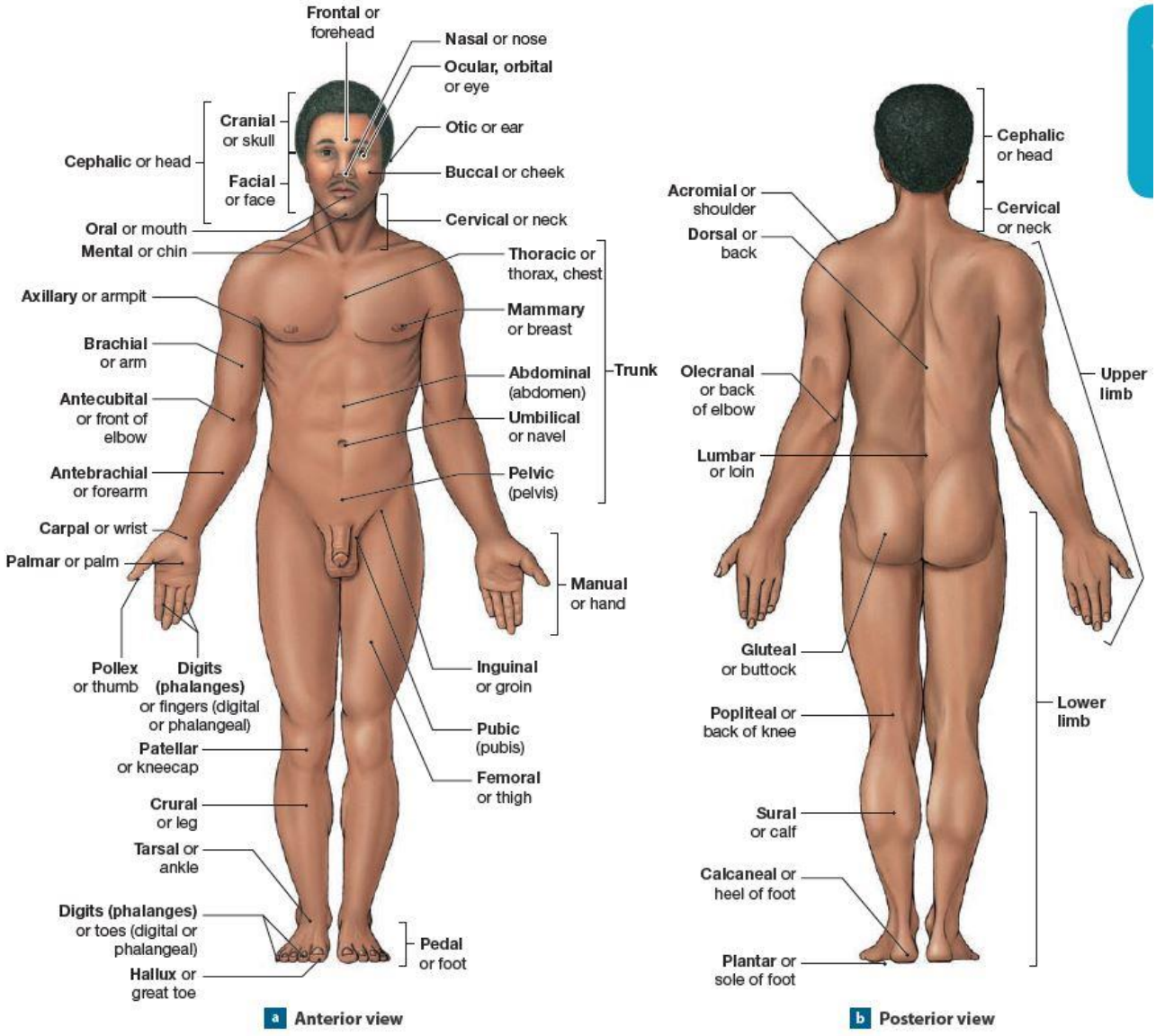


Figure 3: Anatomical Landmarks. Anatomical terms are shown in bold face type and common names are in plain type

2. Directional Terms

Directional terms are generally grouped in pairs of opposites. All terms are based on the body being in the anatomical position—standing erect, arms at the sides, with face, palms, and feet facing forward. Keep in mind, too, that the terms *right* and *left* always refer to the *patient's* right and left side.

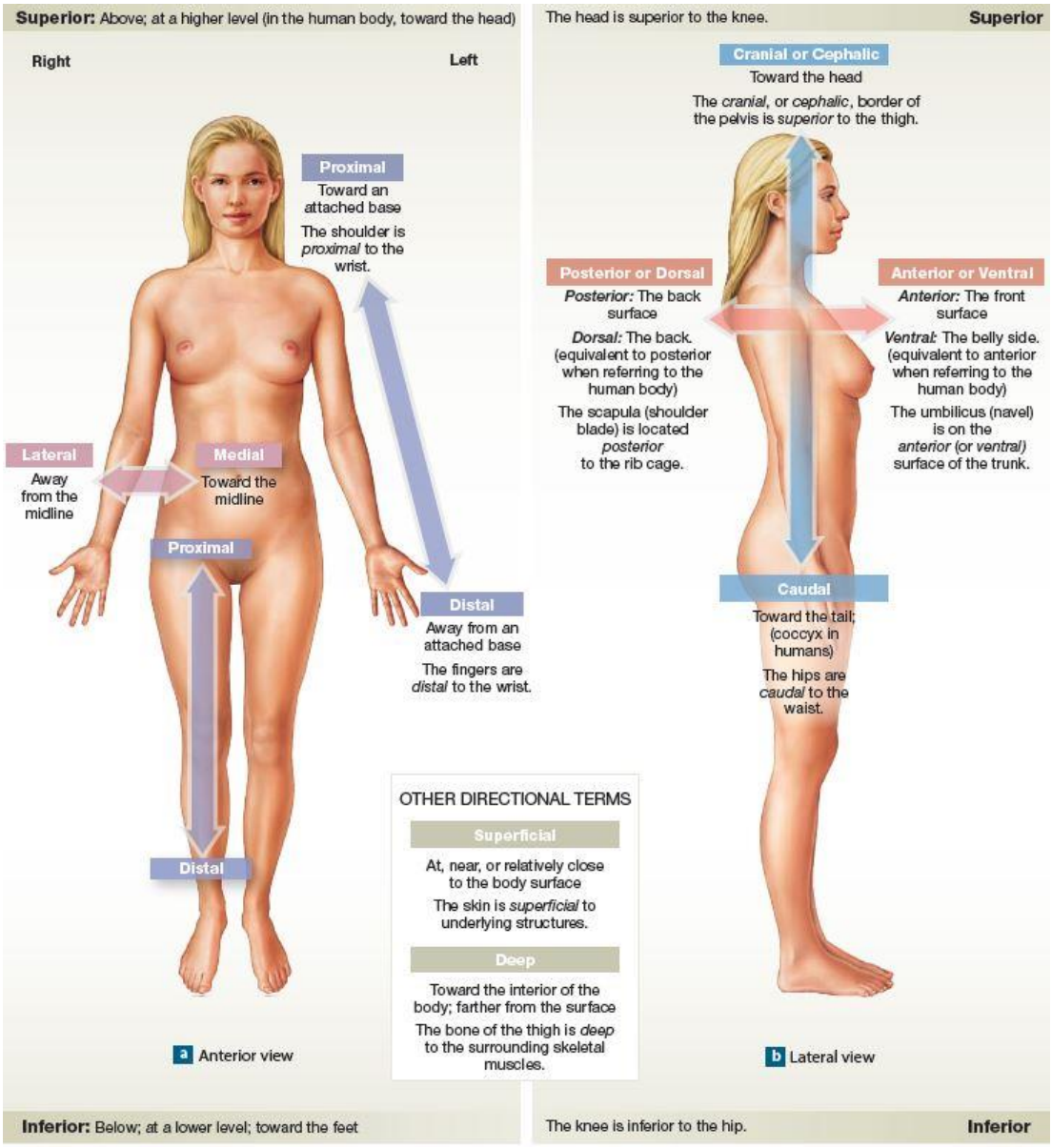


Figure 4: Directional References

3. Body Regions

The illustration shows the terms for the different regions of the body. These terms are used extensively when performing clinical examinations and medical procedures.

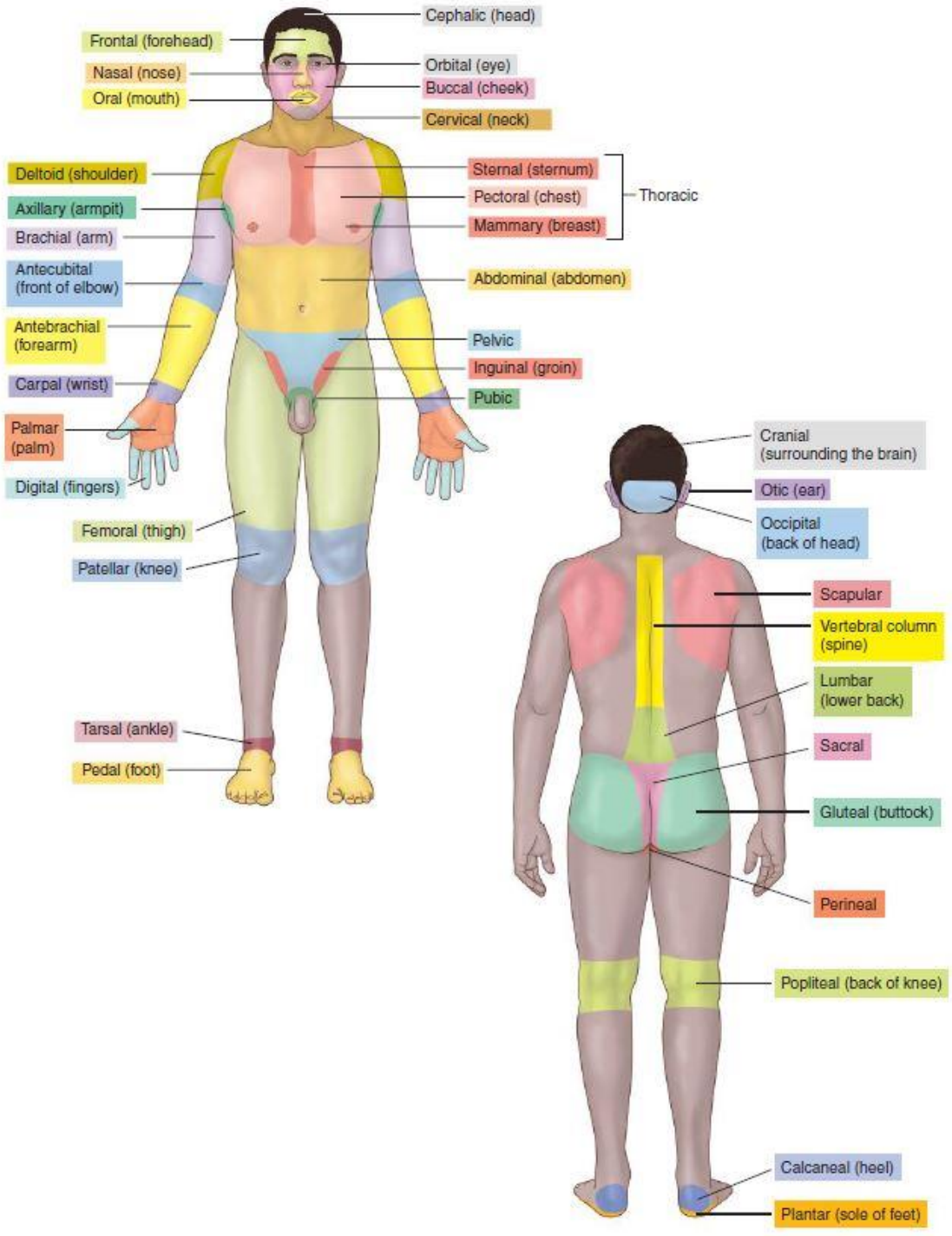


Figure 5: Body Regions

C. Skeletal System

The bones of the skeleton are more than just racks from which muscles hang. They have a variety of vital functions. In addition to supporting the weight of the body, bones work with muscles to maintain body position and to produce controlled, precise movements. Without the skeleton to pull against, contracting muscle fibers could not make us sit, stand, walk, or run.

The 206 bones in the adult human body are actually dynamic living tissue. Bone constantly breaks down and rebuilds itself, not just during the growth phases of childhood, but throughout the life span. Bone is filled with blood vessels, nerves, and living cells; in addition, its interaction with other body systems is necessary not only for movement, but also for life itself.

1. Primary Functions

Your skeletal system includes the bones of the skeleton and, the cartilages, ligaments, and other connective tissues that stabilize or interconnect the bones. This system has five primary functions:

a. *Support.* The skeletal system provides structural support for the entire body. Individual bones or groups of bones provide a framework for the attachment of soft tissues and organs.
b. *Storage of Minerals and Lipids.* Minerals are inorganic ions that contribute to the osmotic concentration of body fluids. Minerals also take part in various physiological processes, and several are important as enzyme cofactors. Calcium is the most abundant mineral in the human body. The calcium salts of bone are a valuable mineral reserve that maintains normal concentrations of calcium and phosphate ions in body fluids. In addition, the bones of the skeleton store energy as lipids in areas filled with *yellow bone marrow.*
c. *Blood Cell Production.* Red blood cells, white blood cells, and other blood elements are produced in *red bone marrow,* which fills the internal cavities of many bones. We will describe blood cell formation when we examine the cardiovascular and lymphatic systems.
d. *Protection.* Skeletal structures surround many soft tissues and organs. The ribs protect the heart and lungs, the skull encloses the brain, the vertebrae shield the spinal cord, and the pelvis cradles digestive and reproductive organs.
e. *Leverage.* Many bones function as levers that can change the magnitude and direction of the forces generated by skeletal muscles. The movements produced range from the precise motion of a fingertip to changes in the position of the entire body.

2. Bone Shapes

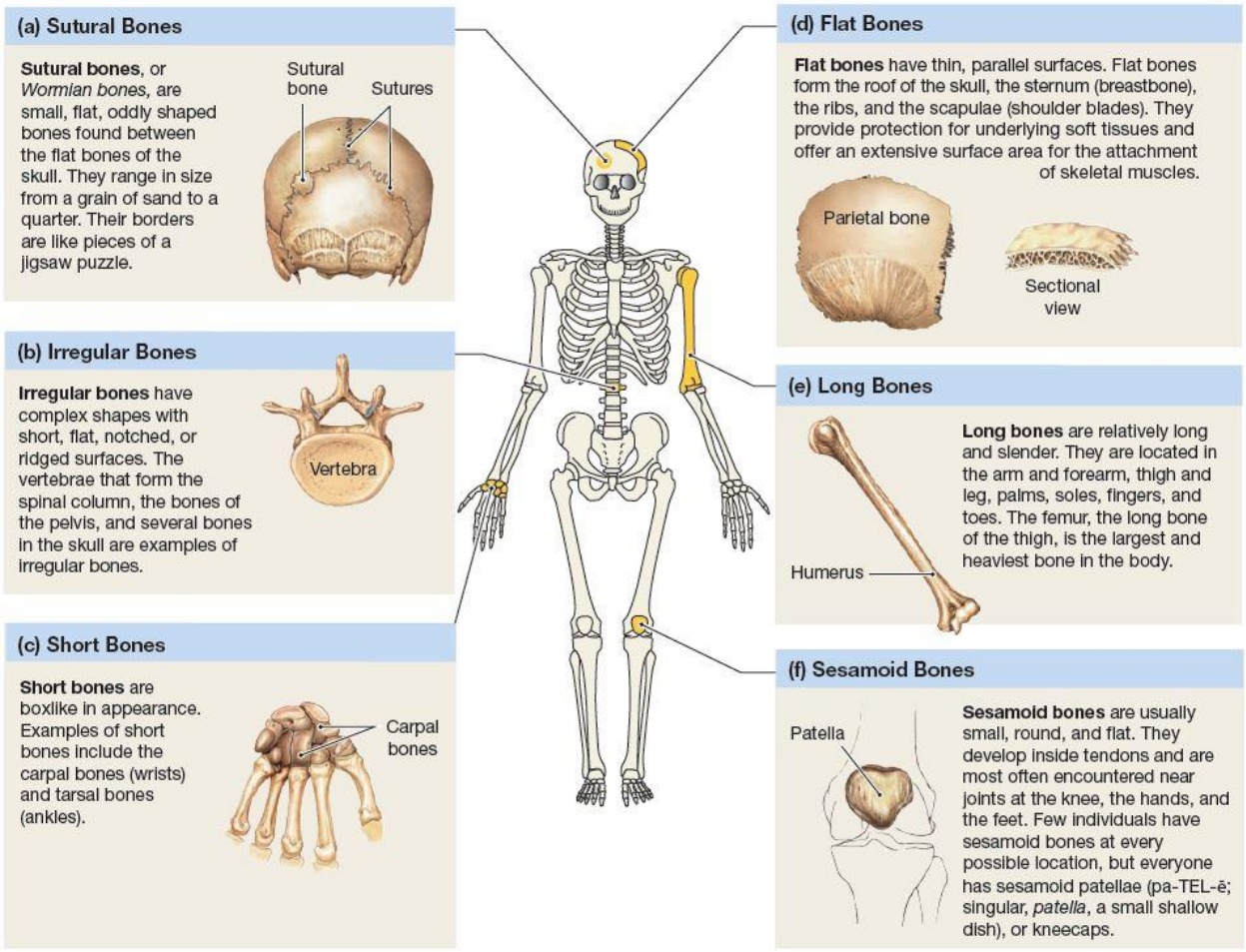


Figure 6: Bone Shapes

3. Bone Development

The growth of the skeleton determines the size and proportions of your body. The bony skeleton begins to form about six weeks after fertilization, when the embryo is approximately 12 mm (0.5 in.) long. (At this stage, the existing skeletal elements are made of cartilage.) During subsequent development, the bones undergo a tremendous increase in size. Bone growth continues through adolescence, and portions of the skeleton generally do not stop growing until about age 25. Ossification or osteogenesis refers specifically to the formation of bone.

4. Axial Skeleton

The axial skeleton forms the longitudinal axis of the body. The axial skeleton has 80 bones, about 40 percent of the bones in the human body: The *skull* (8 *cranial bones* and 14 *facial bones*). Bones associated with the skull (6 *auditory ossicles* and the *hyoid bone*). The *vertebral column* (24 *vertebrae,* the *sacrum,* and the *coccyx*). The *thoracic cage* (the *sternum* and 24 *ribs*).

The axial skeleton provides a framework that supports and protects the brain, the spinal cord, and the thoracic and abdominal organs. It also provides an extensive surface area

for the attachment of muscles that (1) adjust the positions of the head, neck, and trunk; (2) perform respiratory movements; and (3) stabilize or position parts of the appendicular skeleton, which supports the limbs.

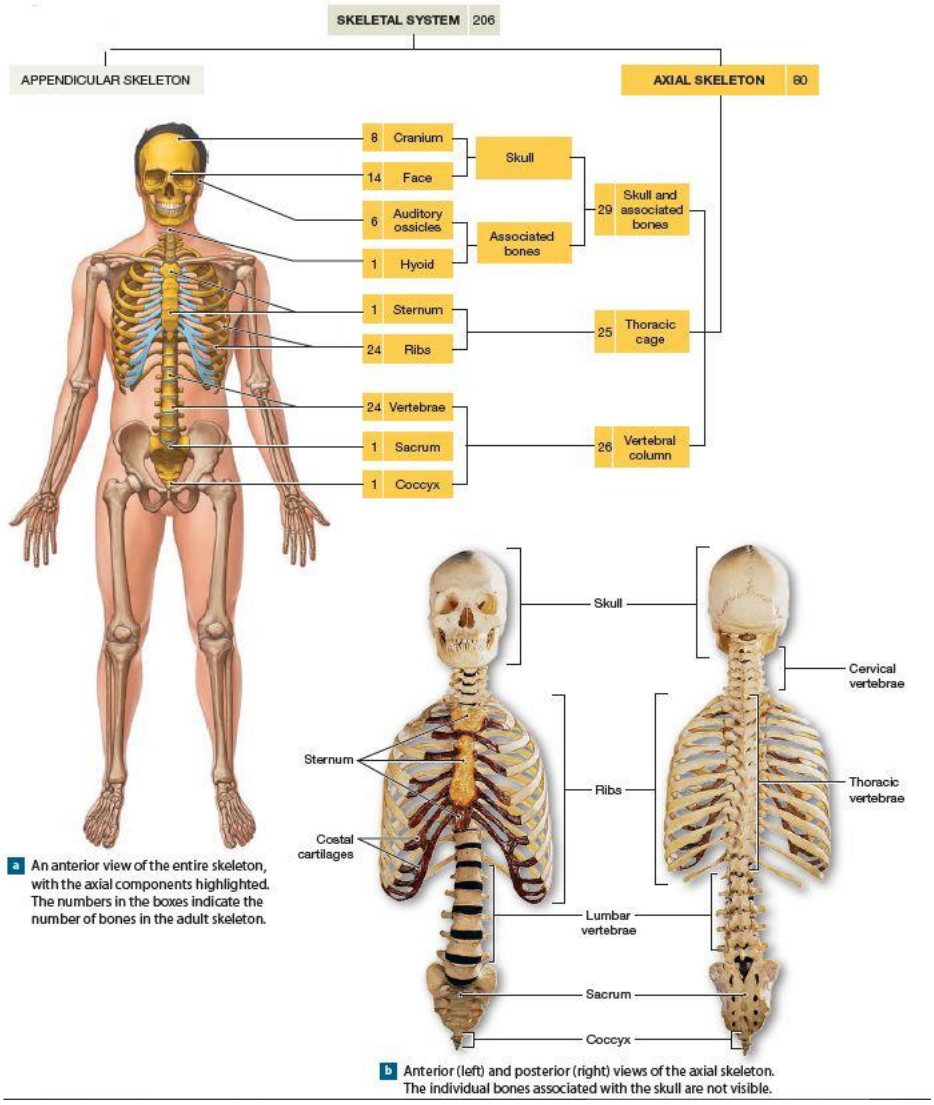


Figure 7: Axial Skeleton

5. Appendicular Skeleton

The appendicular skeleton includes the bones of the limbs and the supporting bone (pectoral and pelvic) girdles that connect them to the trunk). To appreciate the role of the appendicular skeleton in your life, make a mental list of all the things you have done with your arms or legs today. Standing, walking, writing, turning pages, eating, dressing, shaking hands, and texting—the list quickly becomes unwieldy. Your axial skeleton protects and supports internal organs and takes part in vital functions, such as breathing. But your appendicular skeleton lets you manipulate objects and move from place to place.

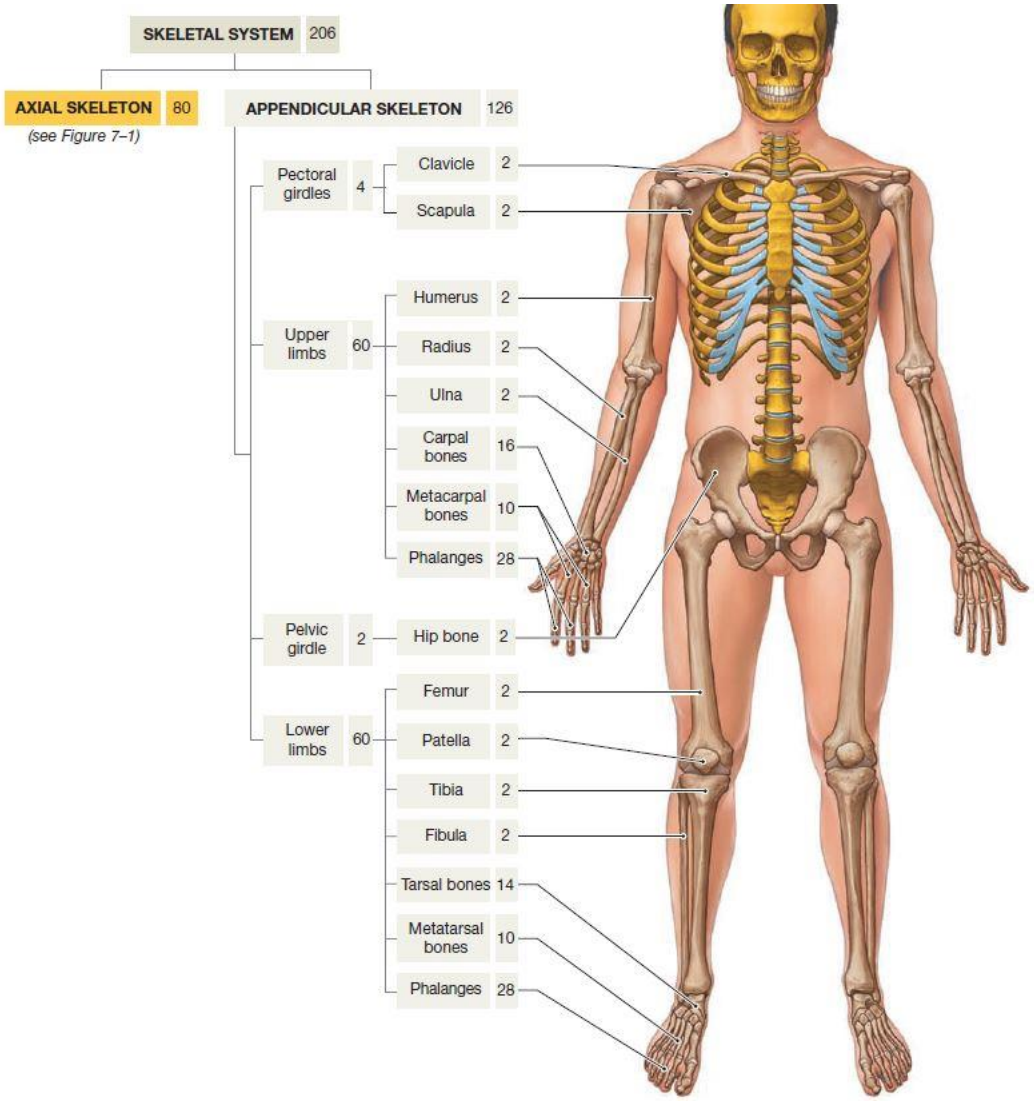


Figure 8: Appendicular Skeleton

D. Muscular System

1. Primary Functions

The muscular system performs six critical functions for the human body. It produces skeletal movement, helps maintain posture and body position, supports soft tissues, guards body entrances and exits, helps maintain body temperature, and stores nutrients.

2. Skeletal Muscles

Muscle tissue consists chiefly of muscle cells that are highly specialized for contraction. Our bodies contain three types of muscle tissue: (1) *skeletal muscle,* (2) *cardiac muscle,* and (3) *smooth muscle.*

3. The nervous system communicates with skeletal muscles at the neuromuscular junction

Skeletal muscle fibers begin contraction with the release of their internal stores of calcium ions. That release is under the control of the nervous system. Communication between a neuron and another cell occurs at a *synapse.* When the other cell is a skeletal

muscle fiber, the synapse is known as a neuromuscular junction (NMJ), or *myoneural junction.*

The NMJ is made up of an *axon terminal* (synaptic terminal) of a neuron, a specialized region of the sarcolemma called the *motor end plate,* and, in between, a narrow space called the *synaptic cleft.* Motor neurons of the central nervous system (brain and spinal cord) carry instructions in the form of action potentials to skeletal muscle fibers.

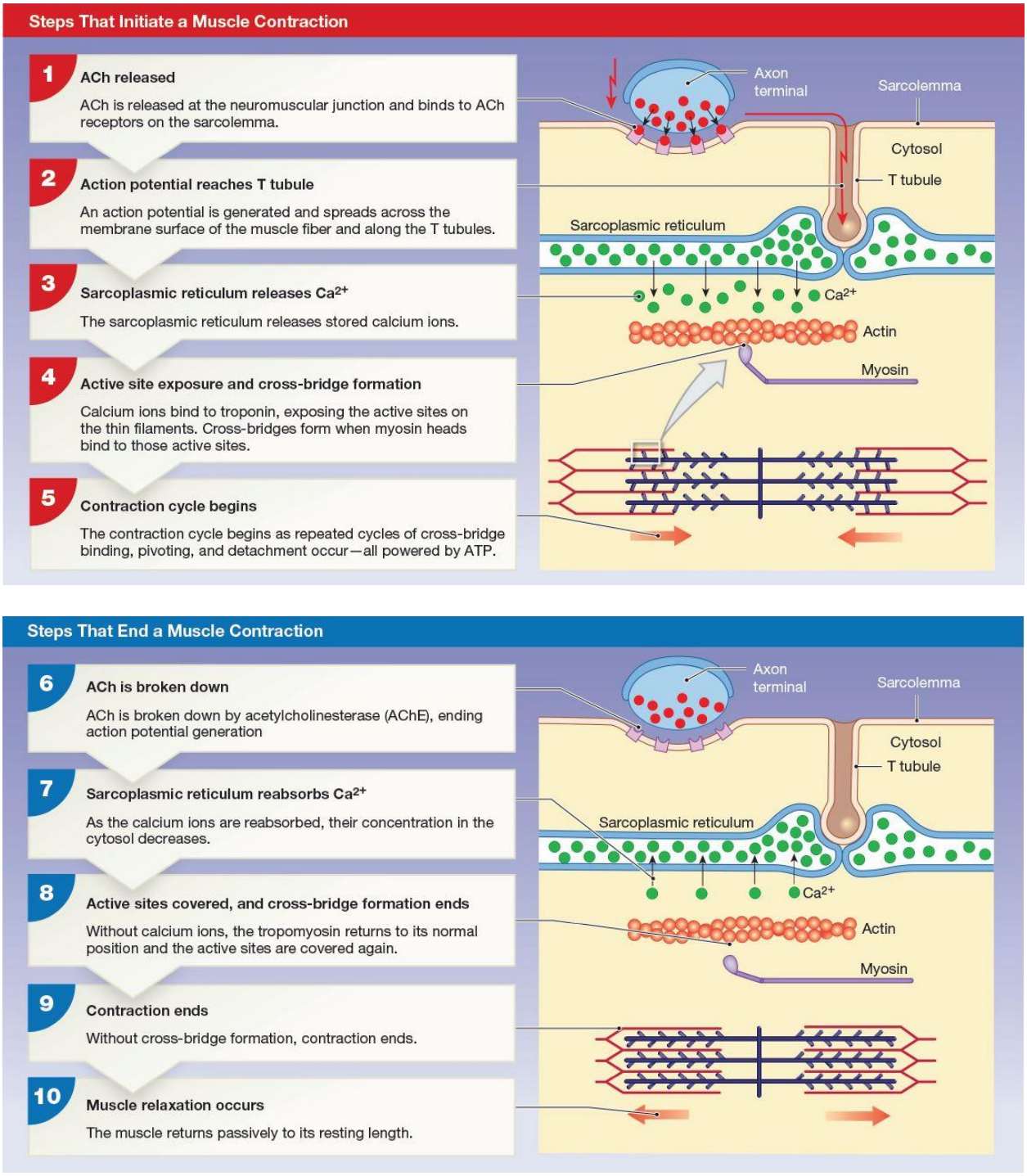


Figure 9: Steps Involved in Skeletal Muscle Contraction and Relaxation.

Skeletal muscles are organs composed mainly of skeletal muscle tissue, but they also contain connective tissues, nerves, and blood vessels. Each cell in skeletal muscle tissue is a single muscle *fiber.* Skeletal muscles attach directly or indirectly to bones.

4. Classification of Skeletal Muscles

a. Parallel muscle – the fascicles are parallel to the long axis of the muscle. Most of the skeletal muscles in the body are parallel muscles.
b. Convergent muscle – muscle fascicles extending over a broad area come together, or converge, on a common attachment site.
c. Pennate muscle – the fascicles form a common angle with the tendon. Because the muscle fibers pull at an angle, contracting pennate muscles do not move their tendons as far as parallel muscles do.
d. Circular muscle, or sphincter – the fascicles are concentrically arranged around an opening. When the muscle contracts, the diameter of the opening becomes smaller.

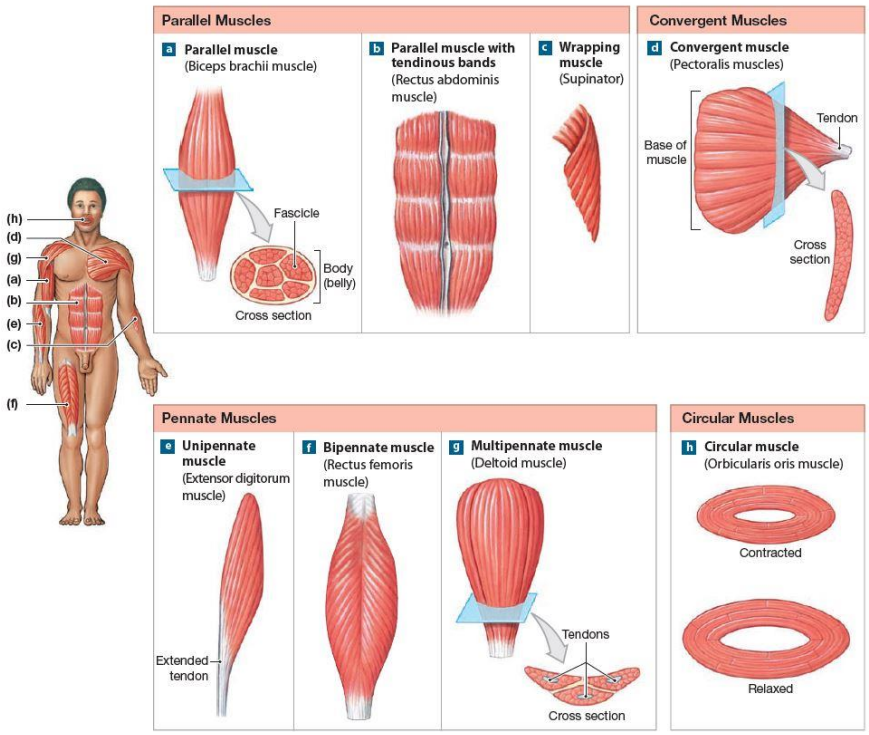


Figure 10: Muscle Type Based on Pattern of Fascicle Organization

5. Axial and Appendicular Muscles

The separation of the skeletal system into axial and appendicular divisions serves as a useful guideline for subdividing the muscular system as well:

- The axial muscles arise on the axial skeleton. This category includes approximately 60 percent of the skeletal muscles in the body. They position the head and spinal column and also move the rib cage, assisting in the movements that make breathing possible. They do not play a role in movement or support of either the pectoral or pelvic girdle or the limbs.
- The appendicular muscles stabilize or move components of the appendicular skeleton. These muscles include the remaining 40 percent of all skeletal muscles.

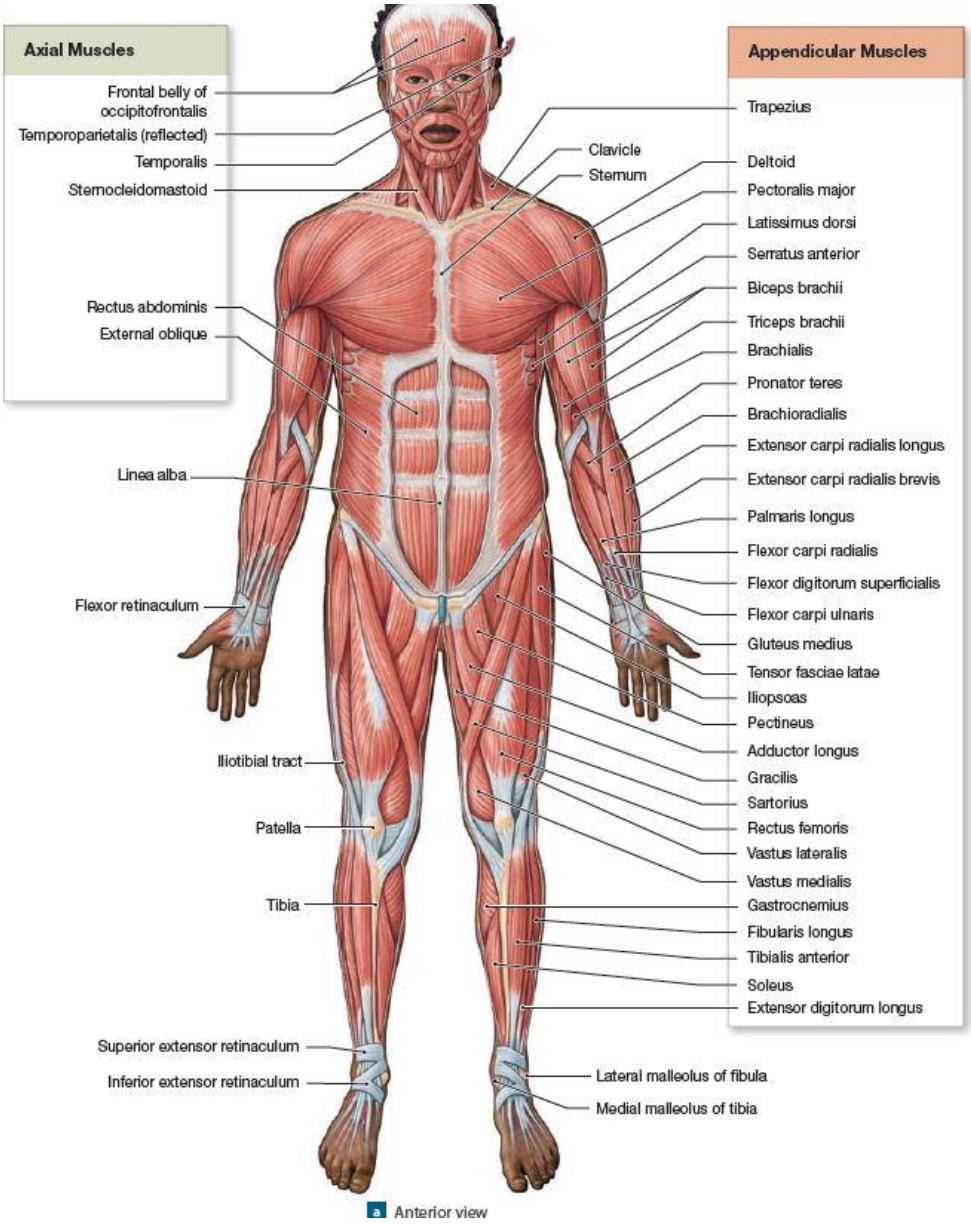


Figure 11: Anterior View of Major Skeletal Muscles

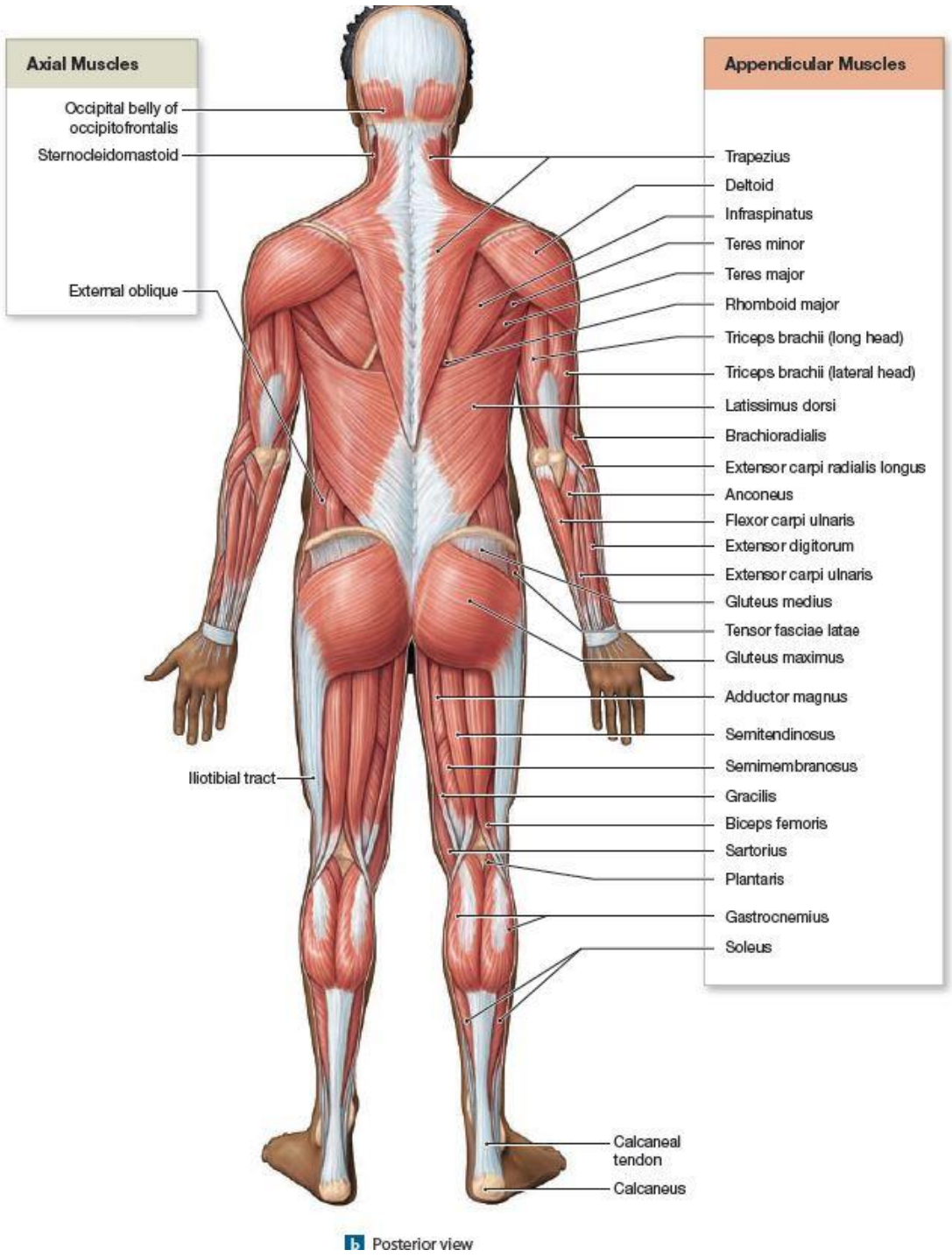


Figure 12: Posterior View of Major Skeletal Muscles

E. JOINTS AND BODY MOVEMENTS

Think of your activities in a typical day: You breathe, talk, walk, sit, stand, and change positions countless times. In each case, your skeleton is directly involved. Movements can occur only at joints, or articulations, where two bones meet, because the bones of the skeleton are fairly inflexible. The characteristic structure of a joint determines the type and amount of movement that may take place. Each joint reflects a compromise between the need for strength and the need for mobility.

1. Joints

We use two classification methods to categorize joints. The first is the one we will use in this chapter. It is a functional scheme because it is based on the amount of movement possible, a property known as the *range of motion* (ROM). Each functional group is further subdivided primarily on the basis of the anatomical structure of the joint

a. An *immovable joint* is a synarthrosis. A synarthrosis can be *fibrous* or *cartilaginous,* depending on the nature of the connection. Over time, the two bones may fuse.
b. A *slightly movable joint* is an amphiarthrosis. An amphiarthrosis is either *fibrous* or *cartilaginous,* depending on the nature of the connection between the opposing bones.
c. A *freely movable joint* is a diarthrosis, or *synovial joint.* Diarthroses are subdivided according to the movement permitted.

The second classification scheme relies solely on the anatomy of the joint, without regard to the degree of movement permitted. Using this framework, we classify joints as *fibrous, cartilaginous, bony,* or *synovial.* Bony joints form when fibrous or cartilaginous joints ossify. The ossification may be normal or abnormal, and may occur at various times in life.

The two classification schemes are loosely correlated. We see many anatomical patterns among immovable or slightly movable joints, but there is only one type of freely movable joint—synovial joints. All synovial joints are diarthroses. We will use the functional classification rather than the anatomical one because our primary interest is how joints work.

Functional Category	Structural Category and Type		Description
SYNARTHROSIS (no movement)			
At a synarthrosis, the bony edges are quite close together and may even interlock. These extremely strong joints are located where movement between the bones must be prevented.	FIBROUS	Suture 	A **suture** (*sutura,* a sewing together) is a synarthrotic joint located only between the bones of the skull. The edges of the bones are interlocked and bound together at the suture by dense fibrous connective tissue.
	FIBROUS	Gomphosis 	A **gomphosis** (gom-FŌ-sis; *gomphos,* bolt) is a synarthrosis that binds the teeth to bony sockets in the maxillae and mandible. The fibrous connection between a tooth and its socket is a *periodontal* (per-ē-ō-DON-tal) *ligament* (*peri,* around + *odontos,* tooth).
	CARTILAGINOUS	Synchondrosis 	A **synchondrosis** (sin-kon-DRŌ-sis; *syn,* together + *chondros,* cartilage) is a rigid, cartilaginous bridge between two articulating bones. The cartilaginous connection between the ends of the first pair of vertebrosternal ribs and manubrium of the sternum is a synchondrosis. Another example is the epiphyseal cartilage, which connects the diaphysis to the epiphysis in a growing long bone.
	BONY	Synostosis 	A **synostosis** (sin-os-TŌ-sis) is a totally rigid, immovable joint created when two bones fuse and the boundary between them disappears. The frontal (metopic) suture of the frontal bone, the fusion of an infant's left and right mandibular bones, and the epiphyseal lines of mature long bones are synostoses.

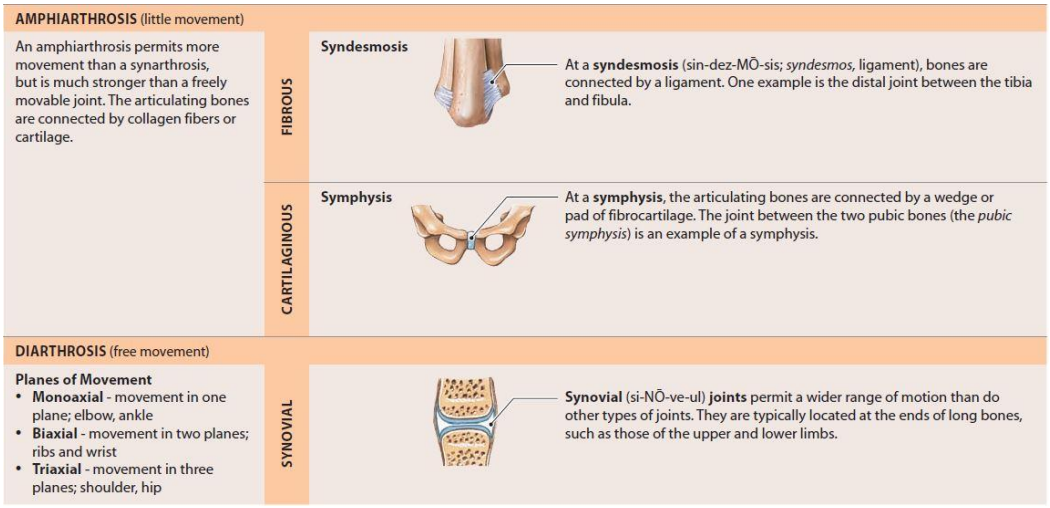


Figure 13: Functional and Structural Classification of Joints

2. Synovial Joints

Synovial joints are freely movable and classified as diarthroses. A two-layered joint capsule, also called an articular capsule, surrounds the synovial joint. Under normal conditions, the bony surfaces at a synovial joint cannot contact one another, because special articular cartilage covers the articulating surfaces.

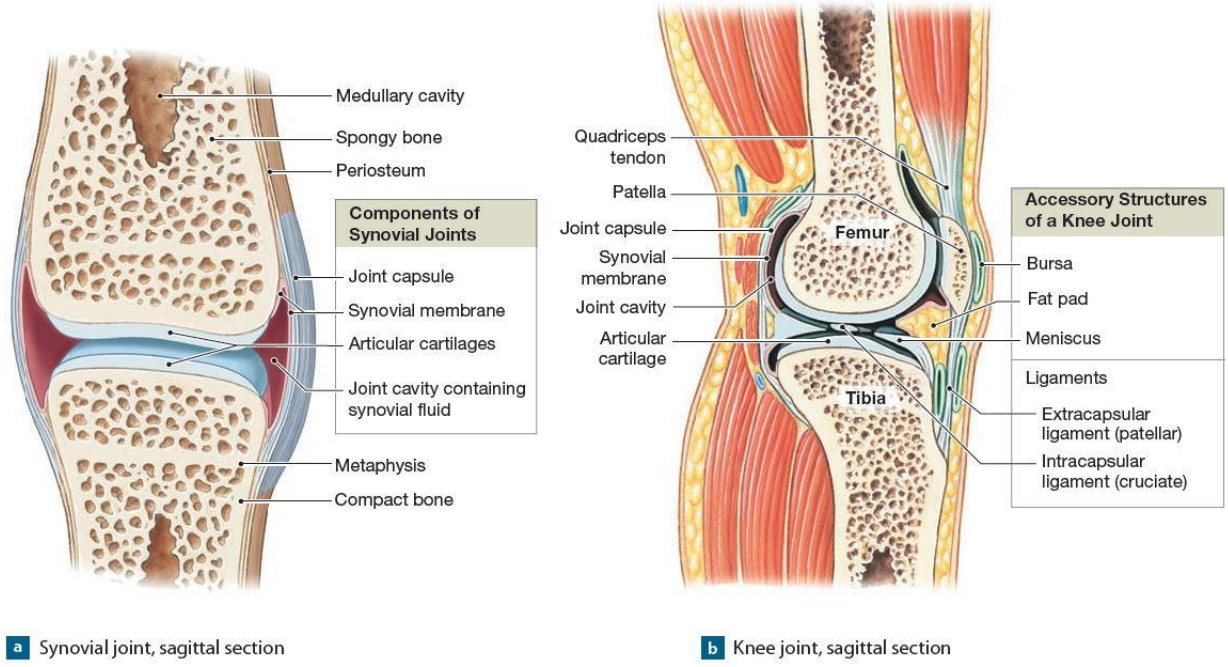


Figure 14: Structure of a synovial joint

3. Synovial Fluid, Ligaments, Tendons and Bursae

Synovial fluid is a clear, viscous solution with the consistency of egg yolk or heavy molasses. Synovial fluid resembles interstitial fluid, but contains proteoglycans with a high concentration of hyaluronan (hyaluronic acid) secreted by fibroblasts of the synovial membrane.

The capsule that surrounds the entire joint is continuous with the periostea of the articulating bones. Accessory ligaments support, strengthen, and reinforce synovial joints. *Capsular ligaments,* or *intrinsic ligaments,* are localized thickenings of the joint capsule. *Extrinsic ligaments* are separate from the joint capsule. These ligaments may be located either inside or outside the joint capsule, and are called *intracapsular* or *extracapsular* ligaments, respectively.

Tendons are not part of the joint itself, but tendons passing across or around a joint may limit the joint's range of motion and provide mechanical support for it. For example, tendons associated with the muscles of the arm help brace the shoulder joint.

Bursae are small, thin, fluidfilled pockets in connective tissue. They contain synovial fluid and are lined by a synovial membrane. Bursae may be connected to the joint cavity or separate from it. They form where a tendon or ligament rubs against other tissues.

4. Types of Movements at Synovial Joints

Gliding Movement

In gliding, two opposing surfaces slide past one another. Gliding occurs between the surfaces of articulating carpal bones, between tarsal bones, and between the clavicles and the sternum. The movement can occur in almost any direction, but the amount of movement is slight, and rotation is generally prevented by the capsule and associated ligaments.

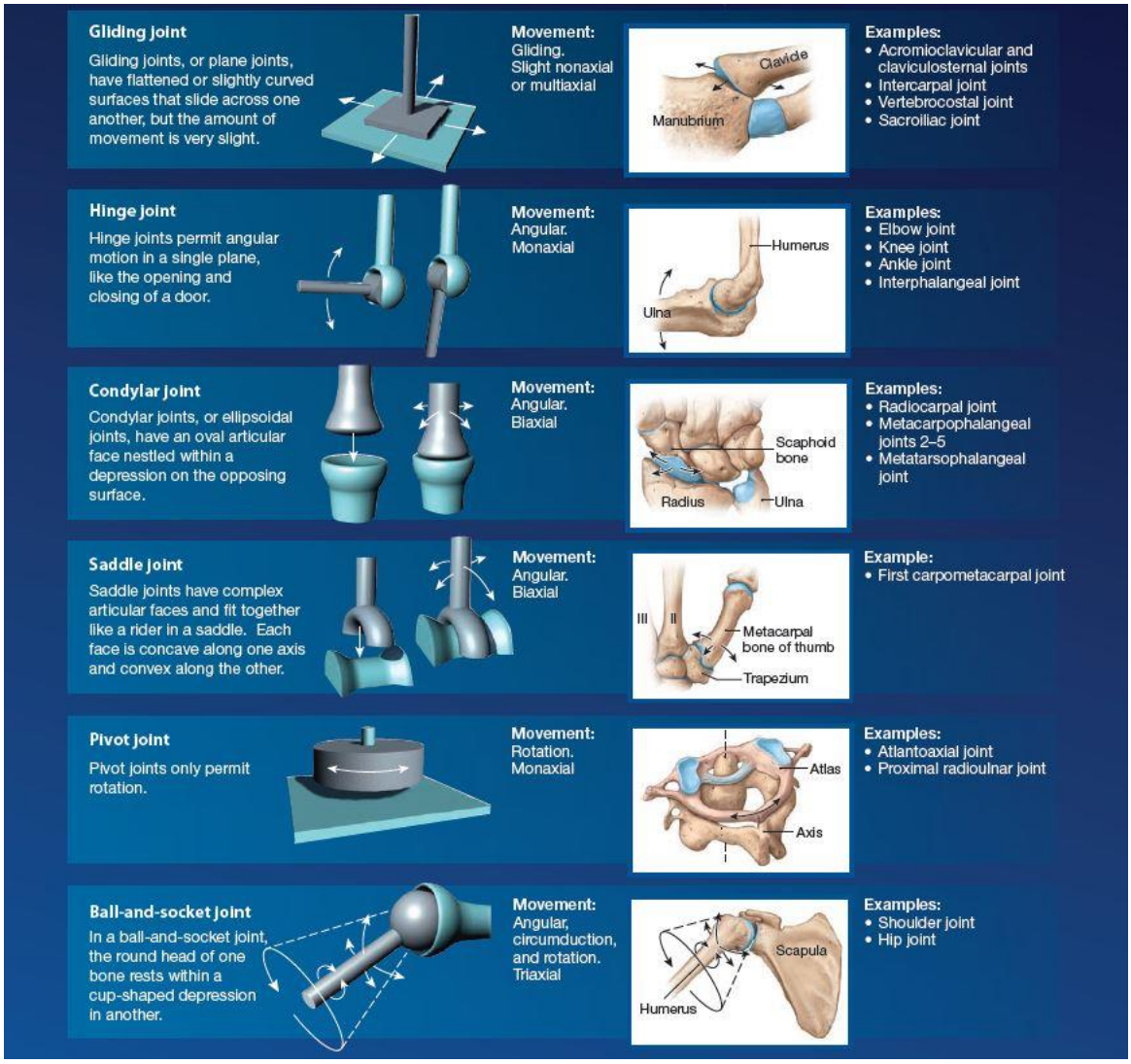


Figure 15: Classification of Synovial Joints

Angular Movement

Examples of angular movement include *flexion, extension, abduction, adduction,* and *circumduction.* Descriptions of these movements are based on reference to an individual in the anatomical position.

- **Flexion and Extension.** Flexion is movement in the anterior–posterior plane that decreases the angle between articulating bones. Extension occurs in the same plane, but it increases the angle between articulating bones

- **Abduction and Adduction.** Abduction is movement *away* from the longitudinal axis of the body in the frontal plane. For example, swinging the upper limb to the side is abduction of the limb. Moving it back to the anatomical position is adduction. Adduction of the wrist moves the heel of the hand and fingers *toward* the body, whereas abduction moves them farther away.

- **Circumduction.** Recall the special type of angular movement, circumduction, from our model. Moving your arm in a loop is circumduction, as when you draw a large circle on a whiteboard. Your hand moves in a circle, but your arm does not rotate.

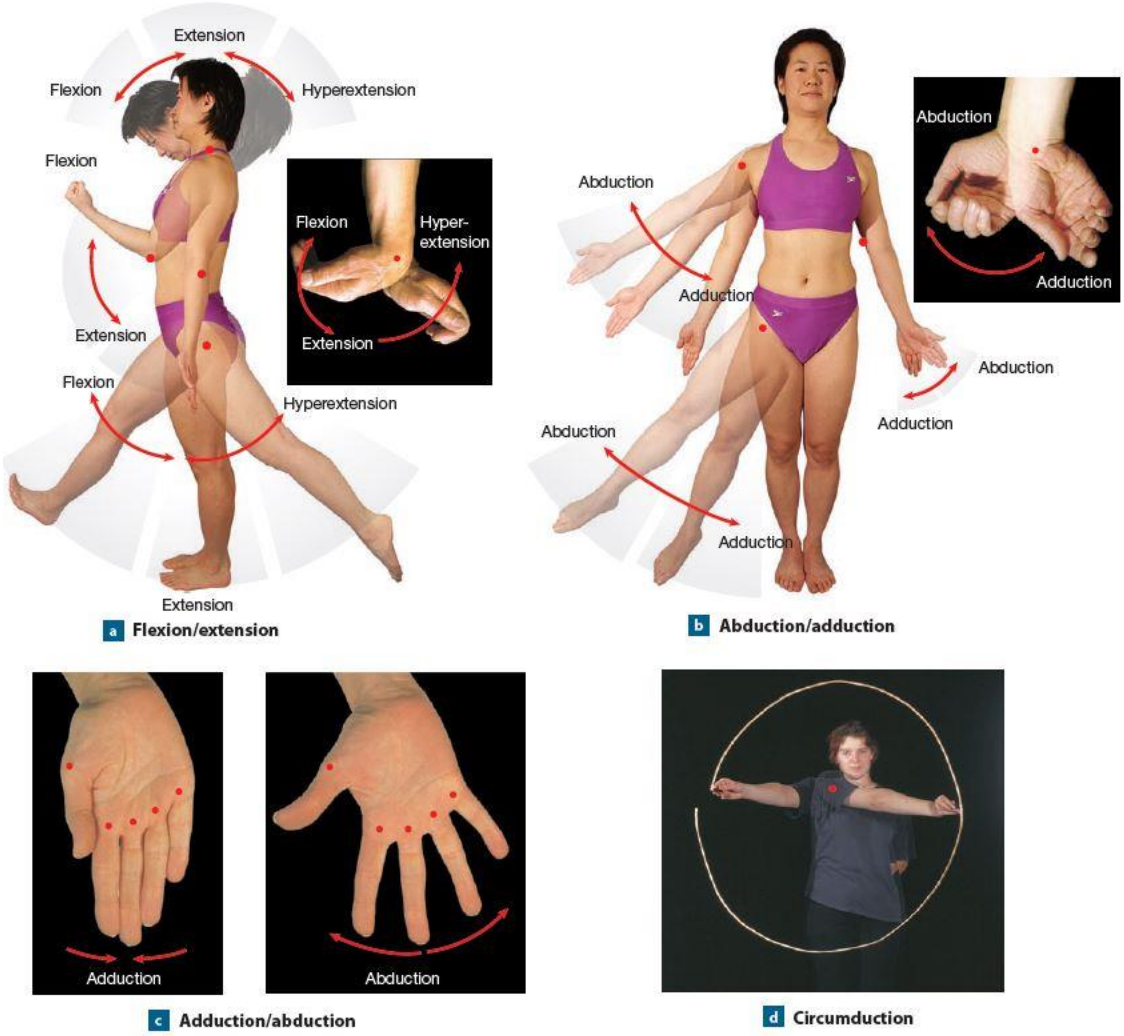
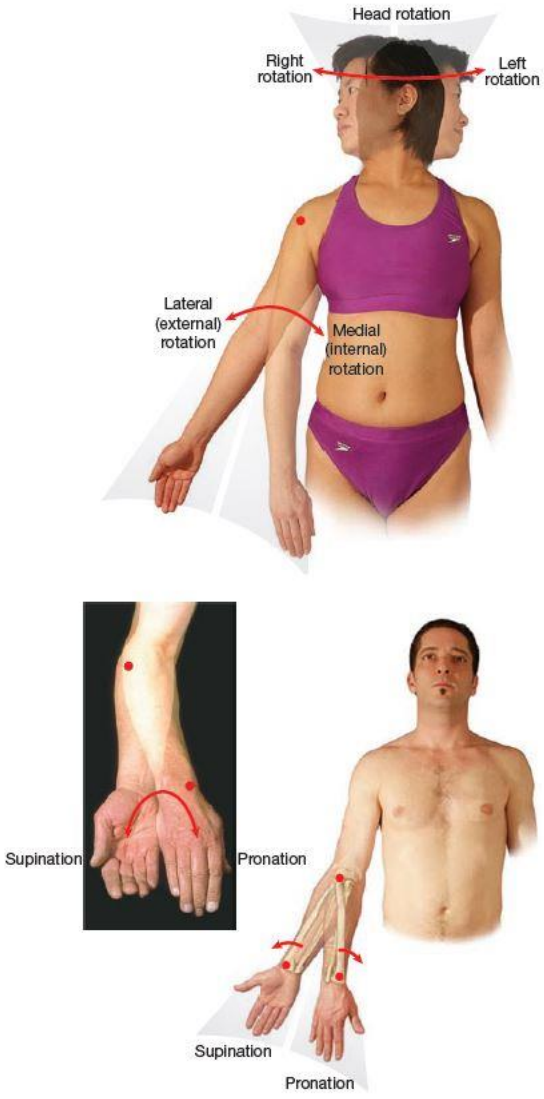


Figure 16: Angular Movements

Rotational Movement



Rotation of the head may involve left rotation or right rotation. Limb rotation by reference to the longitudinal axis of the trunk. During medial rotation, also known as *internal rotation* or *inward rotation,* the anterior surface of a limb turns toward the long axis of the trunk. The reverse movement is called lateral rotation, *external rotation,* or *outward rotation.*

The proximal joint between the radius and the ulna permits rotation of the radial head. As the shaft of the radius rotates, the distal epiphysis of the radius rolls across the anterior surface of the ulna. This movement, called *pronation*, turns the wrist and hand from palm facing front to palm facing back. The opposing movement, in which the palm is turned anteriorly, is *supination***.** The forearm is supinated in the anatomical position.

Figure 17: Rotational Movements

Special Movements

- *Inversion* is a twisting movement of the foot that turns the sole inward, elevating the medial edge of the sole. The opposite movement is called *eversion*.
- *Dorsiflexion* is flexion at the ankle joint and elevation of the sole, as when you dig in your heel. *Plantar flexion*, the opposite movement, extends the ankle joint and elevates the heel, as when you stand on tiptoe. However, it is also acceptable (and simpler) to use “flexion and extension at the ankle,” rather than “dorsiflexion and plantar flexion.”
- *Opposition* is movement of the thumb toward the surface of the palm or the pads of other fingers. Opposition enables you to grasp and hold objects between your thumb and palm. *Reposition* is the movement that returns the thumb and fingers from opposition.
- *Protractio*n is moving a body part anteriorly in the horizontal plane. *Retracti*on is the reverse movement. You protract your jaw when you put your chin forward, and you retract your jaw when you return it to its normal position.

- *Elevation and depression* take place when a structure moves in a superior or an inferior direction, respectively. You depress your mandible when you open your mouth, and you elevate your mandible as you close your mouth. Another familiar elevation takes place when you shrug your shoulders.
- *Lateral flexion* occurs when your vertebral column bends to the side. This movement is most pronounced in the cervical and thoracic regions.

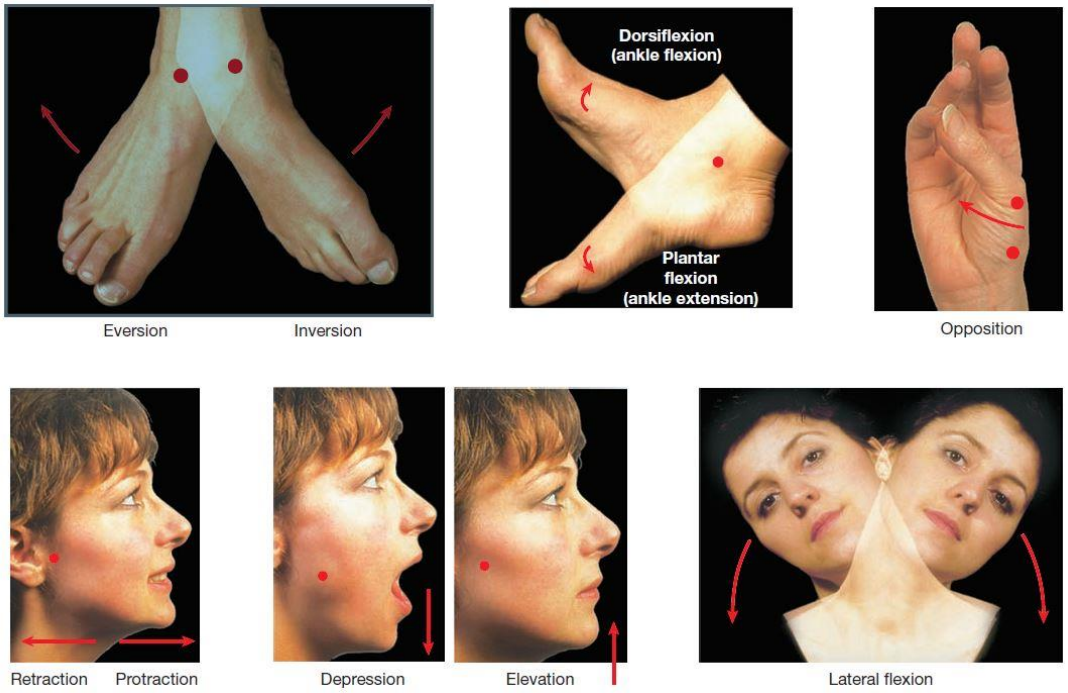


Figure 18: Special Movements

Suggested Media at Home

Topic	Website/s
Basics of Anatomy and Physiology	• https://www.khanacademy.org/science/health-and-medicine/human-anatomy-and-physiology?fbclid=IwAR0IFYSEdB4cQ4ZR3xLwuPR_AE093hnwda__0_niEncfr3iyeLhhz6Aw9Uo • https://www.khanacademy.org/science/biology/principles-of-physiology
Anatomical Position and Directional Terms	• https://www.youtube.com/watch?v=pQUMJ6Gh9Bw
Body Regions	• https://www.youtube.com/watch?v=qDoKJO3M9So
Organ Systems of the Human Body	• https://www.youtube.com/watch?v=gEUu-A2wfSE
Skeletal System	• https://www.youtube.com/watch?v=f-FF7Qigd3U • https://www.innerbody.com/image/skelfov.html • https://www.healthline.com/human-body-maps/skeletal-system#conditions • https://www.khanacademy.org/science/high-school-biology/hs-human-body-systems/hs-the-musculoskeletal-system/v/skeletal-structure-and-function
Muscular System	• https://www.innerbody.com/image/musfov.html • https://www.khanacademy.org/science/high-school-

	biology/hs-human-body-systems/hs-the-musculoskeletal-system/v/anatomy-of-a-muscle-cell-1 • https://www.youtube.com/watch?v=jFTPG9LQ7Ps • https://www.youtube.com/watch?v=LK51apYOdtc
Joints & Body Movements	• https://www.visiblebody.com/learn/skeleton/joints-and-ligaments • https://www.khanacademy.org/science/high-school-biology/hs-human-body-systems/hs-the-musculoskeletal-system/v/ligaments-tendons-and-joints • https://www.youtube.com/watch?v=aRaLjN2cTYo • https://www.youtube.com/watch?v=X5RUFXZZBH4 •

IV. ACTIVITIES/ASSESSMENTS

In this section, a variety of suggested activities and assessments were provided. It provides activities to assess the appreciation (reflective questions) and understanding (knowledge check) of the students on the topics provided. It also suggests options for practical application (a performance task) by providing a performance activity that includes the rubrics or criteria for rating. Remember that as a teacher, you have the academic freedom to choose what activities you would like to assign to your students, modify some of the tasks, or provide additional or other activities that you think will be more suitable for our dear students, recognizing the different capabilities and situations they have right now.

Activity Worksheet 3. Reflective Questions

Name:_____ **Score:** _____
Course, Year and Section: _____ **Equivalent:** _____

A. Reflection

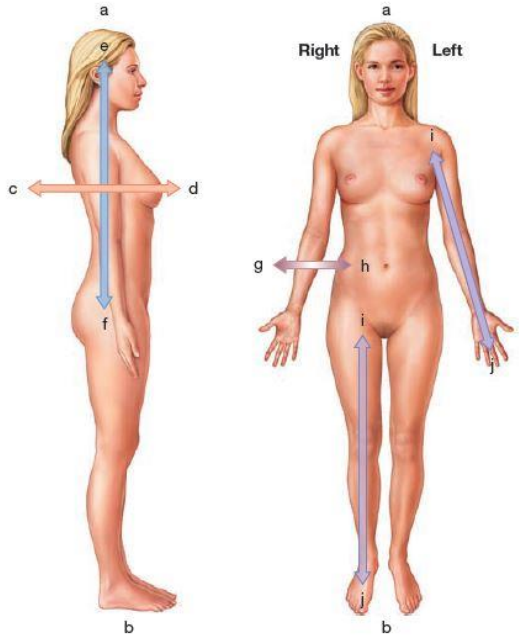
As a student, why is learning about basic anatomy important? What do you think are the advantages of having anatomical knowledge in movements, exercises and injuries? (10 points)

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Activity Worksheet 4. Knowledge Check

Name:_____ **Score:** _____
Course, Year and Section: _____ **Equivalent:** _____

B.1 Label the Directional Terms



a. _____
b. _____
c. _____
d. _____
e. _____
f. _____
g. _____
h. _____
i. _____
j. _____

B.2 Match each numbered item with the most closely related lettered item. Use letter for answers in the space provided

____1. total number of adult human bones
____2. heel of foot
____3. adduction
____4. anatomy
____5. Pennate muscle
____6. carpal
____7. cardiovascular system's major organ
____8. neck
____9. axillary
____10. supine

(a) armpit
(b) heart
(c) face up position
(d) movement *away* from the longitudinal axis of the body in the frontal plane
(e) the fascicles form a common angle with the tendon
(f) 206
(g) calcaneal
(h) wrist
(i) cervical
(j) study of internal and external body structures

Performance Task 3

Movement Analysis

Name:_____ **Score:** _____
Course, Year and Section: _____ **Equivalent:** _____

Sports, Bones, Muscles & Movements. Choose one (1) sport of your choice and list down two (2) skills. Try to perform the skill and answer the following:

- What are the MAJOR MUSCLES involved in the movement?
- Describe the anatomical movements applied to execute the skill

(10 points each)
Name of Sport: _____

1. Skill No. 1: _____

_____
_____
_____
_____
_____
_____
_____
_____
_____

2. Skill No. 2: _____

_____
_____
_____
_____
_____
_____
_____
_____

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Chapter 3

EXERCISE PRESCRIPTION

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. demonstrate understanding on basic principles on exercise prescription;
2. apply MFIT formula on the creation of exercise program;
3. determine their own target cardir fitness zone; and
4. create and perform exercise workout that are relatively safe to perform.

Course Outline:

- Overview
- Discussion
    - Physical Activity Pyramid
    - Three Phases of Exercise Program
    - Principles of Training Exercise
    - The FITT Formula
    - How to Determine Cardio Respiratory Training Zone
    - Exercise Prescription Guidelines
- Summary
- Suggested Media at Home
- Activities and Assessments
    - Reflective Questions
    - Knowledge Check
    - Performance Task
- References

II. OVERVIEW

According to World Health Organization (WHO) at least 60% of the world's population are in active and does not achieve the recommended amount of daily physical activities. And one way to start being active is to carefully monitor your daily physical activity. Other than monitoring the time spent in doing exercise or engaging in physical activity, a good way to monitor physical activity is using pedometer. It is a small and handy device that detects vertical movement at the hip and measure number of footsteps in a travel distance. Though it is not 100% accurate and being questioned of its reliability, it is a great motivational tool to help assess, maintain, and increase daily physical activity involvement. It is ideal to use in physical activities that involve related lower-body movements such as walking, running, and jogging. A total of 10,000 steps per day, with at least 10-30minutes of physical activity are the recommended daily steps for adults. Please refer to table below for specific ratings.

Adult Activity Levels Based on Total Number of Steps	
Steps per Day	**Category**
Highly Active	≥12,500
Active	10,000–12,499
Somewhat active	7,500–9,999
Low Active	5,000-7,499
Sedentary Lifetysle	< 5,000

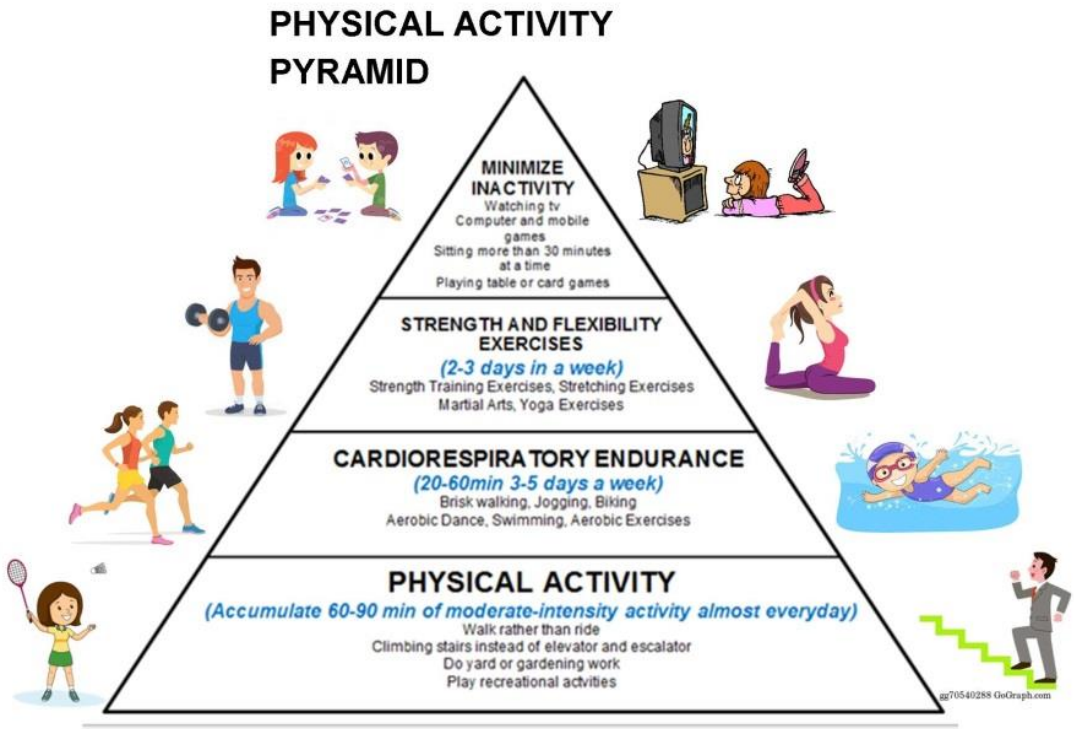
Source: C. Tudor-Locke and D. R. Basset, "How Many Steps/Day Are Enough? Preliminary Pedometer Indices for Public Health," Sports Medicine 34 (2008): 1–8

There is no doubt that regular physical activity and exercise have a lot of significant benefits to be stepped aside. But it is a must to have it as part of your daily lifestyle. On the contrary, huge percentage of the population does not exercise regularly and only few are able to reach high physical fitness level. While among others, were not able to sustain the exercise

program up to the end. If you are not yet into active life style, why not give exercise a try? And to guide you about engaging in physical activity and exercise, please see figure below.

III. COURSE MATERIALS: DISCUSSION

A. Physical Activity Pyramid



This pyramid shows different types of physical activities and exercise that contribute to the optimum development of health, wellness, and physical fitness of individuals.

To make it simple, the bottom of the pyramid is the daily activities that should be done at least a total of 60-90 minutes of moderate activity near every day. Doing household chores, gardening, climbing stairs, and participating in recreational activities such as badminton, and traditional and street games are some of the activities that you may consider to have and active life style.

On the second level, if you decided to engage in cardiorespiratory exercises like brisk walking, jogging, biking, and other aerobic exercise it is recommended to have a moderate intensity of 20-60 minutes, 3-5 days a week program.

Placed on the third level are the strength and flexibility exercises which recommend having exercise program every 2-3 days in a week. Exercise principles should be carefully take into consideration in designing the program. Some of the exercises are yoga, martial arts, and resistance training. While on top, are the activities that should be done in minimal number of minutes. Sedentary lifestyle must be avoided, since it contributes for having non-communicable diseases such as heart diseases, stroke, and diabetes.

Exercise prescription refers to way of designing a program that is individualized, safe, and effective. It embraces MFIT and basic exercise principles in order to maintain or improve physical fitness level, lessen the risk for lifestyle and other diseases, and improve the quality of life.

B. Three Phases of Exercise Program

For you to be able to have an effective and safe exercise program, following the three phases of exercise if highly essential.

1. **Warm – Up.** This prepares the body especially the heart and circulatory system for physical activity or exercise. According to the American College of Sports Medicine (ACSM) warm-up must include a 5-10 minutes of low to moderate cardiovascular and muscular endurance activities like walking, jogging or running for the first phase and a gentle static stretching of the major large muscle group for the second phase. Stretching exercises in this phase is highly recommended to reduce risk of possible injuries.

2. **Work – Out or Conditioning Phase.** Refers to the key component of a fitness program or your chosen physical activity (e.g. resistance training, aerobic dancing, swimming, bicycling) This phase should follow the principle of training or the MFIT Formula with regards to mode, frequency, intensity, and time of physical activity.

3. **Cool – Down.** This promotes an effective and gradual recovery from physical activity. The transition in the cool-down is now from higher intensity of workout/conditioning phase back toward to resting status, allowing a gradual slowing of the heart rate, systolic blood pressure, and metabolic end products (e.g. lactate). The first phase also includes moderate to low cardiovascular and muscular endurance activities at least 5-10 minutes. If exercise is stopped abruptly, this may result to post exercise hypotension (low blood pressure) and possible dizziness. This may happen due to blood is left in the area of the working muscle and has no way to get back to the heart and brain. The last part involves stretching exercises that can relieve spasms in fatigued muscles and improves flexibility. According to ACSM, the cool-down stretch should last 5 minutes or more.

C. Principles of Training Exercise

There are several important principles of exercise that provide the basis of determining the amount of physical activity necessary for developing and maintaining good health, wellness and fitness.

1. Overload Principle. The most basic of all physical activity principle that indicates doing **"more than normal"** to result in an improved physical fitness or health benefits. (e.g. to increase flexibility, a muscle must be stretched longer than is normal)

2. Principle of Progression. Refers to the corollary of overload principle that should done in a **gradual manner** rather than in major burst. (e.g. training sessions when done in gradual progression becomes more challenging over time; low -moderate-high intensity; basic-intermediate-advance level)

3. Principle of Specificity. To develop a certain body parts or fitness components, **specific type** of exercise should be performed to achieve good results. (e.g. to improve core muscles, different core strengthening exercises you might consider such as; plank, bird dog, dead bug series, and alike. Another example: for you to improve your cardiorespiratory fitness, cardio exercises such as jogging, cycling, aerobic dance exercises and alike are highly recommended)

4. Principle of Reversibility. It means that disuse or inactivity results in loss of benefits achieved as a result of overload. This principle can be sum up or simplify by the fact that "**if you don't use it, you will lose it."**e.g. if you do not adhere with your physical activity like 30 minutes running everyday, the benefit you attained like weight loss will gradually erode

5. Principle of Rest and Recovery. Pertains to the amount of rest needed to allow the body to adapt to and recover from exercise. It is important to give our body a **time to recuperate** after overload. (e.g. taking a break between heavy training workout or recreation and sport activities).

6. Principle of Individuality. We should be guided by this principle and be constantly reminded that the **impact and benefits of physical activity or workout are different** from one person to another because of individual differences. (e.g. the effect and amount of weight loss in running varies as to the body composition, metabolism of an individual, and lifestyle habit).

D. The FITT Formula

MFIT (Mode, Frequency, Intensity, and Time) refers to the four important variables needed in the application of the different principles of exercise and creation of comprehensive exercise program.

Frequency (how often) Pertains to the number days of a person engaging in a physical activity or exercise program in a week. It should be performed at least 3 up to 5 days per week so as to attain most benefits or depends on the person's specific benefit desired . It must be performed in a regular basis to be effective.

Intensity (how hard) Refers to the required exertion of intense physical activity to be performed or worked against a load greater than normal (overload). To determine appropriate exercise intensity various methods, including heart rate reserve (HRR), percentage of age-predicted maximal heart rate (%HR_{max}), oxygen update reserve (VO2R), and perceived exertion can be utilized.

Time/Duration (how long) This is the length of time the activity session is performed. At least 30 minutes of moderate intensity exercise is recommended per day on at least 5 days a week (150mins per week) or at least 20-25 minutes of vigorous intensity exercise is recommended per day on at least 3 days a week (75mins per week).

Type (Mode) Refers to the type of physical activity you perform. Example of aerobic activities include walking, jogging, running, aerobic dance, swimming and biking. Sports related activities can also be done such as basketball, volleyball, badminton, and nontraditional games.

E. Exercise Prescription Guidelines

Table 1: Exercise Prescription Guidelines

Cardiorespiratory	
Mode	Moderate or vigorous intensity aerobic activity (examples: walking, jogging, stair climbing, aerobics, water aerobics, swimming, basketball, soccer, and alike).
Intensity	30%-90% of heart rate reserve (the training intensity is based on age, health status, initial fitness level, exercise tolerance, and exercise program goals).
Frequency	3 to 5 days per week for vigorous-intensity aerobic activity at least 75 minutes per week, or 5 days per week of moderate-intensity aerobic activity for a minimum total of 150 minutes weekly.
Time/Duration	At least 20 minutes of continues vigorous-intensity or 30 minutes of moderate-intensity aerobic activity (the latter may be accumulated in segments of at least 10 minutes in duration each over the course of the day)

How to Determine Cardio Respiratory Training Zone

Muscles have to be overload for them to develop. CR develops when the heart is working between 30 percent and 90 percent of heart rate reserve (HRR). Health benefits achieved when training is working at a lower exercise intensity that is between 30 percent and 60 percent of the person's HRR. Even greater health and cardioprotective benefits, and higher and faster improvement in CR fitness (VO2max), however, are achieved primarily through vigorous-intensity programs, that is, at the intensity greater that 60 percent. For this reason, many experts prescribed exercise between 60 percent and 90 percent. Intensity of exercise can be calculated easily, and training can be monitored by checking your pulse. To determine the intensity of exercise or cardiorespiratory training zone, follow these steps.

1. Estimate your maximal heart arte (MHR) according to the following formula:
MHR = 207 – (.7 x age).

2. Check your resting rate (RHR) sometime in the evening after you have been sitting quietly for 15-20 minutes. You may take your pulse for 30 seconds and multiply by 2, or take it for a full minute. You can check your pulse on the wrist by placing two or three fingers over the radial artery or on the neck by placing your fingers over the carotid artery.

3. Determine the heart rate reserve (HRR) by subtracting the resting heart rate from maximal heart rate (HRR = MHR –RHR).

4. Calculate the training intensities (Tis) at 30, 40, 50, 60, 70, 90 percent. Multiply the heart rate reserve (HRR) by the respective .30, .40., .50, 60., .70, .80, and .90., and then add the resting heat rate.

   Example. The 30, 40, 50, 60, 70, and 90 percent Tis for a 20 year-old with resting heart rate of 68 beats per minute (bpm) would be as follows:

   MHR: 207 – (.70 X 20) = 193 bpm
   RHR: = 68bpm
   HRR: 193 - 68 = 125 beats
   30% TI = (125 X .30) + 68 = 106 bpm
   40% TI = (125 X .40) + 68 = 118 bpm
   50% TI = (125 X .50) + 68 = 131 bpm
   60% TI = (125 X .60) + 68 = 143 bpm
   70% TI = (125 X .70) + 68 = 155 bpm
   90% TI = (125 X .90) + 68 = 181 bpm

   Light-intensity CR training zone: 106 to 118 bpm
   Moderate-intensity CR training zone: 118-143 bpm
   Vigorous-intensity CR training zone: 143-181 bpm

When you exercise to improve the CR system ideally you should maintain rate between the 60-90 percent training intensities to obtain the best development. If you have been physically inactive, you should train at around 30 to 40 percent intensity during the 2 to 4 weeks of the exercise program. You may increase to a 50 to 60 percent training intensity for the next four weeks, thereafter you should exercise between the 60 to 90 percent training intensity.

Flexibility Exercise Prescription Guidelines	
Mode	Static, dynamic, or proprioceptive neuromuscular facilitation (PNF) stretching to include all major muscle/tendon groups of the body
Intensity	To the point of mild tension or limits of discomfort
Frequency	At least 2 or 3 days per week. Ideally, 5 to 7 days per week
Time/Duration	Repeat each exercise 2 to 4 times, holding the final position between 10 and 30 seconds per repetition, with a cumulative goal of 60 seconds per exercise
Muscular Fitness Exercise Prescription Guidelines	
Mode	Select 8 to 10 dynamic strength-training exercises that involve the body's major muscle groups and include opposing muscle groups (chest and upper back, abdomen and lower back, front and back of the legs).
Intensity (resistance)	Sufficient resistance to perform 8 to 12 repetitions maximum for muscular strength and 15 to 25 repetitions to near fatigue for muscular endurance. Older adults and injury prone individuals should use 10 to 15 repetitions with moderate resistance (50% to 60% of their 1 RM)
Frequency	2 to 3 days per week on nonconsecutive days. More frequent training can be done if different muscle groups are exercised on different days. (Allow at least 48 hours between strength-training sessions of the same muscle group.)
Time/Duration (Sets)	2 to 4 sets per exercise with 2 to 3 minutes recovery between sets for optimal strength development. Less than 2 minutes per set if exercises are alternated that require different muscle groups (chest and upper back) or between muscular endurance sets.

Source: Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness,Fourteenth Edition. Cengage Learning, USA

Summary

At this time of digital age, there are lots of free fitness workouts, fitness challenge, full body workout available in different digital forms such as cellphone apps, social media, television, and more. But always keep in mind that "no one size fits all". It simply means that, not because those workouts are effective to them, it is effective to anyone, or to you. But learning, understanding, and applying the basic fitness concepts and principles on exercise prescription, a higher chance for you to be successful and safely accomplish your exercise program.

Suggested Media at Home

- **FITT Principle**
  https://www.youtube.com/watch?v=cVVqy6aUvK0
- **Principles of training**
  https://www.youtube.com/watch?v=eNcxtSF2-Ac
- **Quick guide to heart rate training | Polar**
  https://www.youtube.com/watch?v=xPFq9EpovNw
- **Physical Activity and Exercise Prescription**
  https://www.physio-pedia.com/Physical_Activity_and_Exercise_Prescription
- **4 General Principles of Exercise Prescription**
  https://www.chp.gov.hk/archive/epp/files/DoctorsHanbook_ch4.pdf
- **HIIT VS CIRCUIT (which one is better for you?)**
  https://www.youtube.com/watch?v=QKGOvb8CiGE

IV. ACTIVITIES/ASSESSMENTS

In this section, a variety of suggested activities and assessments were provided. It provides activities to assess the appreciation (reflective questions) and understanding (knowledge check) of the students on the topics provided. It also suggests options for practical application (a performance task) by providing a performance activity that includes the rubrics or criteria for rating. Remember that as a teacher, you have the academic freedom to choose what activities you would like to assign to your students, modify some of the tasks, or provide additional or other activities that you think will be more suitable for our dear students, recognizing the different capabilities and situations they have right now.

Activity Worksheet 5. *Reflective Question*

Name:_____ Section:_____ Date:_____ Group no._____

In doing some of your exercise workouts or programs, do you consciously consider and follow significant principles such as FITT, exercise training principles, three phases of exercise program? Why? Kindly explain your answer.(10points)

Learning the different exercise prescription for cardiorespiratory, muscular, and felxibility fitness, what are your realization and future plans? .(10points)

Activity Worksheet 6. Knowledge Check

Name:_____ Section:_____ Date:_____ Group no._____

I. Encircle the letter of the best answer that corresponds to the following questions or statements.

1. What is the recommended duration for cardio respiratory exercise per session?
   a. 10-20 minutes b. 15-30 minutes c. 20-60 minutes d. 60-20 minutes

2. Rheena Lyz is currently aiming to improve her cardiorespiratory fitness. In order to achiever best development on here goal, she needs to participate in a cardio workout maintaining _____ training intensities.
   a. 30-40% b. 40-60% c. 60-90% d. 70-90%

3. One good simple physical activity that we can consider to have and active lifestyle is walking. What is the recommended number of steps per day of an individual is needed to achieve this goal?
   a. 1,000-5,000 b. 5,000-8,000 c. 8,000-9,000 d. 10,000-13,000

4. It refers to s light to moderate activity done prior to a work-out.
   a. Guidelines b. Cool-down c. Warm-up d. Pre-test

5. Basic principle in doing exercise that means performing “more than the normal”
   a. Progression b. Overload c. Reversibility d. Specificity

6. This principle of exercise can be sum up or simplify by the fact that “if you don’t use it, you will lose it.”
   a. Specificity b. Overload c. Reversibility d. Rest and Recovery

7. Jasper is teaching in special education school. He was assigned by his principal to come up with an exercise program to develop the overall fitness of the students with special needs. What is/are the primary consideration/s that he need/s to consider?
   a. Principle of progression b. Principle of individuality
   c. Principle of specificity d. All of the above

8. In executing stretching exercises, the degree of stretch should be…
   a. to about 80 percent of capacity.
   b. to mild tension at the end of the range of motion.
   c. applied until the muscle(s) start shaking.
   d. progressively increased until the desired stretch is attained.

9. The recommended number of repetitions for strength exercises for better health is between_____.
   a. 1 and 6 reps.
   b. 4 and 10 reps.
   c. 8 and 12 reps.
   d. 10 and 25 reps.

10. Choose the letter that does not belong to the group.
    a. Deep full squat, hero, and donkey kick
    b. Knee pull down, knee to chest, swan stretch
    c. Abdominal crunch, one leg stretch, quad stretch
    d. Neck circling, hurdle stretch, standing toe touch

Performance Task 4

Simple Exercise Workout

Group Leader :_____ Section: _____
Members: _____ Date: _____
_____ Group no. _____
_____

Instruction: The class will be divided into 4-5 groups or as prescribed by the teacher. The student/s will perform in a virtual platform a 10minute-exercise routine consisting: (1) 3min-active/dynamic warm-up; (2) 5-main workout; and (3) 2min-Cool-down. In warm-up, locomotor movements are needed, while on the cool-down proper stretching is advised.

Enough time will be given for the students to practice as a group and may consider using social media to collaborate with other members. Take note to consider the capabilities and health concerns of each member. The said routine will be submitted through online or in other form like usb or compact disk on the given date. For students who will not be able to work collaboratively, individual performance is advised.

The student/s will be assessed based on the following rubrics:

Criteria	Excellent	Good	Average	Fair	Needs Improvement
	5	**4**	**3**	**2**	**1**
Phases of Exercise Program	Routine has complete phases of exercise program (warm-up/workout/cool down) and has executed it properly all throughout the performance	Routine has complete phases of exercise program (warm-up/workout/cool down) and has executed it properly most of the time of the performance	Routine has complete phases of exercise program (warm-up/workout/cool down) but executed it with few errors during the performance	Routine has incomplete phases of exercise program and executed it with several errors during the performance	Routine did not follow the three phases of exercise program and did not execute it properly
	5	**4**	**3**	**2**	**1**
Energy and Effort	Consistently performed the movements with energy and effort at all times	Movements were performed with energy and effort most of the time.	Movements were sometimes performed with energy and effort. Frequently makes excuses.	Movements were rarely performed with energy and effort. Always makes excuses	Movements were performed without energy and effort.
	5	**4**	**3**	**2**	**1**
Application of Principles of Training Exercise	The routine is performed correctly and considered the complete application of the principles of exercise such as: • Progression -Low to high impact. Simple to complex steps • Individuality -consider individual skills • Rest and recovery- proper routine design that provides recovery	The routine is performed correctly but considered the application of only two principles of exercise	The routine is performed correctly but considered the application of only one principle of exercise	The routine is performed incorrectly and considered the application of only two principles of exercise	The routine is performed incorrectly and did not consider the application of the principles of exercise
		Total Score:		**Rating:**	

Rating:

13-15	*Excellent*
10-12	*Good*
7- 9	*Average*
4- 6	*Fair*
3 and below	*Needs Improvement*

Conforme:

_____ Signature over printed name/ Date
Evaluator

_____ Signature over printed name/ Date
Student

Comments/Suggestions:_____

_____
_____
_____
_____
_____
_____

Performance Task 5

Cardiorespiratory Exercise Prescription

Name:_____ Section:_____ Date:_____ Group no._____

Direction: Compute your Cardio Respiratory Training Zone and come up with your own cardio exercise prescription based on the result applying the MFIT principle.

Intensity

1. Estimate your own maximal heart rate (MHR) MHR = 207 − (.70 × age)
   MHR = 207 − (.70 ×_____) = _____ bpm
2. Resting heart rate (RHR) = _____ bpm
3. Heart rate reserve (HRR) = MHR – RHR
   HRR = _____bpm - _____ bpm = _____ beats
4. Training intensity (TI) = HRR × % TI + RHR 30% TI = × .30 + bpm 40% TI = × .40 + bpm 60% TI = × .60 + bpm 90% TI = × .90 + bpm
5. Cardiorespiratory training zone:

   30% TI = (_____ X .30) + _____ = _____ bpm
   40% TI = (_____ X .40) + _____ = _____ bpm
   50% TI = (_____ X .50) + _____ = _____ bpm
   60% TI = (_____ X .60) + _____ = _____ bpm
   70% TI = (_____ X .70) + _____ = _____ bpm
   90% TI = (_____ X .90) + _____ = _____ bpm

Light-intensity cardiorespiratory training zone (30% to 40% TI): _____ to _____ bpm
Moderate-intensity cardiorespiratory training zone (40% to 60% TI): _____ to _____ bpm
Vigorous-intensity cardiorespiratory training zone (60% to 90% TI): _____ to _____ bpm

Mode/Type: List any activity or combination of aerobic activities that you will use in your cardiorespiratory training program:

_____
_____
_____
_____

C. The Exercise Program (Frequency, Intensity and Time)

The following is your weekly program for development of cardiorespiratory endurance.

- If you are in the **poor or fair** cardiorespiratory fitness category, start with a light intensity training zone (30-40%)
- If you are in the **average category**, you may start with a moderate intensity training zone (40-60%)
- If you are already active and in the **good or excellent category**, you may start with a moderate intensity training zone (40-60%) or you may opt to start with high intensity training zone (60%-90%)

For the purpose of the accomplishment of the task, you **(students)** are required to implement at least 1-2 weeks of the exercise program. But it is encouraged to continue with the 12-week program,.

Time/Duration: Indicate the length of your exercise sessions:_____ minutes

_____
_____
_____

Frequency: Indicate the days you will exercise:

_____

_____

D. Reflective Questions

After accomplishing the task (1-2 weeks of cardio program), kindly share briefly your experiences.

_____
_____
_____
_____
_____
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_____

Did you accomplish the task smoothly or did you encounter any unexpected problems along the way? How did you solve the problem. Explain your answer.

_____
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_____

Any realization or lesson that you learned from the task? Kindly share.

_____
_____
_____
_____
_____
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_____

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Chapter 4

PHYSICAL FITNESS AND SELF-TESTING ACTIVITIES

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. identify the different health-related fitness tests;
2. explain the importance of physical fitness assessment;
3. demonstrate proper skills and confidence on administering physical fitness test;
4. establish baseline data (pre-test score); and
5. display acceptance and understanding on their strength and weaknesses.

Course Outline:

- Overview
- Discussion
    - Physical Fitness
    - Importance of Physical Fitness Assessment
    - Things to Consider in Self-Testing Activity
    - Measuring Health-Related Physical Fitness

- Suggested Media at Home
- Activities and Assessments
    - Reflective Questions
    - Knowledge Check
    - Performance Task
- References

II. OVERVIEW

Though most of the people usually aim to be physically fit, to be healthy it is not the only basis to have a better quality of life. Thus it should be looked into a bigger picture of developing the different dimension of wellness to have a successful and happy life. In this chapter, review on the importance of wellness in one's life will be tackeled as well as the eight dimensions of wellness. In addition the importance and the proper way of establishing base line data through safe self-administered fitness testing will be the highlight of this section. It will provide baseline data as basis for developing future fitness program plans.

III. COURSE MATERIALS: DISCUSSION

A. Physical Fitness

As the fitness concept grew at the end of the last century, it became clear that several specific components contribute to an individual's overall level of fitness. Physical fitness is classified into health-related, skill-related, and physiological fitness.

Each of the components can affect our daily activities. For example, cardiovascular fitness is needed when walking a certain distance, climbing stairs, doing yard work (a.k.a. gardening, for those of us who have a garden), cleaning your room (or apartment or house, depending on where you live). In addition, leisure and social activities, such as a weekend bike, or hike, or sports, or a night out dancing, can become more enjoyable with good cardiorespiratory fitness. In addition, there is a reduced risk of hypokinetic disease. Hypokinetic diseases are those diseases that stem from lack of physical activity. Such diseases include hypertension, heart diseases, chronic low back pain, and obesity.

Health-related Fitness is related to the ability of the person to perform activities of daily living without undue fatigue and has a low risk of premature sedentary lifestyle diseases. There are five health-related fitness components: cardiorespiratory fitness (aerobic), muscular strength, muscular endurance, body composition and flexibility.

1.1. Body Composition- relates to the relative amounts of muscle, fat, bone, and other vital parts of the body. There are a variety of other methods assessing body composition, including formulas to calculate body mass index (BMI). Body composition is the only nonperformance component of health-related physical fitness.
1.2. Flexibility- relates to the range of motion available at the joint. Flexibility is specific to each joint of the body. Therefore, there is no general test of flexibility unlike for the cardiorespiratory fitness where there is a standard test.
1.3. Cardiorespiratory Fitness- relates to the ability of the circulatory and respiratory systems to supply oxygen during sustained physical activity. Cardiorespiratory fitness is also known as *cardiovascular fitness, cardiovascular endurance, or aerobic fitness.*
1.4. Muscular Strength- relates to the ability of the muscle to exert force. Muscular strength is also specific in nature.
1.5. Muscular Endurance- relates to the muscle's ability to continue to performing without fatigue. Like flexibility, muscular endurance is specific to each major muscle group of the body.

B. Importance of Physical Fitness Assessment

According to Institute of Medicine (2012) there is an association between low health related fitness to cardiovascular diseases and diabetes. Likewise, the World Health Organization (2000) also declared obesity as a global epidemic. Therefore, the importance of assessing and monitoring your health related fitness is very essential to each and every one's health. Moreover, health-related fitness testing:

- educate participants about their current health-related status;
- provide data-base that are helpful on developing exercise prescriptions/programs for each individual;
- monitor changes in health-related fitness status of participants and populations with time;
- motivate participants to attain reasonable fitness goals; and
- modify physical education curricula if necessary.

C. Things to Consider in Self-Testing Activity

Be mindful of safety practices and precautions in performing the fitness tests such as the following:

- ✓ Consider health concerns or exemptions in some of the tests. Guidance from the teacher is essential or approval from the doctor if needed.
- ✓ Make sure that the Physical Activity Readiness Questionnaire and You (PARQ&You) was properly accomplished, signed by the student and parents/guardian, and submitted to the teacher.
- ✓ Be sure to warm-up before and cool down after the tests.
- ✓ Ensure a safe and conducive fitness testing area.
- ✓ Wear appropriate clothing and footwear to avoid possible injuries.
- ✓ Maintain proper hydration at all times.
- ✓ Strictly follow the procedure or mechanics of each test to ensure reliable result.

D. Measuring Health-Related Physical Fitness

1. Component: Body Composition

Test: Body Mass Index (BMI)

The body mass index (BMI) is defined as the ratio of body weight (measured in kilograms) and the square of the height (measured in meters). The body mass index is determined as follows"

$$BMI = \frac{\text{Bodyweight (kg)}}{\text{Height (m)}^2}$$

The following steps can be used with this formula (Adams and Adams, 2009)

1. Weight in pounds divided to 2.2 = Weight in Kilogram (kg)
2. Height in inches X 0.0254= Height in Meters (m)
3. Height in meters X height in meters= height in meters squared ($m^{2)}$
4. Step 1 divided by step 3= BMI

Body Composition Rating Scale

BMI	Disease Risk	Classification
< 18.5	Increased	Underweight
18.5-21.99	Low	Acceptable
22.0-24.99	Very Low	Acceptable
25.0-29.99	Increased	Overweight
30.0-34.99	High	Obesity I
35.0-39.99	Very High	Obesity II
≥ 40	Extremely High	Obesity III

Source: Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness,Fourteenth Edition. Cengage Learning, USA

Anthropometric Measurement

A series of quantitative measurements of the muscle, bone, and adipose tissue used to measure the composition of the body. The main components of anthropometry are height, weight, body circumferences (waist, hip, and limbs) body mass index (BMI), and skinfold thickness. There are significant measurements as it signify for probable weight problem (underweight/obesity) and can be utilized as a reference point for physical fitness program.

2. Component: Flexibility

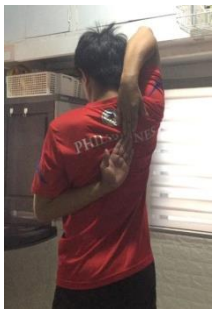
2.1 Test: Modified Sit-and-Reach

Purpose: To measure flexibility of hamstrings

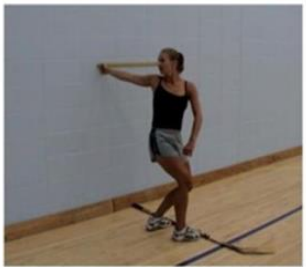
Equipment: Sit-and-reach flexibility tester, or place a yardstick on top mark at 9 inches on a box approximately 12 inches high.



Modified Sit & Reach



Zipper Test Full



Body Trunk Rotation

e.com/watch?v=sFi_RgN1kGs
https://www.pinterest.ph/pin/253327547759287251/
https://slideplayer.com/slide/4173224/

Procedures:

- Remove shoes and sit on the floor. Place the sole of the foot of the extended leg flat against a box or bench. Bend opposite knee and place the head, back, and hips against a wall with a 90-degree angle at the hips.
- Place one hand over the other and slowly reach forward as far as you can with arms fully extended. Keep head and back in contact with the wall. A partner will slide the measuring stick on the bench until it touches the fingertips.
- With the measuring stick fixed in the new position, reach forward as far as possible, three times, holding the position on the third reach for at least 2 seconds while the partner records the distance on the ruler. Keep the knee of the extended leg straight (see illustration).
- Repeat the test a second time and average the scores of the two trials.

2.2. Test: Zipper Test
Purpose: To measure shoulder flexibility
Equipment: Ruler, Measuring Tape

Procedures:

- Raise your arm, bend your elbow, and reach down across your back as far as possible.
- At the same time, extend your left arm down and behind your back, bend your elbow up across your back, and try to cross your fingers over those of your right hand as shown in the accompanying illustration.
- Measure the distance to the nearest half-inch. If your fingers overlap, score as a plus. If they fail to meet, score as a minus. Score as a zero if your fingertips just touch.
- Repeat with your arms crossed in the opposite direction (left arm up). Most people will find that they are more flexible on one side than the other.

2.3. Test: Trunk Rotation (Optional)
Purpose: To measure trunk flexibility
Equipment: Ruler, Measuring Tape

Procedures:

- Tape two yardsticks to the wall at shoulder height, one right side up and the other upside down.
- Stand with your left shoulder an arm's length (fist closed) from the wall. Toes should be on the line, which is perpendicular to the wall and even with the 15-inch mark on the yardstick.
- Drop the left arm and raise the right arm to the side, palm down, and fist closed.
- Without moving your feet, rotate the trunk to the right as far as possible, reaching along the yardstick, and hold it 2 seconds. Do not move the feet or bend the trunk. Your knees may bend slightly.
- A partner will read the distance reached to the nearest half-inch. Record your score. Repeat two times and average your two scores.
- Next, perform the test facing the opposite direction. Rotate to the left. For this test, you will use the second yardstick (upside down) so that, the greater the rotation, the higher the score. If you have only one yardstick, turn it right side up for the first test and upside down for the second test.

Flexibility Rating Scale

	Modified Sit and Reach		Zipper Test				Trunk Rotation	
Fitness Category	**Men**	**Women**	**Men**		**Women**		**Men**	**Women**
			Right Up	Left Up	Right Up	Left Up		
High Performance zone	16+	17+	5+	4+	6+	5+	20+	20.5+
Good Fitness Zone	13-15	14-16	1-4	1-3	2-5	2-4	16-19.5	17-20
Marginal Zone	10-12	11-13	0	0	1	1	13.5-15.5	14.5-16.5
Low Fitness Zone	≤ 9	≤ 10	<0	<0	<1	<1	<13.5	<14.5

Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness and Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York

3. Component: Cardiorespiratory Fitness
Test: 3 Minute-Step Test
Equipment: between 12-16 1/4 inches step box or bench, stopwatch, metronome or cadence tape

Procedures:

- Allow the subject to practice the stepping to the metronome cadence, which is set at 96 beats per minute for men and 88 beats per minutes for women.
  88 beats per minute timer https://www.youtube.com/watch?v=Ln9rNK1eeBg
  96 beats per minute timer https://www.youtube.com/watch?v=SZOvr7mNUtM



- The student steps up and down on the platform at the given rate for a total of 3 minutes.
- The student immediately stops on completion of the test and remain on a standing position. Starting within 5 to 20 seconds recovery, take the heart rate for 15 seconds. Convert recovery heat rate to beats per minute. (multiply 15 seconds heart rate by:
  Scoring: Maximal oxygen uptake (VO_{2max}) in mL/kg/min is estimated according to the following equation:

  Men:
  VO_{2max} = 111.33 – (0.42 x recovery heart rate in bpm)
  Women:
  VO_{2max} = 65.81 – (0.1847 x recovery heart rate in bpm)
  Example: The recovery 15-second heart rate for male following the 3-minute step test is found to be 39 beats. His VO_{2max} is estimated as follows:
  15-second heart rate = 39beats
  Minute heart rate = 39 x 4 = 156bpm
  VO_{2max} = 111.33 – (0.42 x 156) = 45.81 mL/kg/min

Fitness Classification
(Based on VO2ma in mL/kg/min)

Male					
Age	Excellent	Good	Average	Fair	Poor
<29	>53	44-52.9	34-43.9	25-33.9	<24.9
30-39	>50	42-49.9	31-41.9	23-30.9	<22.9
40-49	>45	39-44.9	27-38.9	20-26.9	<19.9
50-59	>43	38-42.9	25-37.9	18-24.9	<17.9
60-69	>41	36-40.9	23-35.9	16-22.9	<15.9
≥70	≥38	33-37.9	21-32.9	13-20.9	≤12.9
Women					
Age	Excellent	Good	Average	Fair	Poor
<29	>49	39-48.9	31-38.9	24-30.9	<23.9
30-39	>45	37-44.9	28-36.9	20-27.9	<19.9
40-49	>42	35-41.9	25-34.9	17-24.9	<16.9
50-59	>40	34-39.9	22-33.9	15-21.9	<14.9
60-69	>37	33-36.9	21-32.9	13-20.9	<12.9
≥70	≥35	31-34.9	20-30.9	12-19.9	≤11.9

Source: Hoeger, Werner W.K., Hoeger, Sharon A., Hoeger, Cherie I., and Fawson, Amber L., (2018) Principles and Labs for Fitness & Wellness,Fourteenth Edition. Cengage Learning, USA

4. Component: Muscular Strength and Endurance

Muscular strength refers to the ability of a muscle to exert maximum force against resistance while muscular endurance is the ability of a muscle to exert submaximal force repeatedly over time.

4.1 Test: Flexed Arm Support (Static)

Purpose: To measure muscular strength and endurance of arms

Equipment: Mat or carpet



Female



Male

Procedures:

Women: Support the body in a push-up position from the knees. The hands should be outside the shoulders, the back and legs straight. Lower the body until the upper arm is parallel to the floor or elbow is flexed at 90 degrees.

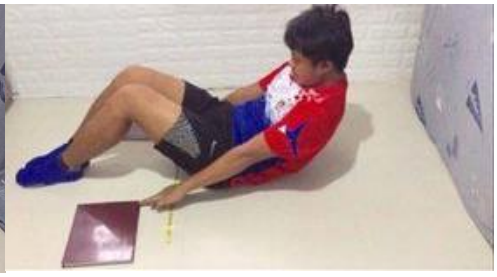
Men: Use the same procedure as for women except support the push-up position from the toes instead of the knee. (same position as for 90 degree push-up.) Hold the 90-degrees position as long as possible, up to the 35 seconds.

Static Endurance Rating Scale (Flexed Arm Support)

Rating	Score in Seconds
High Performance zone	30+
Good Fitness Zone	20-29
Marginal Zone	10-19
Low Fitness Zone	<10

Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness and Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York

4.2 Test: Curl Up (Dynamic)
Purpose: To measure the muscular strength and endurance of abdominal muscle.
Equipment: Mat or carpet

Procedures:

- Sit on a mat or carpet with your legs bent more than 90 degrees so your feet remain flat on the floor (about half way between 90 degrees and straight).
- Make two tape marks 4 ½ inches apart or lay a 4 ½-inch strip of paper so that the finger tips touch one tape mark (or other side of the paper).
- Keeping your heels in contact with the floor, curl the head and shoulders forward until your fingers reach 4 ½ inches (other side of the strip)
- Lower slowly to beginning position. Repeat one curl-up every 3 seconds. Continue until every 3 seconds.
- Two partners may be helpful. One stands on the cardboard strip (to prevent movement) if one is used. The second assure that the head returns to the floor after each repetition.

Dynamic Muscular Endurance Rating Scale

	Age: 17-26		Age:27-39	
Rating	**Men**	**Women**	**Men**	**Women**
High Performance zone	35+	25+	34+	24+
Good Fitness Zone	24-34	18-24	23-33	17-23
Marginal Zone	15-23	10-17	14-22	9-16
Low Fitness Zone	<15	<10	<14	<9

Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York

Suggested Media at Home

- **Fitnessgram Back Saver Sit & Reach Test (using ordinary box and ruler)**
  https://www.youtube.com/watch?v=sFi_RgN1kGs
- **Fitnessgram Back Saver Sit & Reach Test**
  https://www.youtube.com/watch?v=sFi_RgN1kGs
- **The 5 Components of Health Related Fitness**
  https://www.youtube.com/watch?v=eCJovVdQgVw
- **The complete FITNESS TEST list**
  https://www.topendsports.com/testing/tests/

IV. ACTIVITIES/ASSESSMENTS

In this section, a variety of suggested activities and assessments were provided. It provides activities to assess the appreciation (reflective questions) and understanding (knowledge check) of the students on the topics provided. It also suggests options for practical application (a performance task) by providing a performance activity that includes the rubrics or criteria for rating. Remember that as a teacher, you have the academic freedom to choose what activities you would like to assign to your students, modify some of the tasks, or provide additional or other activities that you think will be more suitable for our dear students, recognizing the different capabilities and situations they have right now.

Activity Worksheet 7. Reflective Questions

Name:_____ Section:_____ Date:_____ Group no._____

Direction: Read the questions below carefully and write your answer on the space provided.

On a scale of 1-10 kindly rate your onw wellness in terms of eight dimensions? Explain your answer. (10points)

What do you think is the significance of physical fitness testing in your own personal life, most especially during this time of pandemic? (10points)

Activity Worksheet 8. Knowledge Check

Name:_____ Section:_____ Date:_____ Group no._____

I. Direction. Encircle the letter of the best answer that corresponds to the following questions or statements.

1. It a series of quantitative measurements of the muscle, bone, and adipose tissue used to measure the composition of the body
   a. Fitness Test b. Body Composition c. Assessment d. Anthropometry

2. In order for an individual to reduce risk of serious health problems, you have to achieve and maintain ideal Body Mass Index (BMI) which ranges _____ to _____.
   a. 30.0 to 34.99 b. below 18.5 c. 18.5 to 24.99 d. 25.0 to 29.99

3. Jennifer is a high school student who wants to learn how to swim. What health related components are primarily needed to develop to achieve her goal?
   a. Body Composition and Flexibility b. Coordination and Balance
   c. Power and Speed d. All of the above

4. In performing the different health related fitness tests, what is the top priority before proceeding to its actual performance?
   a. Warm-up b. Cool-down c. Proper Clothing d. Safety Precautions

5. In order to be successful in a triathlon competition, you must have high level of:
   a. Cardiorespiratory Fitness b. Health-related Fitness
   c. Skill-related Fitness d. b and c
   e. All of the above

6. Part of the human body that flexes the forearm at the elbow.
   a.deltoid b. trapezius c. biceps brachii d. deltoid and biceps

7. Skeletal mucle that draws thigh backward and flexes the knee.
   a.deltoid b. trapezius c. biceps brachii d. deltoid and biceps

The following physical fitness tests are done to measure:

8. Flexibility of the lower back and hamstring muscles on the back of the thighs.
   a. Push-up b. Coin Catch c. Curl-up d. Sit and Reach

9. Abdominal strength and endurance
   a. Standing long jump b. Crunches c. Coin Catch d. Shuttle run

10. Anthropometric measurements
    a. Height b. Waist Hip Ratio c. Arm Span d. All of the above

Performance Task 6

Health-Related Fitness Tests (Pre-Test)

Name:_____ Section:_____ Date:_____ Group no._____

Procedures

1. Read the direction of each of the health-related fitness described in the guide manual given.
2. Take tests that are possible to be conducted given the time, equipment, and the space available.
3. Be sure to warm-up before and cool down after the tests.
4. It is alright to practice the tests before trying them. However, you should decide ahead of time which trial you will use to test your health-related fitness.
5. After completing the tests, write your scores in the appropriate places in the results section.
6. Determine your rating for each of the tests from rating charts in the source materials by shading the circle that corresponds to your score.

Body Composition

Body Mass Index	**Health Risk**	**Increased**	**Low**	**Very Low**	**Increased**	**High**	**Very High**	**Extremely High**
	Classification	**Underweight**	**Acceptable**	**Acceptable**	**Overweight**	**Obesity I**	**Obesity II**	**Obesity III**
	Score _____ kg/m2	○	○	○	○	○	○	○

		Fitness Category				
CARDIO RESPIRATORY FITNESS		**Excellent**	**Good**	**Average**	**Fair**	**Needs Improvement**
Step Test						
15 sec recovery heart rate	_____ beats					
VO_{2max}	_____ mL/kg/min	○	○	○	○	○
FLEXIBILITY	**Score**	**High**	**Good**	**Marginal**	**Poor**	
Modified Sit and Reach						
Left	_____ inches	○	○	○	○	
Right	_____ inches	○	○	○	○	
Zipper Test (Finger Touch)						
Left	_____ inches	○	○	○	○	
Right	_____ inches	○	○	○	○	
Trunk Rotation (Optional)						
Left	_____ inches	○	○	○	○	
Right	_____ inches	○	○	○	○	
MUSCULAR STRENGTH & ENDURANCE		**High**	**Good**	**Marginal**	**Poor**	
Flexed-arm support *(static): Women in knee position and men in full support position*	_____ seconds	○	○	○	○	
Curl-up/Crunches *Abdominal Muscles*	_____ repetitions	○	○	○	○	

This is to acknowledge and certify that the data provided herein are true and correct.

Signature over printed name/ Date
Student

Signature over printed name/ Date
Witness/Guardian

The Four Fitness Zones

High-Performance Zone Reaching this zone provides additional health benefits and is important to high-level performance. However, high performance scores are hard for some people to achieve, and for many people high-level performance is not important. So reaching this zone may be more important to some than others.

Good Fitness Zone If you reach the good fitness zone, you have enough of a specific fitness component to help reduce health risk. However, staying active (in addition to reaching this fitness zone) is important.

Marginal Zone Marginal scores indicate that some improvement is in order, but you are nearing minimal health standards set by experts.

Low-Fit Zone If you score low in fitness, you are probably less fit than you should be for your own good health and wellness.

In two or three paragraphs, kindly describe your feelings about your current health-related fitness status. Comment on the areas in which you did well or did not well and the meaning of these findings. Moreover, briefly discuss your plan on how you will improve your current fitness level by the end of this semester.

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Chapter 5

FUNDAMENTAL MOVEMENT PATTERNS

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. Identify the different types of locomotor and non-locomotor movements;
2. Perform the types of locomotor and non-locomotor movements scientifically; and
3. Design a specific activity using the given movement.

Course Outline:

- Overview
- Discussion
    - Non-Locomotor Movements
    - Locomotor Movements
- Summary
- Suggested Media at Home
- Activities and Assessments
    - Reflective Questions
    - Knowledge Check
    - Performance Task
- References

II. OVERVIEW

Fundamental movement skills are a specific set of skills that involve different body parts such as feet, legs, trunk, head, arms and hands. These skills are the “structure blocks” for more compound and specific skills that kids will need throughout their lives to competently participate in different games, sports and recreational activities.

III. COURSE MATERIALS: DISCUSSION

Fundamental Movements

A **non-locomotor** movement is a fundamental body movement that does not require moving in any direction. These stability skills include movement of limbs and parts of the body or whole-body parts. These movements done in a stationary position.A **locomotor** movement is another fundamental body movement that requires travel through space or carry weight from one location to another.

Below are the fundamental movements that you can apply for your daily activities.

A. Non-locomotor movements

1. Bracing the core-, this movement is very important to stabilize your spine and it’s a form of exercise to gradually improve your core group muscles.

Mechanics

1. Perform a supine lying position
2. Put your one hand to your abdomen and the other one in your chest.



3. Brace or contract your abdominal muscles slowly
4. Keepyour lower back in inward position.
5. Elevate your pelvic floor muscles
6. Maintain your pelvic floor and lower abdominal muscles contracted together
7. Maintain the contraction of your abdominal muscles and breath normally
8. Inhale and exhale for three to four seconds and repeat ten times.

2. Dead bug – is a simple exercise that strengthens and stabilizes the core muscles. It helps to build a strong core to protect your spine.



Mechanics

1. Lying on your back
2. Knees bent at 90-degrees and feet are flat on the floor
3. Alongside your body, rest your arms.
4. Shoulders and your lower back to fall heavy to the floor.
5. Draw your shoulders down away from your ears. From a starting position, raise your hands so that your elbows are above your shoulder while your fists are facing in towards each other.
6. Lift your legs so your knees are directly over your hips.
7. On exhale, slowly lower your left leg and right arm until they are just above the floor.
8. On inhale, put them back to original position.
9. Repeat on the other side.

3. Rolling- refers to the exercise by turning your back over your stomach from the lower body.



Mechanics

1. Lie down with your back lying flat on the floor and your arms extended upward while your legs are stretch downward.
2. Maintain such position as if you are paralyzed from the waist up.
3. To start rolling, put your left hand across and down your body and try to reach your opposite pocket.

Note: Rolling on the right side of the body would mean reaching across with your left arm. Rolling on the left would require reaching the right arm.

4. Continue to reach with your arm, head, and shoulders until you achieve lift and are able to flip yourself onto your stomach without any assistance from the lower body.

4. Bird dog series- this aims to develop your core muscle using your body weight.



Mechanics

1. Place your knees under your hips and your hands under your shoulder touching the ground just like a table position. The abdominals should be brace in this position.
2. Raise your right arm and left leg parallel to the ground. Lengthen your back of your neck and tuck your chin.
3. Hold the position for a few seconds and back to starting position.
4. Raise your left arm and right leg parallel to the ground. Lengthen your back of your neck and tuck your chin.
5. Hold this position for a few seconds and return to starting position. This is one round.
6. Do this 2 -3 Sets 10 -12 repetitions.

5. Press up –this is scapular protraction and retraction movements.



Mechanics

1. Place your hands shoulder-width apart with arms straight and so your shoulders, elbows and wrists line up.
2. Your fingers should be slightly spread and pointing forward, with your hands forming right angles with your forearms.
3. Initiate the move by bending your elbows to lower your chest towards the ground.
4. Your hips should stay in line with your shoulders for the entire set.
5. Keep your feet hip-width apart. Placing them farther apart makes it easier and bringing them together makes it harder.
6. The closer you can get your chest to the floor the better because this makes the working muscles move through a full range of motion.

6. Cobra Stretch- is one of the most important stretching for abdominal muscles especially if you spend all day long at you desk. This type of movement helps you to toned your glutes muscles and makes you muscles stronger especially at your spine muscles, and it provides the lenghtening of anterior neck muscles, shoulder, chest and abdominals.



Mechanics

1. Assume a prone position while your hands palm down to the floor beneath on your shoulder.
2. Lift your chest off the ground by straightening your arms.
3. Take a good look upward and and keep your abdominal muscles engaged.

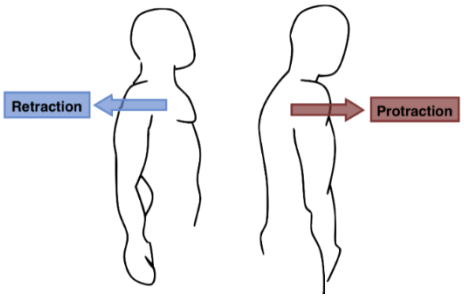
7. Squat- also known as an air squat or the most common squat. It aims to develop your lower body muscles such as gluteus muscles (buttocks), quadriceps (front of thigh), harmstring (back of thigh), adductor (groin), and calves.



Mechanics

1. Start with your feet slightly wider than hip-width apart.
2. Keep your chest up, engage your abdominals, and shift your weight onto your heels as you push your hips back into a sitting position.
3. Lower your hips until your thighs are parallel or almost parallel to the floor.
4. You should feel the squat in your thighs and glutes.
5. Pause with your knees over, but not beyond, your toes.
6. Exhale and push back up to the starting position.

8. Scapular retraction and protraction- type movement that helps to restore the proper alignment and strength of the muscles to support the shoulder.



Mechanics

Protraction is a forward movement of your mandible jaw or shoulder, while the retraction is the opposite of protraction by returning your mandible jaw or shoulder backward

Scapular retraction

1. Stand up straight with feel-shoulder width apart
2. Pull your shoulder blades (scapulae) towards the spine or toward each other

Scapular protraction
1. It is the opposite of scapular retraction, wherein your shoulder blades moves away from each other and your arms sags forward.

9. **Stationary lunge**- exercise that helps you to strengthen your quads, glutes, and hamstring muscles.



Mechanics:

1. Start with feet shoulder width apart. Put your right foot in front and must be flat on the ground, and your left foot should be up on its toes. Both legs are in 90 degrees position.
2. Bend your knees and lunge, stopping when your right thigh is parallel to the ground.
3. Push up through your right foot to return to the starting position. Repeat for desired number of reps, and then switch legs.

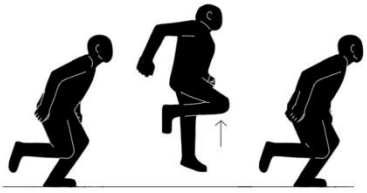
B. Locomotor Movements



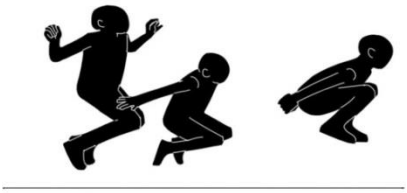
1. Walk– series of steps to any direction



2. Run–move fast by using the feet, with one foot off the ground at any given time



3. Hop– spring on one foot and landing on the same foot.



4. Jump–spring on one or both feet and landing on both feet.



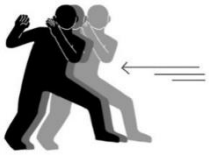
5. Leap– spring on one foot and landing on the other foot.



6. Lateral/Shuffle- step siderward& close with the trail foot, step & close... (same foot always leads).



7. Skip– stepping from one foot to another with hop or bounce.



8. **Slide**–lead foot glides forward or forward while the other foot follows.

Summary

- Two types of basic movements: locomotor and non-locomotor.
- Fundamental movement skills are a specific set of skills that involve different body parts such as feet, legs, trunk, head, arms and hands.
- Basic movements are the foundation of human movements, which help you to move efficiently and effectively.
- Application of locomotor and non- locomotor in your daily activities

Suggested Media at Home

- Visit this site and discuss the importance of fundamental movements to children https://www.youtube.com/watch?v=swVMaPv5EJQ

IV. ACTIVITIES/ASSESSMENTS

In this section, a variety of suggested activities and assessments were provided. It provides activities to assess the appreciation (reflective questions) and understanding (knowledge check) of the students on the topics provided. It also suggests options for practical application (a performance task) by providing a performance activity that includes the rubrics or criteria for rating. Remember that as a teacher, you have the academic freedom to choose what activities you would like to assign to your students, modify some of the tasks, or provide additional or other activities that you think will be more suitable for our dear students, recognizing the different capabilities and situations they have right now.

Activity Worksheet 9. Reflective Questions

Name:_____ Section:_____ Date:_____ Group no._____

Before this lesson, did you have background knowledge about the topic? If you did, what new things did you learn? If this is all new to you, what did you gain from the lesson? (10points)

_____
_____
_____
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Activity Worksheet 10. Knowledge Check

I. MATCHING TYPE: Match column A to column B. Write the letter of the correct answer on the space provided before the number. Answer may be repeated more than once.

Write A if the statement belongs to the Locomotor movements
Write B if the statement belong to the Non Locomotor movements.

_____1. Hop
_____2. Run
_____3. Walk
_____4. Skip
_____5. Leap
_____6. Plank
_____7. Crawl
_____8. Squat
_____9. Lateral
_____10. Bracing
_____11. Sliding
_____12. Bird dog
_____13. Press Up
_____14. Dead bug
_____15. Back pedal
_____16. Twist
_____17. Rolling
_____18. Plank
_____19. Jump
_____20. Gallop

A. Locomotor
B. Non Locomotor

Performance Task 7

Non-Locomotor Movements *(Midterm Performance)*

Group Leader :_____ Section: _____
Members: _____ Date: _____
_____ Group no. _____
_____

Instruction: The class will be divided into 4-5 groups or as prescribed by the teacher. Representative from each group will pick a number (1-3) from a bowl in which designated non-locomotor movements are grouped. The student/s then will perform in a virtual platform the assigned non-locomotor movements. Enough time will be given for the students to practice as a group and may consider using social media to collaborate with other members. Take note to consider the capabilities and health concerns of each member. The said routine will be submitted through online or in other form like flashdrive or compact disk on the given date. **For students who will not be able to work collaboratively, individual performance is advised.**

Non-Locomotor Movements		
1. Bracing the core Squat series Lunges	**2.** Rolling Plank series Bird dog series	**3.** Cobra Stretch Dead bug series Press up, scapular protraction & retraction

The student/s will be assessed based on the following rubrics:

Criteria	Excellent	Good	Average	Fair	Needs Improvement
	5	**4**	**3**	**2**	**1**
Phases of Exercise Program	Routine has complete phases of exercise program (warm-up/workout/cool down) and has executed it properly all throughout the performance	Routine has complete phases of exercise program (warm-up/workout/cool down) and has executed it properly most of the time of the performance	Routine has complete phases of exercise program (warm-up/workout/cool down) but executed it with few errors during the performance	Routine has incomplete phases of exercise program and executed it with several errors during the performance	Routine did not follow the three phases of exercise program and did not execute it properly
	5	**4**	**3**	**2**	**1**
Movement Competency	Consistently and independently demonstrate superior ability in applying the required concepts and skills	Usually and independently demonstrate superior ability in applying the required concepts and skills	Sometimes demonstrates the ability to apply the required concepts and skills with assistance	Seldom demonstrates the ability to apply the required concepts and skills	Did not demonstrate any ability to apply the required concepts and skills
	5	**4**	**3**	**2**	**1**
Energy and Effort	Consistently performed the movements with energy and effort at all times	Movements were performed with energy and effort most of the time.	Movements were sometimes performed with energy and effort. Frequently makes excuses.	Movements were rarely performed with energy and effort. Always makes excuses	Movements were performed without energy and effort.
		Total Score:		**Rating:**	

Rating:	
13-15	*Excellent*
10-12	*Good*
7- 9	*Average*
4- 6	*Fair*
3 and below	*Needs Improvement*

Conforme:

_____
Signature over printed name/ Date

_____

Evaluator

Signature over printed name/ Date
Student

Comments/Suggestions:_____

_____

_____

_____

_____

_____

_____

_____

Performance Task 8

Locomotor Movements

Group Leader :_____ Section: _____
Members: _____ Date: _____
_____ Group no. _____
_____

Instruction: The class will be divided into 4-5 groups or as prescribed by the teacher. The students are expected to create dance (5-8minutes) routine using basic locomotor movements. Enough time will be given for the students to practice as a group and may consider using social media to collaborate with other members. Take note to consider the capabilities and health concerns of each member. The said routine will be submitted through online or in other form like flashdrive or compact disk on the given date. **For students who will not be able to work collaboratively, individual performance is advised.** The student/s will be assessed based on the following rubrics:

RUBRICS					
LEVEL OF PERFORMANCE					
CRITERIA	**5 EXCELLENT**	**4 VERY SATISFACTORY**	**3 SATISFACTORY**	**2 NEEDS IMPROVEMENT**	**1 POOR**
Proper Mechanics	Precisely followed the mechanics of the skill	Followed the mechanics of the skill but some errors were evident	Showed average mastery of the mechanics	Showed little mastery of the mechanics	Poorly executed the skills
Execution	Executed the skill with exceptional ability	Executed the skill correctly but some errors were evident	Showed an average level of accuracy in executing the skill	Showed little level of accuracy in executing the skill	Poorly executed the skill
Good form	Executed the high level of balance, control and ease of movement	Executed the skill with high level of balance, control and ease of movements	Executed the skill with and average level of balance, control and ease of movements	Executed the skill with little level of balance, control and ease movements	Executed the skill lousily
Enthusiasm	Showed a very high level of interest in executing the skill	Showed a high level of interest in executing the skill	Quite eager in executing the skill	Showed little interest in executing the skill	Did not show interest in executing the skill

Conforme:

_____
Signature over printed name/ Date
Evaluator

_____
Signature over printed name/ Date
Student

Comments/Suggestions:_____
_____
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References:

Harris-Fry, N. (2017, March 10). How To Do The Dead Bug Exercise. Retrieved June 14, 2019, from https://www.coachmag.co.uk/abs-workouts/6384/the-dead-bug-exercise

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Petrucci, K., & Flynn, P. (n.d.).4 Types of Simple Rolling Movements for Paleo Fitness. Retrieved June 16, 2019, from https://www.dummies.com/health/exercise/4-types-of-simple-rolling-movements-for-paleo-fitness/

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Chapter 6

BASIC STRENGTH TRAINING AND OTHER HOME EXERCISES

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. Identify the different physical activities at home and its benefits
2. Discuss the importance of physical activities at home.
3. Perform properly the different exercises at home to enhance core muscles, upper body muscles and lower body muscle

Course Outline:

- Overview
- Discussion
    - Health Benefits of Muscle Fitness Exercise
    - Basic Guidelines for A Safe and Effective Muscular Strength Exercises
    - Sample Strength Training Exercises
        - Upper Extremity
        - Lower extremity
        - Core Exercises
    - Other Training Methods and Types of Equipment
    - Physical Activities at home
    - Other Suggested Exercises
- Summary
- Suggested Media at Home
- Activities and Assessments
    - Reflective Questions
    - Knowledge Check
    - Performance Task
- References

II. OVERVIEW

The information in this section can be used to improve your level of fitness and health. Strength training exercises will help you to improve the strength and tone your muscles. Strengthening your core muscles, it helps to improve your posture and mobility because it will stabilize your spine. Strengthening your upper and lower body muscles make you upper body strong. In order to active this; you must need to do it a regular basis.

III. COURSE MATERIALS: DISCUSSION

A. Health Benefits of Muscle Fitness Exercise

Good muscle fitness and regular muscle fitness exercise contribute to the prevention of chronic lifestyle diseases and early death. Much of the early research focused on the health benefits of moderate to vigorous aerobic activities. In recent years, more and more evidence has shown that muscle fitness exercise and good muscle fitness have important health benefits. The list below summarizes guidelines, reports, and position statements of the ACSM, the AHA, the Institute of Medicine, and the National Strength and Conditioning Association (NSCA) regarding the reduced risk of chronic disease and early death from good muscle fitness.

· A lower risk of all-cause mortality

· Fewer heart attacks; a better heart disease risk profile
· A lower incidence of high blood pressure (hypertension)
· Reduced risk of some cancers
· A healthier blood lipid profile
· Better insulin sensitivity and improved blood glucose
· A reduced risk of metabolic syndrome
· A better body composition profile (e.g., less body fat and more lean muscle)
· Greater bone mass and less risk of osteoporosis
· A lower risk for osteoarthritis and musculoskeletal disorders

B. Basic Guidelines for A Safe and Effective Muscular Strength Exercises

1. Warm-up and warm-down for 5 to 10 minutes. (Walking for warm-up and stretching for warm-down)
2. Focus on form. Follow the mechanics properly. Align your body correctly and move smoothly through each exercise.
3. Pay attention on your breathing. Exhale as you work against, and inhale as you release.
4. Stick with your routine.
5. Get rest. Get your muscle to recover. At least 48 hours before the next training session.

C. Sample exercises to strengthen core muscles, upper body muscles and lower body muscles

Several different strategies for strength training are now available in the in the field of industry. A designed program specifically for an athlete is highly different from sedentary, middle-aged adult or regular college students. Thus, there is no single best strategy or technique for muscular strength and endurance training. Always consult with your medical doctor before engaging in any strength and endurance training program

How often to do it

If you are beginner in this type of exercises, it is advisable to do it on a regular basis which 3 training session per week so that you can gain the most benefits, or 2 sessions per week is still very good to start. It is recommended that you do not to train the same muscles on consecutive days.

I. Upper Body Muscles Exercises

A. Push-up - an exercise forearm muscles performed pushing the body up and down with the arms while lying face down with the palms on the floor. Major muscles involve in this exercise are deltoids, biceps, triceps, pectoralis muscles, erector spinae and rectus abdominis.





Mechanics:

1. Execute the prone lying position to the floor.
2. Keep your body straight line from head to feet without arching at your back.
3. Press up into a full elbow and knee extension, with hands placed slightly wider of the shoulder.
   (3 sets by 12 repetitions, 45 secs rest between sets)

B. Knee push-ups- it is also known as beginner push-ups. Helps you to strengthen your core group muscles and upper extremities muscles.





Mechanics:

1. Get into a high plank position from your knees.
2. Maintaining a straight line from your head to your knees, bend your elbows to lower yourself down to the ground. Keep your elbows at a 45-degree angle.
3. Push back up to start.
   (3 sets by 12 repetitions, 45 secs rest between sets)

C. Bicep Curl- an exercise using a dumbbell (bottled water) that held in hand in hand with a motion of flexion and extension.

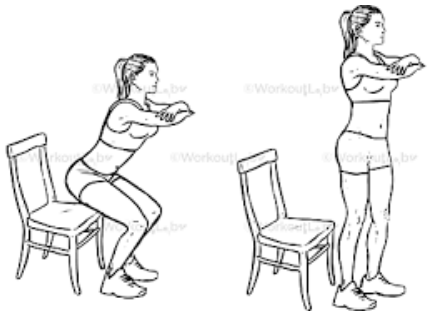


Mechanics:

1. The exercise requires you to stand up straight chest up shoulders back and neck in a neutral position.
2. Feet shoulder width apart.
3. The exercise begins with the dumbbells in hand with a supinated position, arms fully extended and close to the torso.
4. Flex the arm until the dumbbell is about 2-3 inches from the chest and shoulder area.
5. Ensure that the flexion is in controlled form.
6. You may alternate arms or you may lift one at a time.
7. To complete the exercise you must extend the arms back to the starting position.
8. Slow and controlled motion is advised.
   (3 sets by 12 repetitions, 45 secs rest between sets)

II. Lower Body Muscles Exercises

A. Chair squat- helps to strengthen your legs and core muscles, which helps you to move easier every day.



Mechanics:

1. Stand in front of the chair with your feet shoulder-width apart (athletic stance), toes pointed slightly out.
2. Hinging at your hips and bending your knees, lower back and down until your bottom touches the chair, allowing your arms to extend out in front of you.
3. Push up through your heels and return to the starting position.
   (3 sets by 12 repetitions, 45 secs rest between sets)

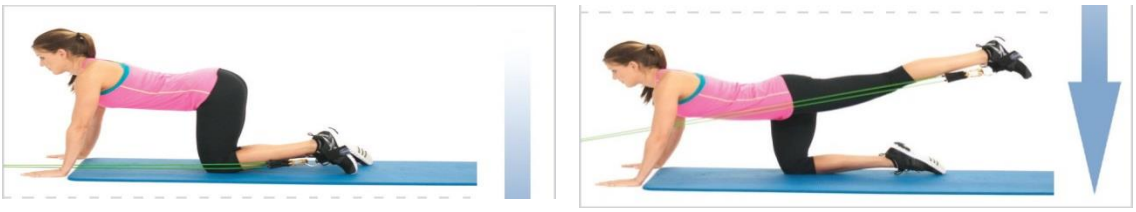
B. Stationary lunge- exercise that helps you to strengthen your quads, glutes, and hamstring muscles.



Mechanics:

4. Start with feet shoulder width apart. Put your right foot in front and must be flat on the ground, and your left foot should be up on its toes. Both legs are in 90 degrees position.
5. Bend your knees and lunge, stopping when your right thigh is parallel to the ground.
6. Push up through your right foot to return to the starting position. Repeat for desired number of reps, and then switch legs.
   (3 sets by 12 repetitions, 45 secs rest between sets)

C. Straight-leg donkey kick- primary muscle involve in this exercise are the glutes group muscles.



Mechanics:

1. Get on all fours, with your hands aligned with your shoulders and your knees aligned with your hips.
2. Keeping your back straight, push your right foot out to the imaginary wall behind you while keeping your leg straight.
3. Your foot should remain flexed (toes pointing down to the floor) throughout. Take care to keep your hips square to the ground. Squeeze your buttocks at the top.
4. Return to the starting position. Repeat for the desired number of reps. Repeat on the other leg.
   (3 sets by 12 repetitions, 45 secs rest between sets)

D. Tip Toe Walk. An exercise for the gastrocnemius and soleus.



Mechanics

1. Stand on your tip toes then walk
2. Do not allow your heels to touch the ground at any given time.
   (3 sets, 1 minute per set, 45 secs rest between sets)

III. Core Muscle Group Exercises

A. Bridge – activate your core. It is good example of warm-up exercise.



Mechanics:

1. Laying on back with your knees bent shoulder width apart, feet flat on the floor, and your arms extended by your sides.
2. Pushing through your feet and bracing your core, elevate your hips off the ground until your hips knees and shoulders are in straight line, squeezing your glutes at the top.
3. Slowly return to the starting position and repeat.
   (2 sets of 10 bridges, 45 sec rest between set)

B. Forearm plank- simple and effective body



weight exercises. Primarily it strengthens your core, muscles that connect your upper body and lower body, also it includes arms, shoulders, and gluteus.

Mechanics:

1. Assume a plank position on your forearms. Your body should form a straight line from head to feet.
2. Ensure your lower back and hips don't sag. Hold the position for 30 seconds to 1 minute. (3 sets of 30 seconds per set, 45 secs rest between rest)

C. Plank to Single Arm Reach - exercise for your lower and abdominal muscles to improve core stability and balance.





Mechanics:

1. Begin in a plank position with your shoulders over your wrists. Keep your body in a straight line form.

2. Raise your right arm in front while keeping your abdominal muscle tight. Repeat on the other side.

(2 sets of 10 plank to single arm reach, 45 secs rest between set)

C. Sit-ups – it is great core workout, which can be done even without equipment.





Mechanics:

1. Laying on your back on the floor.
2. Bend your knees or hook up on a secure brace. If you can't find, try to look a partner to hold you're both feet while doing the exercise.
3. Put your hand on your chest or on your ears.
4. To raise your body from the ground, you need to bend your hips and waist.
5. Lastly, lower your body back to starting position.
   (3 sets by 12 repetitions, 45 secs rest between sets)

Sample Strength Training Workout

Day 1

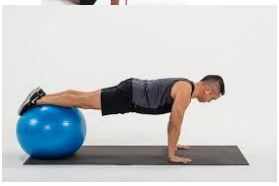
Exercise	Sets	Repetitions	Rest
Knee Push Ups	2	8-12	1 min.
Biceps Curl	2	8-12	1 min.
Chair Squat	2	8-12	1 min.
Straight Leg donkey Kick	2	8-12	1 min.
Scapular Retraction and Protraction	2	8-12	1 min.
Press Up	2	8-12	1 min.
Bird Dog Series	2	8-12	1 min.

Day 2

Exercise	Set/s	Repetition	Rest
Bracing the Core	1	8-12	1 min.
Dead Bug Series	1	8-12	1 min.
Sit-Ups	1	8-12	1 min.
Forearm Plank	1	30 seconds (duration)	1 min.
Plank to Single Arm Reach	1	8-12	1 min.

D. Other Training Methods and Types of Equipment





•Resistance Band Exercises. Exercise bands or elastic bands are elastic strips or tube of rubber material, light weight, inexpensive, and portable being used for resistance training exercises. A lot of free weight exercises can be adapted for resistance bands such as front squat, bicep curls, and other core exercises.

•Stability Balls. It is an extra-large inflatable ball use to create exercises for the entire body, but commonly used for core strengthening (abdomen, back, and chest) that are essential in avoiding back problems.



•Pilates. It was developed by Joseph Pilates, a German gymnast and boxer in the early 20th century. It is a method of exercise that involves muscular strength and endurance movements and low-impact flexibility exercises. It focuses on strengthening the core muscles, proper postural alignment, and muscle balance. It also emphasizes concentration, control, movement flow, and breathing.

E. Physical Activities at Home and Other Suggested Exercises

Physical activities at home will help you to improve your health and to reduce the risk of developing diseases. It also improves the quality of life. First, you need to choose activities that you enjoy. You should a consistent physical activity each week in order to improve the ability to do daily activities. Physical activities at home will help you also to improve your core, upper extremities and lower extremities muscles. Also in this chapter, you will learn the importance of strength training exercises for core, upper extremities, and lower extremities.

Gardening - practice of growing or culturing plants at home.

Household Chores – typically considered as task such as seeping the floor, washing dishes, doing laundry, feeding pets, preparing meals, cleaning bathrooms, and dusting.

Summary

1. Physical activities at home can helps you to improve your health and to reduce the risk of developing disease.
2. Physical activities at home also strengthen your core, upper extremities and lower extremities muscles.
3. Common physical activities at home includes gardening and doing household chores.

Suggested Media at Home

Visit this site https://virtualhealthpartners.com/benefits-of-exercising-from-home/ and list down the benefits of exercising at home

IV. ACTIVITIES/ASSESSMENTS

In this section, a variety of suggested activities and assessments were provided. It provides activities to assess the appreciation (reflective questions) and understanding (knowledge check) of the students on the topics provided. It also suggests options for practical application (a performance task) by providing a performance activity that includes the rubrics or criteria for rating. Remember that as a teacher, you have the academic freedom to choose what activities you would like to assign to your students, modify some of the tasks, or provide additional or other activities that you think will be more suitable for our dear students, recognizing the different capabilities and situations they have right now.

Activity Workseet 11. Reflective Question

Name:_____ Section:_____ Date:_____ Group no._____

What did you learn about yourself as you worked on this activity?

_____
_____
_____
_____
_____
_____
_____
_____
_____
_____

Activity Worksheet 12. Knowledge Check

Multiple Choice: Encircle the letter of the correct answer.

1. What is the main reason why physical activities at home are essential to a healthy life?
   a. Reduce the risk of developing disease
   b. Improve the ability to do daily activities
   c. Strengthen your core group muscles
   d. All of the above

2. What is the common exercise to improve pectoralis muscles?
   a. forearm plank
   b. sit-ups
   c. push-ups
   d. stationary lunge

3. Tip-toe walk is a good exercise to improve speed. Why?
   a. Develops the quadriceps
   b. Develops the soleus
   c. Develop the whole thigh muscles
   d. Develops gluteal muscles

4. Why do you need to perform the athletic stance in squat?
   a. Maintain balance
   b. Make exercise easier
   c. Prevent injuries
   d. Gives power

5. Chair squat helps you to move easier every day. Why?
   a. Develops your legs and core muscles
   b. Develops your Soleus
   c. Develops your rectus abdominis
   d. Develops your whole thigh muscles

Performance Task 9

Basic Strength Training Workout

Group Leader :_____ Section: _____
Members: _____ Date: _____
_____ Group no. _____
_____

Instruction: The class will be divided into 4-5 groups or as prescribed by the teacher. The student/s will perform in a virtual platform a 10-15 minute basic full body workout consisting: (1) Upper body; (2) core; and (3) lower body. Enough time will be given for the students to practice as a group and may consider using social media to collaborate with other members. Take note to consider the capabilities and health concerns of each member. The said routine will be submitted through online or in other form like usb or compact disk on the given date. For students who will not be able to work collaboratively, individual performance is advised.

Note: Warm-up and cool down is mandatory as part of the program, but no longer required to be part of the submission.

The student/s will be assessed based on the following rubrics:

Criteria	Excellent	Good	Average	Fair	Needs Improvement
	5	**4**	**3**	**2**	**1**
Movement Competency	Consistently and independently demonstrate superior ability in applying the required concepts and skills	Usually and independently demonstrate superior ability in applying the required concepts and skills	Sometimes demonstrates the ability to apply the required concepts and skills with assistance	Seldom demonstrates the ability to apply the required concepts and skills	Did not demonstrate any ability to apply the required concepts and skills
	5	**4**	**3**	**2**	**1**
Energy and Effort	Consistently performed the movements with energy and effort at all times	Movements were performed with energy and effort most of the time.	Movements were sometimes performed with energy and effort. Frequently makes excuses.	Movements were rarely performed with energy and effort. Always makes excuses	Movements were performed without energy and effort.
	5	**4**	**3**	**2**	**1**
Application of Principles of Training Exercise	The routine is performed correctly and considered the complete application of the principles of exercise such as: • Progression -Low to high impact. Simple to complex steps • Individuality -consider individual skills • Rest and recovery- proper routine design that provides recovery	The routine is performed correctly but considered the application of only two principles of exercise	The routine is performed correctly but considered the application of only one principle of exercise	The routine is performed incorrectly and considered the application of only two principles of exercise	The routine is performed incorrectly and did not consider the application of the principles of exercise
		Total Score:		**Rating:**	

Rating:

13-15	*Excellent*
10-12	*Good*
7- 9	*Average*
4- 6	*Fair*
3 and below	*Needs Improvement*

Conforme:

_____	_____
Signature over printed name/ Date	Signature over printed name/ Date
Evaluator	***Student***

Comments/Suggestions:_____

_____

_____

_____

_____

_____

_____

Performance Task 10

Skill Test. Video Presentation

Group Leader :_____ Section: _____
Members: _____ Date: _____
_____ Group no. _____
_____

Instruction: The class will be divided into 4-5 groups or as prescribed by the teacher. The students are expected to perform the prescribed exercises provided by the faculty. Enough time will be given for the students to practice as a group and may consider using social media to collaborate with other members. Take note to consider the capabilities and health concerns of each member. The said routine will be submitted through online or in other form like flashdrive or compact disk on the given date. **For students who will not be able to work collaboratively, individual performance is advised.**

The student/s will be assessed based on the following rubrics:

RUBRICS					
LEVEL OF PERFORMANCE					
CRITERIA	**5 EXCELLENT**	**4 VERY SATISFACTORY**	**3 SATISFACTORY**	**2 NEEDS IMPROVEMENT**	**1 POOR**
Proper Mechanics	Precisely followed the mechanics of the skill	Followed the mechanics of the skill but some errors were evident	Showed average mastery of the mechanics	Showed little mastery of the mechanics	Poorly executed the skills
Execution	Executed the skill with exceptional ability	Executed the skill correctly but some errors were evident	Showed an average level of accuracy in executing the skill	Showed little level of accuracy in executing the skill	Poorly executed the skill
Good form	Executed the high level of balance, control and ease of movement	Executed the skill with high level of balance, control and ease of movements	Executed the skill with and average level of balance, control and ease of movements	Executed the skill with little level of balance, control and ease movements	Executed the skill lousily
Enthusiasm	Showed a very high level of interest in executing the skill	Showed a high level of interest in executing the skill	Quite eager in executing the skill	Showed little interest in executing the skill	Did not show interest in executing the skill

Conforme:

_____
Signature over printed name/ Date
Evaluator

_____
Signature over printed name/ Date
Student

Comments/Suggestions:_____
_____
_____
_____
_____
_____
_____
_____

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Chapter 7

The FITNESS CHALLENGE

I. LEARNING OBJECTIVES:

At the end of this chapter, the students are expected to:

1. come up with a smart exercise program based on their goals and objectives supported by healthy eating habits;
2. improve fitness level based on the baseline (pre-test) score; and
3. demonstrate physical literacy in performing the different activities.

Course Outline:

- Introduction
- Discussion
    - General Guidelines on Fitness Plan Creation and Implementation
    - Setting Smart Goals
    - Suggested Fitness Workouts
    - Comprehensive Personal Fitness Worksheet
- Suggested Media at Home
- References

II. OVERVIEW

On this final section you are expected to create your own exercise program based on SMART Goals. You will be working with your teammates/buddy as you will be monitoring and assessing your program and activities along the way. You are about to apply all the basic concepts, principles, and safety measures independently.

III. COURSE MATERIALS: DISCUSSION

A. General Guidelines on Fitness Plan Creation and Implementation

You are highly advised to create program that adheres to the following instructions below:

- ✓ Consider health condition in crafting your program. Guidance from the teacher and parents is essential or approval from the doctor if needed.
- ✓ Make sure that the Physical Activity Readiness Questionnaire and You (PARQ&You) was properly accomplished, signed by the student and parents/guardian, and submitted to the teacher given on the beginning of the session.
- ✓ Be sure to apply proper exercise prescriptions that correspond to your goals. Review the pass lessons if needed.
- ✓ Ensure safe and conducive activity area. It is highly recommended to perform activities that can be done at home due to pandemic.
- ✓ Wear appropriate clothing and footwear to avoid possible injuries.
- ✓ Maintain proper hydration at all times.
- ✓ Strictly follow the minimum guidelines set by the government and WHO such as; maintaining proper physical distancing, frequent washing of hands, using of 70% alcohol, wearing of mask when outside, and alike.
- ✓ According to WHO, it is not advisable to wear mask when doing exercise as mask may reduce the ability to breathe and promotes the growth of microorganism. The important preventive measure during exercise is to maintain not less than one meter way from others. So it is advisable to do activities inside your home.

- ✓ Note: since the government imposed to wear mask outside, even if you are doing exercise routine, take your own risk and necessary precautions in wearing your mask in the outdoors.

B. Setting SMART Goals

Have you decided that it's time to make a change but aren't sure how to get started? Or have you already set more goals for yourself than you care to admit-but keep failing to reach them? It's time for a new way of thinking. No matter how big or small your goal-whether it's losing 5 or 50 pounds, walking a mile or running your first marathon-making change requires planning and SMART goal setting.

1. Specific. Your goal should be clear and easy to understand.
    - A common goal, "get healthy," is too general. There are so many ways to get healthy. How do you want to do it? Is it losing weight? Start exercising? Stop smoking? Break it down and it will be easier to manage.
    - Let's pick weight loss and make a SMART goal out of it together. For example, "I will lose weight."
2. Measurable. A goal to "lose weight" is not enough. How will you track your progress and how you will know when you have reached your goal? Making your goal measurable means adding a number.
3. Attainable. Before you can add a number, you have to know how high or low you want to go. It's good to 'shoot for the stars', but don't be too extreme. Likewise, a goal that is too easy is also not very motivating. Only you know your limits.

    - Let's take our goal above. What percentage is attainable for you? Research suggests that a 5-10% weight loss is attainable for most overweight people.
    - A measurable, attainable goal could be, "I will lose 7% of my body weight."

1. Relevant. Set goals that are important to where you are in your life right now. Don't set a goal that someone else is pressuring you to attain-that isn't very motivating.

    - Examine our goal so far. Does it seem relevant to you? If so, let's keep going. If you are not concerned about weight loss or this is not a good time in your life to focus on that, choose something that IS motivating to you.

2. Time-bound. Include an end-point. Knowing that you have a deadline motivates you to get started.

    - Since healthy weight loss is about 1-2 pounds per week, set your deadline accordingly. For our example we can use 3 months. "I will lose 7% of my body weight in 3 months."

Now we have a SMART goal! With a goal like this, it's a good idea to set a few more action-oriented SMART goals so that you have a game plan. Here are a few examples:

1. I will walk 5 days every week for 30 minutes each.
2. I will drink water instead of soda every day this week.
3. I will bring my lunch to work instead of eating out 4 days this week.

Becoming a better version of you starts by being SMART!

C. Suggested Fitness Workouts

- **Walking Program: 10,000 Steps Challenge**
    - Walking is one of the simplest and least expensive options to increase your physical activity level and improve your overall health. It is a weight-bearing exercise that helps

maintain bone density and is easy on joints. Walking at a brisk pace is considered moderate-intensity physical activity, and doing this most days of the week for 30 minutes or more enables you to meet the criteria for physical activity for health benefits recommended by ACSM and the American Heart Association.

Please see link for full details: https://www.acsm.org/docs/default-source/files-for-resource-library/starting-a-walking-program.pdf?sfvrsn=85e9d2d9_2

- **Aerobic Dance Fitness**
  - Choreographed series of dance steps and exercise done with music. Varied dance forms for this activity like jazz, latin, ballroom and hip-hop are available for different interests and abilities. A routine may be consists of stepping, walking, stepping, jogging, kicking and arm swinging movements. It promotes and develops cardiorespiratory fitness and at the same time, fun way to exercise and work out.

    For sample dance work out: ***30 Mins Aerobic Dance Workout - Bipasha Basu Break free Full Routine - Full Body Workout*** *https://www.youtube.com/watch?v=tj9d6aBOzDo*

- **Body Weight Exercises**

No Equipment Calisthenics These are exercises that use body weight as resistance for strength training. Some common exercises include chair-dips, curl-ups, lunges, pull-ups, push-ups, and squats

For sample HIIT work out: ***BEST 15 min Beginner Workout for Fat Burning (NO JUMPING HIIT!!!)*** *https://www.youtube.com/watch?v=bleOTMDa3_4*

- **Yoga Exercises**

Yoga is a systematic practice of physical exercise, breath control, relaxation, diet control, and positive thinking and meditation aimed at developing harmony in the body, mind, and environment. The practice entails low-impact physical activity, postures (called asanas), breathing techniques (pranayama), relaxation, and meditation. Most people are familiar with the physical poses or yoga positions but don't know that yoga involves so much more.

In the health fields, yoga techniques are being applied in health promotion programs, substance abuse treatment programs, and as a complementary treatment for diseases such as anxiety disorders, depression, coronary heart disease, cancers, and HIV/AIDS. Yoga is a low-cost self-help approach to well-being.

For sample yoga exercise: ***10 min Morning Yoga Full Body Stretch*** *https://www.youtube.com/watch?v=4pKly2JojMw*

- **Jump Rope Exercise**

Jump rope (American English) or skipping rope (British English) is the primary tool used in the game of skipping played by children and many young adults, where one or more participants jump over a rope swung so that it passes under their feet and over their heads. This may consist of one participant turning and jumping the rope, or a minimum of three participants taking turns, two of whom turn the rope while one or more jumps. This is called long rope. Sometimes the latter is played with two turning ropes; this form of the activity is called Double 54 Dutch and is more difficult. Jump-rope rhymes are often chanted beginning when the skipper jumps in and ending when the skipper messes up.

For sample jump rope routine: Jump Manila. Jump for Health https://www.facebook.com/Jumpmla/

For your additional references, please see/watch:

Flexibility Exercises

Pages 206-210 *Corbin, C.B., Welk, G., Corbin W.R., & Welk, K., (2016) Concepts of Fitness And Wellness: A Comprehensive Lifestyle Approach, Loose Leaf Edition 11th Edition. McGraw-Hill, New York. Pp. 206-210.*

Active Sports and Recreation

MILO Sports Clinic Online

https://www.youtube.com/playlist?list=PLfofdW2KZ3UxhBLilhJvnLTwOf0wTNn8F&fbclid=IwAR0SL2MYBPhl7_7Vmk2il1gqPAOYHZE5VBdzHSXtdRHlDFXNlzQZgFz25Ds

Final Performance Task

Comprehensive Personal Fitness Plan

Name:_____ Section:_____ Date:_____ Group no._____

Purpose: To establish a comprehensive fitness plan of lifestyle activity and to self-monitor progress in your plan. *(note: you may want to read again the basic concept and exercise prescription for before accomplishing this activity).*

I. Exercise Clearance

1. Is it safe for you to participate in an exercise program?
   [ ] Yes
   [ ] No *(If no, please, contact you doctor or instructor for further assistance)*
   [ ] with some limitations
   [ ] others, please specify
   Note: for any queries and assistance don't hesitate to contact yout instructor/teachers for assistance.

2. I fully understood all the exercise principles, basic fitness concepts, and exercise prescription guidelines that I will be utilizing in in planning and implementing my fitness program?
   [ ] Yes
   [ ] No

II. Step 1. Establishing your reasons

Write down some general reasons for developing your physical activity plan. Setting goals requires more specific statements of goals that are realistic and achievable. For people who are engaging in physical activity and exercise but not on a regular basis stage for a specific type of activity, it is recommended that you write only short-term physical activity goals (no more than 4 weeks).

1. _____
   _____
2. _____
   _____
3. _____
   _____

III. Step 2. Identify Your Needs Using Self-Testing Activities and Setting Fitness Goals

Use your results obtained from your previous self-assessment to determine your ratings. If you took more than one self-assessment for one component of physical fitness, select the rating that you think best describes your true fitness for that fitness component. If you were unable to do a self-assessment for some reason, just indicate "No Results".

	Current Status		**Fitness Category Goal**
Components	**Test Result**	**Fitness Category**	**Fitness Goal**
Body Composition			
Cardiorespiratory Fitness			
Flexibility			
Muscular Strength and Endurance			

IV. Step 3. Selecting Activities

In Chart below, indicate the specific activities you plan to perform from each area of the physical activity pyramid. If the activity you expect to perform is listed, note the number of minutes or reps/sets you plan to perform. If the activity you want to perform is not listed, write the name of the activity or exercise in the space designated as "Other." For moderate activities, active aerobics, and active sports and recreation, indicate the length of time the activity will be performed each day. For flexibility, indicate the number of repetitions for each exercise.

Note: Please be mindful of the current situation (Covid-19 Pandemic) in selecting your activities. Choose activities which are applicable given the situation, facilities, materials and most specially it follows guidelines set by the government with this time of pandemic.

Chart 2: Lifetime Physical Activity Selection *(please see the sample below)*

Lifestyle Activities	Min./Day	Active Aerobics	Min./Day	Active Sports and Recreation	Min./Day
Walking		Aerobic Dance Fitness		Badminton	
Yard Work		Biking		Mountain Climbing	
Gardening		Jogging or Running		Karate	
Active House work		Swimming		Arnis	
Biking as means of transportation		Jump Rope Activity		Bowling	
Other:		Stair Climbing		Table Tennis	
Other:		Other:		Other:	
Other:		Other:		Other:	
Other:		Other:		Other:	

Flexibility	Reps/Sets	Muscular Fitness	Reps/Sets
Calf stretch		Upper Extremity	
Hip and thigh stretch		-Push ups	
Groin stretch		- Pull ups	
Hamstring stretch		Lower extremity	
Back stretch (leg hug)		- Lunges	
Trunk twist		- Squats	
Pectoral stretch		Core Exercises	
Arm hug stretch		- Dead Bug Series	
Overhead arm stretch		- Bird Dog Series	
Yoga			
Taichi			
Pilates			
Other:		Other:	
Other:		Other:	
Other:		Other:	

V. Step 4. Making an Activity Plan

In chart below, indicate the activities that you choose from chart no.2. Also, specify time for the day you expect to perform the activity or exercise. Shade the circle once you accomplished the spcific activity

Chart 3: My Personal Physical Activity Plan *(please see the sample below)*

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Week 1	○ *Walking & Jogging 6:30am*	● *Flexibility* & Core Exercises 7:00pm*	● *Walking & Jogging 6:30am*	● *Muscular Fitness* Exercises 6:30am*	● *Flexibility* & Core Exercises 7:00pm*	● Active House Work 8:00am ○ Badminton 6:30am	● *Walking 6:30am* *Rest Day*
	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks
	I was not able to perform the planned exercise due to bad weather. I just perform core exercises.					*I was not able to play badminton due to other responsibility*	*We had a family day with kids at the park*

** Perform the specific exercise you checked in chart 2*

VI. Step 5. Monitoring and Evaluation

Make copies of ***My Personal Physical Activity Plan*** chart (one for each week that you plan to keep records). Each day, make a check by the activities you actually performed. From time to time check your goals to see if they have been accomplished. At some point, it will be necessary to reestablish your goals and create a revised activity plan. From time to time give

feedback, updates, and motivation to your teammates/group mates for them to achieve their goals. Positive reinforcement is highly needed.

Results

After performing your plan for a specific period of time, answer the question on the space provided.

How long have you been performing the plan? _____

Conclusions and Implications

1. In several sentences, discuss your devotion to your activity plan. Have you been able to stick with it? If so, do you think it is a plan you can do for a lifetime? If not, why do you think you are unable to do your plan?

_____
_____
_____
_____
_____
_____
_____
_____

2. In several sentences, discuss how you might modify your plan in the future.

_____
_____
_____
_____
_____
_____
_____
_____

3. In general, what have you learned on doing this kind of activity? Explain your answer.

_____
_____
_____
_____
_____
_____
_____
_____

This is to acknowledge and certify that the data provided herein are true and correct.

_____
Signature over printed name/ Date
Student

_____
Signature over printed name/ Date
Witness/Guardian

Note: In addition to this document, you are required to submit a narrative report of the summary of your activities from day 1 to the last day. It can be in 2-3 page written narrative report and/or a 2-3 minute video summary to be submitted as final requirement. It can be individual or collaborative work as prescribed by the teacher.

My Personal Physical Activity Plan *(Template)*

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Week 1	O	O	O	O	O	O	O
	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks
	Day 1	**Day 2**	**Day 3**	**Day 4**	**Day 5**	**Day 6**	**Day 7**
Week 2	O	O	O	O	O	O	O
	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks
	Day 1	**Day 2**	**Day 3**	**Day 4**	**Day 5**	**Day 6**	**Day 7**
Week 3	O	O	O	O	O	O	O
	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks
	Day 1	**Day 2**	**Day 3**	**Day 4**	**Day 5**	**Day 6**	**Day 7**
Week 4	O	O	O	O	O	O	O
	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks	Remarks

** Perform the specific exercise you checked in chart 2*

Suggested Media at Home

- **FITT Principle**
  https://www.youtube.com/watch?v=cVVqy6aUvK0
- **Principles of training**
  https://www.youtube.com/watch?v=eNcxtSF2-Ac
- **Quick guide to heart rate training | Polar**
  https://www.youtube.com/watch?v=xPFq9EpovNw
- **Physical Activity and Exercise Prescription**
  https://www.physio-pedia.com/Physical_Activity_and_Exercise_Prescription
- **4 General Principles of Exercise Prescription**
  https://www.chp.gov.hk/archive/epp/files/DoctorsHanbook_ch4.pdf

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Assoc. Prof. John Mark N. Nero is currently the faculty assistant of the Vice President for Student Affairs and Services of the Polytechnic University of the Philippines. A licensed professional teacher, he is teaching various subjects such as Physical Education, Sports, Recreation, Curriculum Development and Human Anatomy under the PUP College of Human Kinetics. A course specialist in the PUP Open University System and an Associate Professorial Lecturer in the Master in Physical Education and Sports Program of the PUP College of Education.

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An active member of numerous educational and research organizations and speaker of different seminars and trainings in local, regional and national setting. He is also an independent distributor, supervisor and wellness coach at Herbalife Philippines.

John Mark N. Nero was raised in San Jose Del Monte City Bulacan. He obtained his Bachelor in Physical Education at Polytechnic University of the Philippines in 2005 where he received Magna Cum Laude citation. He holds Master's Degree in Physical Education and Sports from Polytechnic University of the Philippines and currently taking his Doctor in Educational Management in the same University

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Rhene A. Camarador is a licensed Professional Teacher since 2007. He received his undergraduate and master's degrees in Physical Education and Sports from the Polytechnic University of the Philippines. He has been teaching in several schools from public and private sectors for 15 years now and counting. At present he is a regular full-time faculty member of PUP's College of Human Kinetics and College of Education-Graduate Studies. He attended and presented research papers at various international conferences. A published author in international peer-reviewed and Scopus journal indexed. A member of several organizations such as; Professional Pickleball Association of the Philippines; International Federation of Physical Education, Fitness & Sports Science Association (member), India; and the International Organization for Health, Sports, and Kinesiology (Member). He has been into mountain climbing, backpacking, camping, biking, running, and other outdoor activities for almost 14 years. Also a council member of the Kamandag Barikus Mountaineers' Association since 2007 with more than 40 mountains conquered and 15 advocacy and outreach programs organized with KBMA. With his passion to promote outdoor adventure activities for all, he established Outdoor Adventure Education Philippines (OAEPhil) on April 26, 2021. A non-stock and non-profit advocacy group that promotes and develops outdoor adventure education in the Philippines and the principles of experiential education in the natural setting.

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